Use Cloud Volumes ONTAP

Cloud Volumes ONTAP

NetApp
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Use Cloud Volumes ONTAP

License management

Manage capacity-based licenses

Manage your capacity-based licenses from the BlueXP digital wallet to ensure that your NetApp account has enough capacity for your Cloud Volumes ONTAP systems.

Capacity-based licenses enable you to pay for Cloud Volumes ONTAP per TiB of capacity.

The *BlueXP digital wallet* enables you to manage licenses for Cloud Volumes ONTAP from a single location. You can add new licenses and update existing licenses.

> While the actual usage and metering for the products and services managed in BlueXP are always calculated in GiB and TiB, the terms GB/GiB and TB/TiB are used interchangeably. This is reflected in the Cloud Marketplace listings, price quotes, listing descriptions, and in other supporting documentation

Learn more about Cloud Volumes ONTAP licenses.

How licenses are added to the BlueXP digital wallet

After you purchase a license from your NetApp sales representative, NetApp will send you an email with the serial number and additional licensing details.

In the meantime, BlueXP automatically queries NetApp's licensing service to obtain details about the licenses associated with your NetApp Support Site account. If there are no errors, BlueXP automatically adds the licenses to the digital wallet.

If BlueXP can't add the license, you'll need to manually add them to the digital wallet yourself. For example, if the Connector is installed in a location that doesn't have internet access, you'll need to add the licenses yourself. Learn how to add purchased licenses to your account.

View the consumed capacity in your account

The BlueXP digital wallet shows you the total consumed capacity in your account and the consumed capacity by licensing package. This can help you understand how you're being charged and whether you need to purchase additional capacity.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, keep **Capacity Based Licenses** selected.
3. View the packages summary, which shows you consumed capacity, total precommitted capacity, and total PAYGO capacity.
   - *Total consumed capacity* is the total provisioned capacity of all Cloud Volumes ONTAP systems in your NetApp account. The charging is based on each volume's provisioned size, regardless of local, used, stored, or effective space within the volume.
   - *Total precommitted capacity* is the total licensed capacity (BYOL or Marketplace Contract) that you purchased from NetApp.

- *Total PAYGO* is the total provisioned capacity using cloud marketplace subscriptions. Charging via PAYGO is used only if the consumed capacity is higher than the licensed capacity or if there is no BYOL license available in the BlueXP digital wallet.

  Here's an example of a Cloud Volumes ONTAP packages summary in BlueXP digital wallet:

4. Under the summary, view the consumed capacity for each of your licensing packages.
   - *Consumed capacity* shows you the capacity of the volumes for that package. For more details about a specific package, hover your mouse over the tooltip.

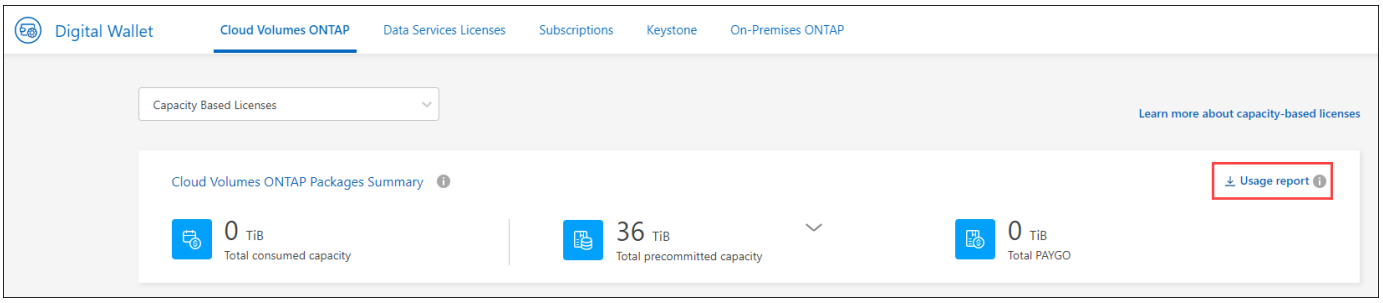
     To better understand the capacities that display for the Essentials package, you should be familiar with how charging works. Learn about charging for the Essentials package.

   - *Precommitted capacity* is the licensed capacity (BYOL or Marketplace Contract) that you purchased from NetApp.
     - *BYOL* shows you the licensed capacity that you purchased from NetApp for this package type.
     - *Marketplace Contracts* shows the licensed capacity that you purchased with a marketplace contract for this package type.
   - *PAYGO* shows you the consumed capacity by license consumption model.

     Here's an example for an account that has several licensing packages:

Download usage reports

Account administrators can download four usage reports from the digital wallet in BlueXP. These usage reports provide capacity details of your subscriptions and tell you how you're being charged for the resources in your Cloud Volumes ONTAP subscriptions. The downloadable reports capture data at a point in time and can be easily shared with others.



The following reports are available for download. Capacity values shown are in TiB.

- **High-level usage**: This report shows you exactly what's in the "Cloud Volumes ONTAP Packages Summary" card in the digital wallet. It includes the following information:
  - Total consumed capacity
  - Total precommitted capacity
  - Total BYOL capacity
  - Total Marketplace contracts capacity

    - Total PAYGO capacity
- **Cloud Volumes ONTAP package usage**: This report shows you exactly what's on the package cards in the digital wallet. It includes the following information for each package except the Optimized I/O package:
    - Total consumed capacity
    - Total precommitted capacity
    - Total BYOL capacity
    - Total Marketplace contracts capacity
    - Total PAYGO capacity
- **Storage VMs usage**: This report shows how charged capacity is broken down across Cloud Volumes ONTAP systems and storage virtual machines (SVMs). This information is not available on any screen in the digital wallet. It includes the following information:
    - Working environment ID and name (appears as the UUID)
    - Cloud
    - NetApp account ID
    - Working environment configuration
    - SVM name
    - Provisioned capacity
    - Charged capacity roundup
    - Marketplace billing term
    - Cloud Volumes ONTAP package or feature
    - Charging SaaS Marketplace subscription name
    - Charging SaaS Marketplace subscription ID
    - Workload type
- **Volumes usage**: This report shows how charged capacity is broken down by volumes in a working environment. This information is not available on any screen in the digital wallet. It includes the following information:
    - Working environment ID and name (appears as the UUID)
    - SVN name
    - Volume ID
    - Volume type
    - Volume provisioned capacity

> FlexClone volumes aren't included in this report because these types of volumes don't incur charges.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, keep **Capacity Based Licenses** selected and click **Usage report**.

   The usage report downloads.

3. Open the downloaded file to access the reports.

Add purchased licenses to your account

If you don't see your purchased licenses in the BlueXP digital wallet, you'll need to add the licenses to BlueXP so that the capacity is available for Cloud Volumes ONTAP.

What you'll need

- You need to provide BlueXP the serial number of the license or the license file.
- If you want to enter the serial number, you first need to add your NetApp Support Site account to BlueXP. This is the NetApp Support Site account that's authorized to access the serial number.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, keep **Capacity Based Licenses** selected and click **Add License**.
3. Enter the serial number for your capacity-based license or upload the license file.

   If you entered a serial number, you also need to select the NetApp Support Site account that's authorized to access the serial number.

4. Click **Add License**.

Update a capacity-based license

If you purchased additional capacity or extended the term of your license, BlueXP automatically updates the license in the digital wallet. There's nothing that you need to do.

However, if you deployed BlueXP in a location that doesn't have internet access, then you'll need to manually update the license in BlueXP.

What you'll need

The license file (or *files* if you have an HA pair).

> For more information on how to obtain a license file, see Obtain a system license file.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, click the action menu next to the license and select **Update License**.
3. Upload the license file.
4. Click **Upload License**.

Change charging methods

Capacity-based licensing is available in the form of a *package*. When you create a Cloud Volumes ONTAP working environment, you can choose from several licensing packages based on your business needs. If your needs change after you create the working environment, you can change the package at any time. For example, you might change from the Essentials package to the Professional package.

Learn more about capacity-based licensing packages.

About this task

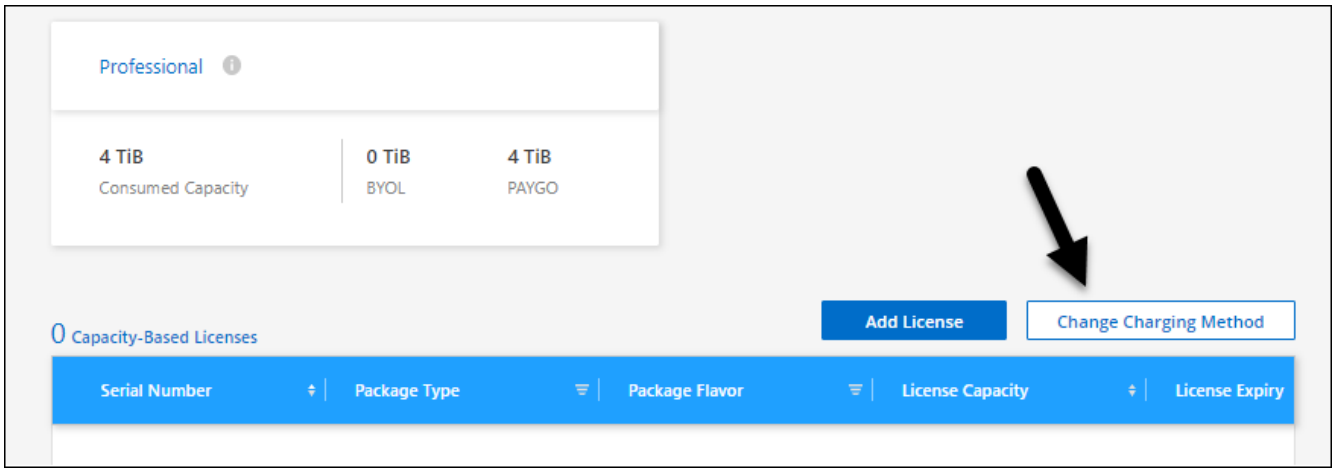
- Changing the charging method doesn't affect whether you're charged through a license purchased from NetApp (BYOL) or from your cloud provider's marketplace (pay as you go).

  BlueXP always attempts to charge against a license first. If a license isn't available, it charges against a marketplace subscription. No "conversion" is required for BYOL to marketplace subscription or vice versa.

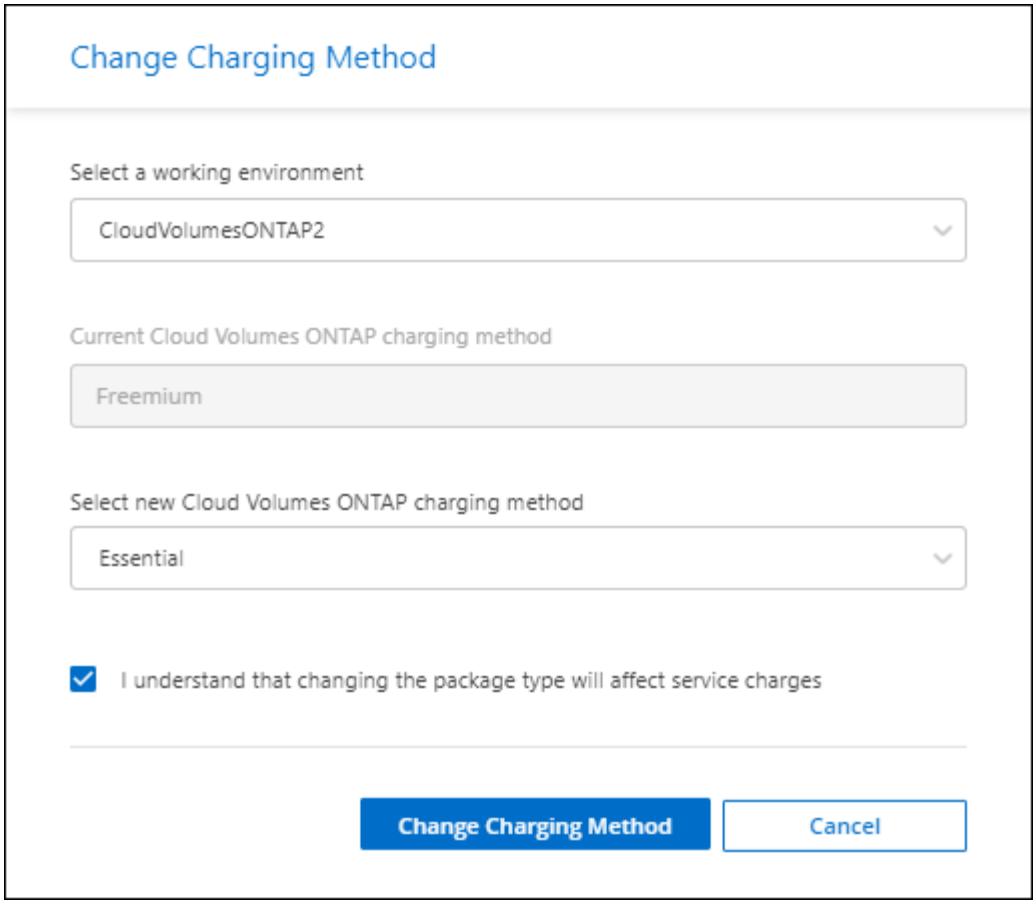
- If you have a private offer or contract from your cloud provider's marketplace, changing to a charging method that's not included in your contract will result in charging against BYOL (if you purchased a license from NetApp) or PAYGO.
- Changing to or from the Edge Cache license isn't supported.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, click **Change Charging Method**.



3. Select a working environment, choose the new charging method, and then confirm your understanding that changing the package type will affect service charges.



4. Click **Change Charging Method**.

Result

BlueXP changes the charging method for the Cloud Volumes ONTAP system.

You might also notice that the BlueXP digital wallet refreshes the consumed capacity for each package type to account for the change that you just made.

Remove a capacity-based license

If a capacity-based license expired and is no longer in use, then you can remove it at any time.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, click the action menu next to the license and select **Remove License**.
3. Click **Remove** to confirm.

Manage Keystone subscriptions

Manage your Keystone subscriptions from the BlueXP digital wallet by enabling subscriptions for use with Cloud Volumes ONTAP and by requesting changes to the committed capacity for your subscription's service levels. Requesting additional capacity for a service level provides more storage for on-premises ONTAP clusters or for Cloud Volumes ONTAP systems.

NetApp Keystone is a flexible pay-as-you-grow subscription-based service that delivers a hybrid cloud experience for customers who prefer OpEx to CapEx or leasing.

Learn more about Keystone

Authorize your account

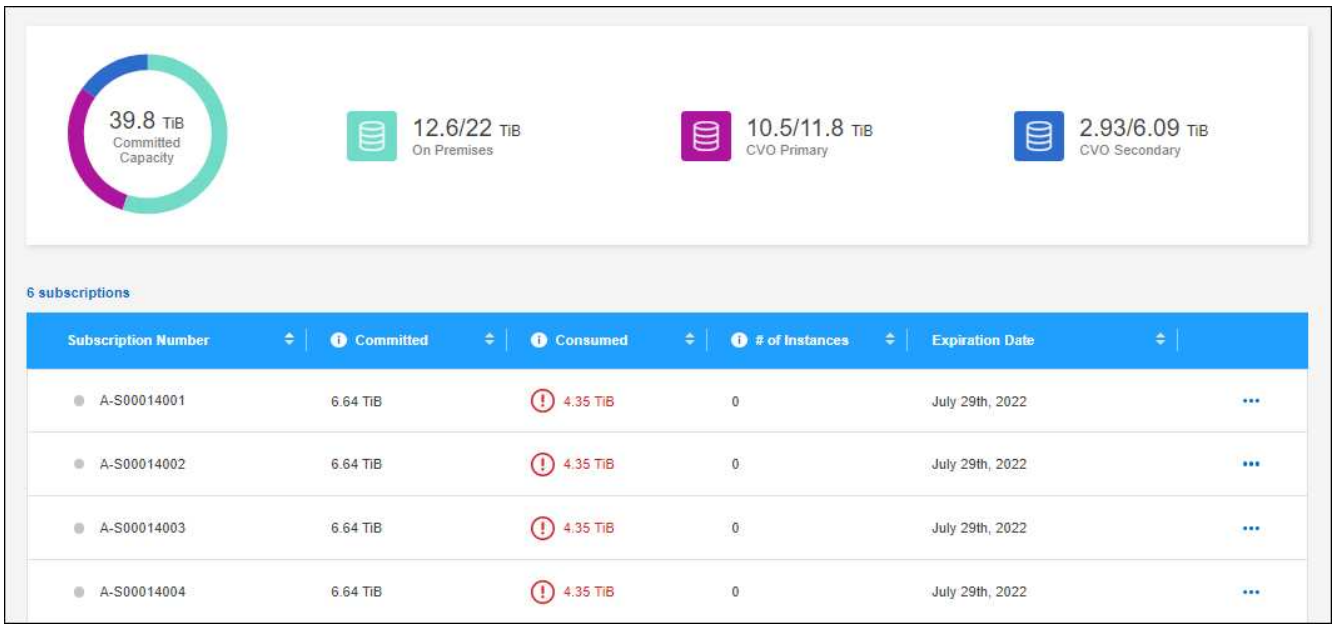
Before you can use and manage Keystone subscriptions in BlueXP, you need to contact NetApp to authorize your BlueXP user account with your Keystone subscriptions.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. Select **Keystone**.
3. If you see the **Welcome to NetApp Keystone** page, send an email to the address listed on the page.

   A NetApp representative will process your request by authorizing your user account to access the subscriptions.

4. Come back to the **Keystone Subscription** to view your subscriptions.

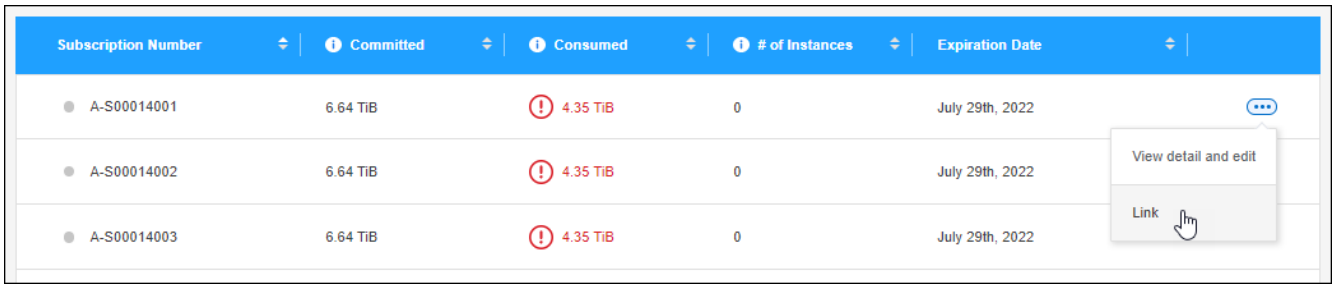


Link a subscription

After NetApp authorizes your account, you can link Keystone subscriptions for use with Cloud Volumes ONTAP. This action enables users to select the subscription as the charging method for new Cloud Volumes ONTAP systems.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. Select **Keystone**.
3. For the subscription that you want to link, click ••• and select **Link**.



Result

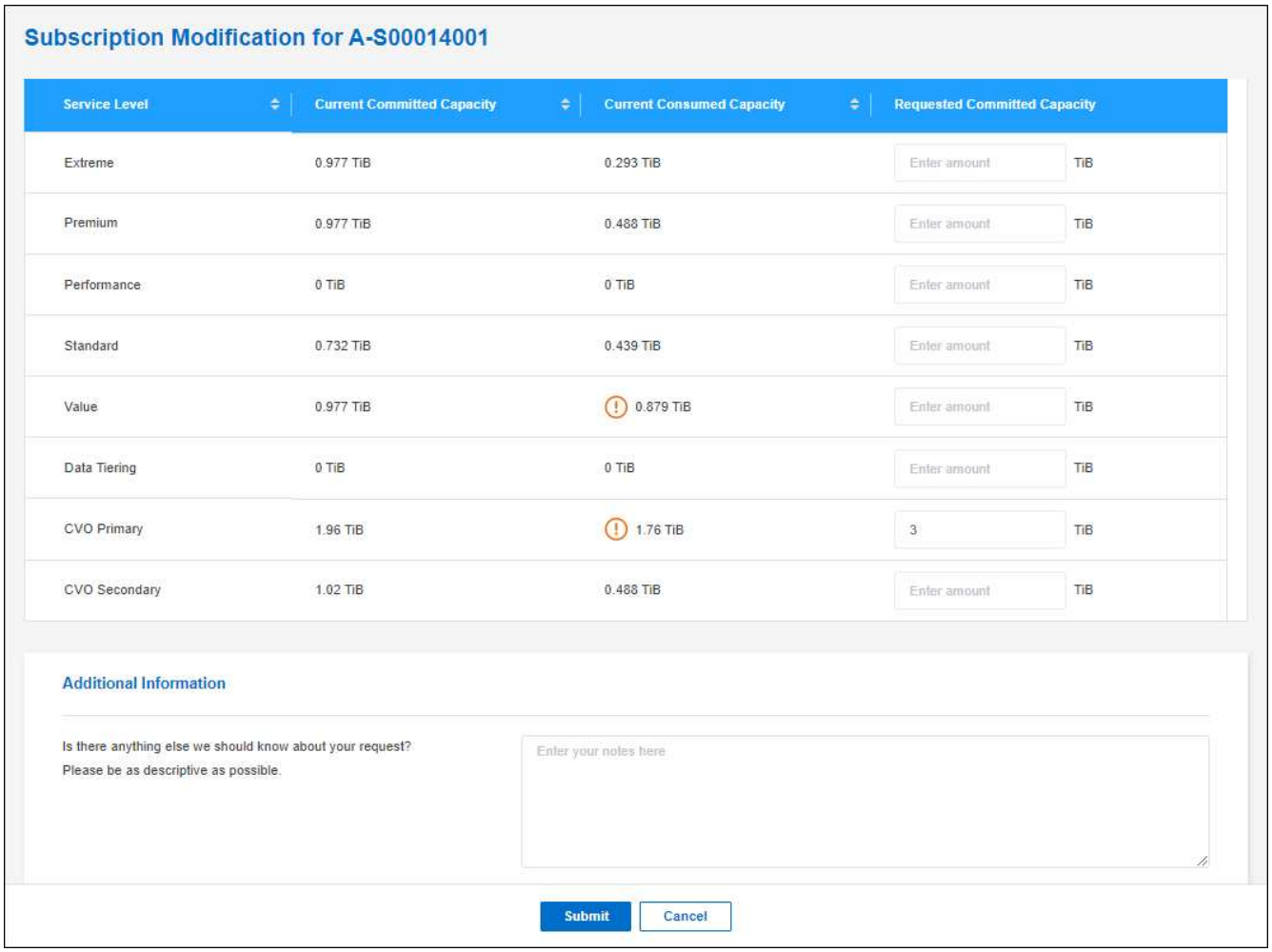
The subscription is now linked to your BlueXP account and available to select when creating a Cloud Volumes ONTAP working environment.

Request more or less committed capacity

If you want to change the committed capacity for your subscription's service levels, you can send a request to NetApp directly from BlueXP. Requesting additional capacity for a service level provides more storage for on-premises clusters or for Cloud Volumes ONTAP systems.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. Select **Keystone**.
3. For the subscription that you want adjust the capacity, click ••• and select **View detail and edit**.
4. Enter the requested committed capacity for one or more subscriptions.



5. Scroll down, enter any additional details for the request, and then click **Submit**.

Result

Your request creates a ticket in NetApp's system for processing.

Monitor usage

The BlueXP digital advisor dashboard enables you to monitor Keystone subscription usage and to generate reports.

Learn more about monitoring subscription usage

Unlink a subscription

If you no longer want to use a Keystone Subscription with BlueXP, you can unlink the subscription. Note that you can only unlink a subscription that isn't attached to an existing Cloud Volumes ONTAP subscription.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. Select **Keystone**.
3. For the subscription that you want to unlink, click ••• and select **Unlink**.

Result

The subscription is unlinked from your BlueXP account and no longer available to select when creating a Cloud Volumes ONTAP working environment.

Manage node-based licenses

Manage node-based licenses in the BlueXP digital wallet to ensure that each Cloud Volumes ONTAP system has a valid license with the required capacity.

Node-based licenses are the previous generation licensing model (and not available for new customers):

- BYOL licenses purchased from NetApp
- Hourly pay-as-you-go (PAYGO) subscriptions from your cloud provider's marketplace

The *BlueXP digital wallet* enables you to manage licenses for Cloud Volumes ONTAP from a single location. You can add new licenses and update existing licenses.

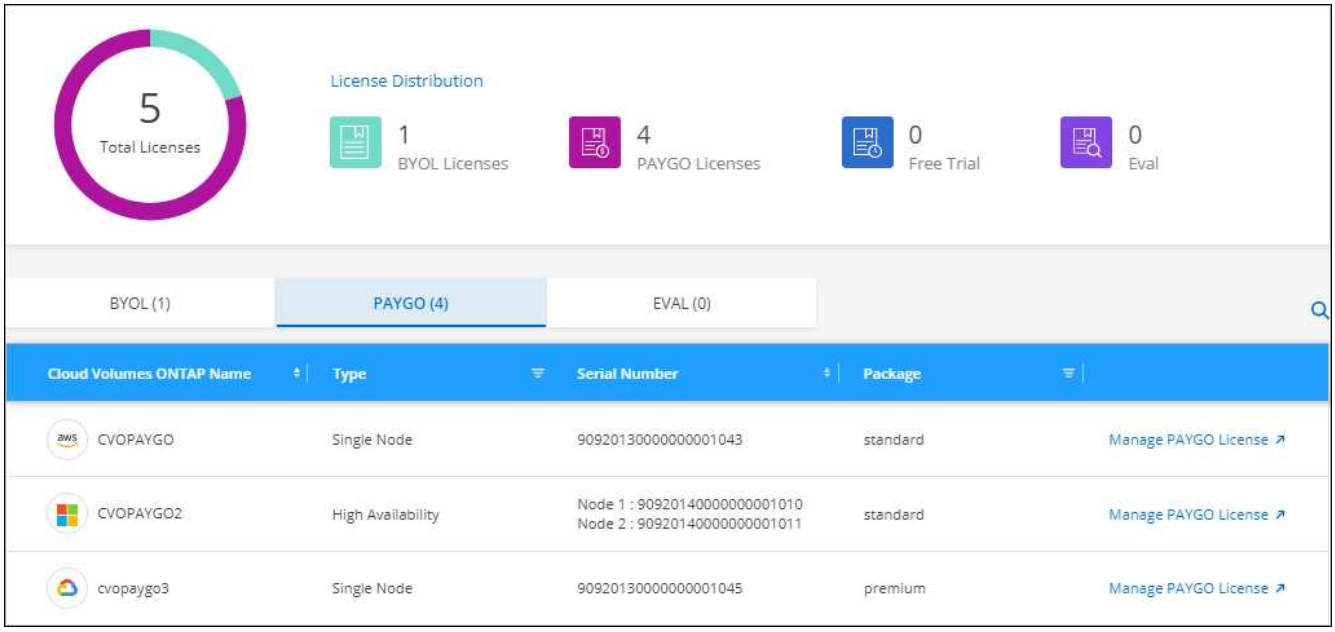
Learn more about Cloud Volumes ONTAP licenses.

Manage PAYGO licenses

The BlueXP digital wallet page enables you to view details about each of your PAYGO Cloud Volumes ONTAP systems, including the serial number and PAYGO license type.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. Click **PAYGO**.
4. View details in the table about each of your PAYGO licenses.



5. If needed, click **Manage PAYGO License** to change the PAYGO license or to change the instance type.

Manage BYOL licenses

Manage licenses that you purchased directly from NetApp by adding and removing system licenses and extra capacity licenses.

Add unassigned licenses

Add a node-based license to the BlueXP digital wallet so that you can select the license when you create a new Cloud Volumes ONTAP system. The digital wallet identifies these licenses as *unassigned*.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. Click **Unassigned**.
4. Click **Add Unassigned Licenses**.
5. Enter the serial number of the license or upload the license file.

   If you don't have the license file yet, refer to the section below.

6. Click **Add License**.

Result

BlueXP adds the license to the digital wallet. The license will be identified as unassigned until you associate it with a new Cloud Volumes ONTAP system. After that happens, the license moves to the **BYOL** tab in the digital wallet.

Exchange unassigned node-based licenses

If you have an unassigned node-based license for Cloud Volumes ONTAP that you haven't used, you can exchange the license by converting it to a BlueXP backup and recovery license, a BlueXP classification license, or a BlueXP tiering license.

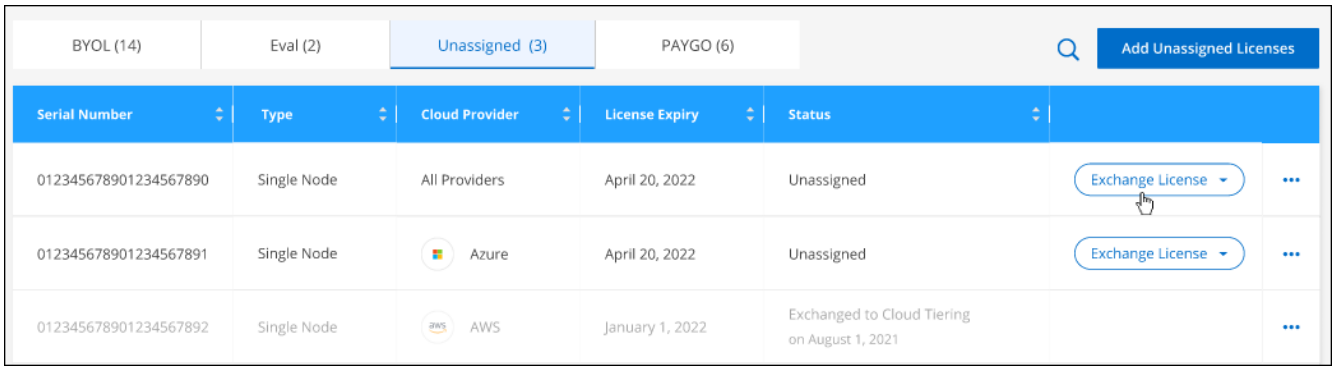
Exchanging the license revokes the Cloud Volumes ONTAP license and creates a dollar-equivalent license for the service:

- Licensing for a Cloud Volumes ONTAP HA pair is converted to a 51 TiB data service license
- Licensing for a Cloud Volumes ONTAP single node is converted to a 32 TiB data service license

The converted license has the same expiry date as the Cloud Volumes ONTAP license.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. Click **Unassigned**.
4. Click **Exchange License**.



5. Select the service that you'd like to exchange the license with.
6. If you're prompted, select an additional license for the HA pair.
7. Read the legal consent and click **Agree**.

Result

BlueXP converts the unassigned license to the service that you selected. You can view the new license in the **Data Services Licenses** tab.

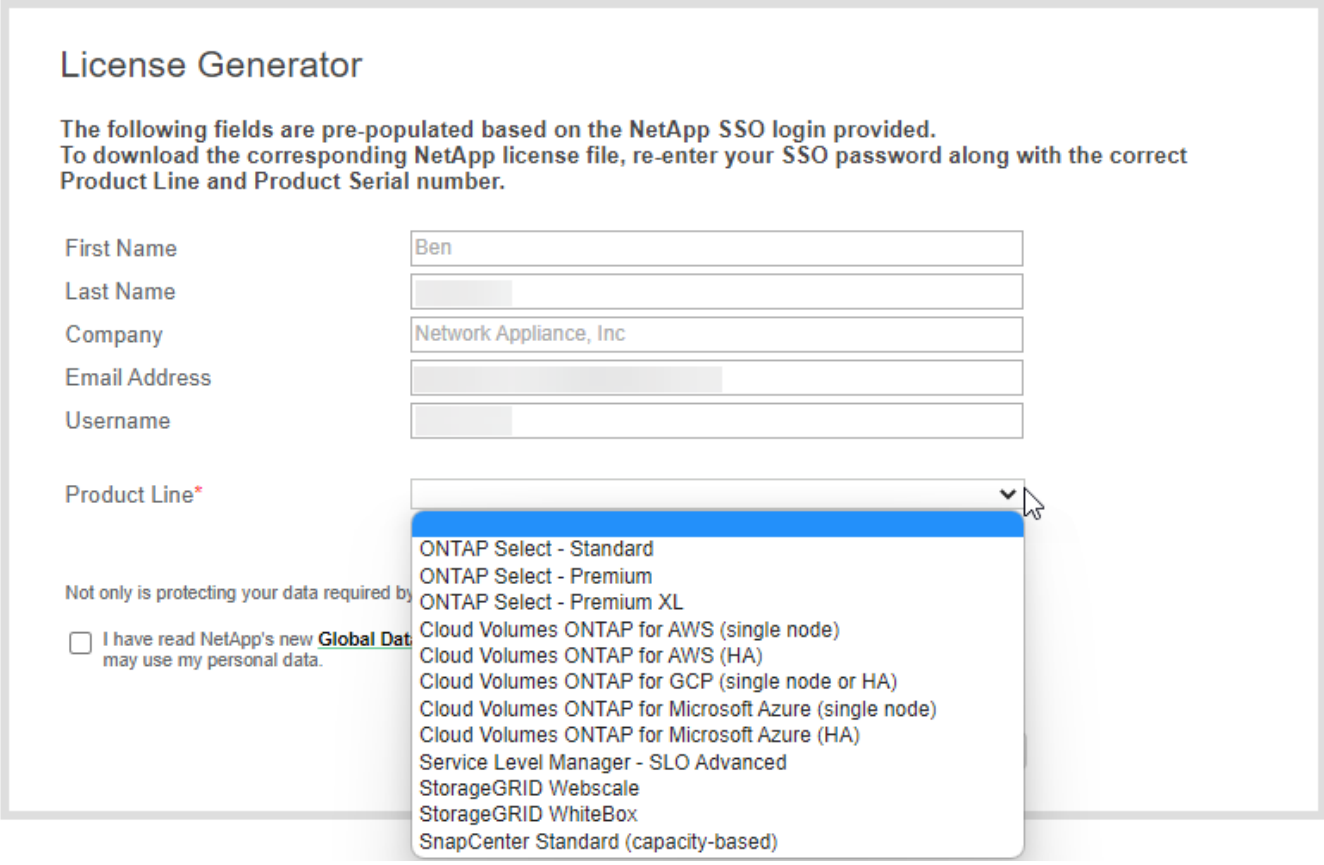
Obtain a system license file

In most cases, BlueXP can automatically obtain your license file using your NetApp Support Site account. But if it can't, then you'll need to manually upload the license file. If you don't have the license file, you can obtain it from netapp.com.

Steps

1. Go to the NetApp License File Generator and log in using your NetApp Support Site credentials.
2. Enter your password, choose your product, enter the serial number, confirm that you have read and accepted the privacy policy, and then click **Submit**.

   Example



3. Choose whether you want to receive the serialnumber.NLF JSON file through email or direct download.

Update a system license

When you renew a BYOL subscription by contacting a NetApp representative, BlueXP automatically obtains the new license from NetApp and installs it on the Cloud Volumes ONTAP system.

If BlueXP can't access the license file over the secure internet connection, you can obtain the file yourself and then manually upload the file to BlueXP.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. In the **BYOL** tab, expand the details for a Cloud Volumes ONTAP system.
4. Click the action menu next to the system license and select **Update License**.
5. Upload the license file (or files if you have an HA pair).
6. Click **Update License**.

Result

BlueXP updates the license on the Cloud Volumes ONTAP system.

Manage extra capacity licenses

You can purchase extra capacity licenses for a Cloud Volumes ONTAP BYOL system to allocate more than the 368 TiB of capacity that's provided with a BYOL system license. For example, you might purchase one extra license capacity to allocate up to 736 TiB of capacity to Cloud Volumes ONTAP. Or you could purchase three

extra capacity licenses to get up to 1.4 PiB.

The number of licenses that you can purchase for a single node system or HA pair is unlimited.

Add capacity licenses

Purchase an extra capacity license by contacting us through the chat icon in the lower-right of BlueXP. After you purchase the license, you can apply it to a Cloud Volumes ONTAP system.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. In the **BYOL** tab, expand the details for a Cloud Volumes ONTAP system.
4. Click **Add Capacity License**.
5. Enter the serial number or upload the license file (or files if you have an HA pair).
6. Click **Add Capacity License**.

Update capacity licenses

If you extended the term of an extra capacity license, you'll need to update the license in BlueXP.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. In the **BYOL** tab, expand the details for a Cloud Volumes ONTAP system.
4. Click the action menu next to the capacity license and select **Update License**.
5. Upload the license file (or files if you have an HA pair).
6. Click **Update License**.

Remove capacity licenses

If an extra capacity license expired and is no longer in use, then you can remove it at any time.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. In the **BYOL** tab, expand the details for a Cloud Volumes ONTAP system.
4. Click the action menu next to the capacity license and select **Remove License**.
5. Click **Remove**.

Convert an Eval license to a BYOL

An evaluation license is good for 30 days. You can apply a new BYOL license on top of the evaluation license for an in-place upgrade.

When you convert an Eval license to a BYOL, BlueXP restarts the Cloud Volumes ONTAP system.

- For a single-node system, the restart results in I/O interruption during the reboot process.
- For an HA pair, the restart initiates takeover and giveback to continue serving I/O to clients.

Steps

1. From the BlueXP navigation menu, select **Governance > Digital wallet**.
2. On the **Cloud Volumes ONTAP** tab, select **Node Based Licenses** from the drop-down.
3. Click **Eval**.
4. In the table, click **Convert to BYOL License** for a Cloud Volumes ONTAP system.
5. Enter the serial number or upload the license file.
6. Click **Convert License**.

Result

BlueXP starts the conversion process. Cloud Volumes ONTAP automatically restarts as part of this process. When it's back up, the licensing information will reflect the new license.

Change between PAYGO and BYOL

Converting a system from PAYGO by-node licensing to BYOL by-node licensing (and vice versa) isn't supported. If you want to switch between a pay-as-you-go subscription and a BYOL subscription, then you need to deploy a new system and replicate data from the existing system to the new system.

Steps

1. Create a new Cloud Volumes ONTAP working environment.
2. Set up a one-time data replication between the systems for each volume that you need to replicate.

   Learn how to replicate data between systems

3. Terminate the Cloud Volumes ONTAP system that you no longer need by deleting the original working environment.

   Learn how to delete a Cloud Volumes ONTAP working environment.

Volume and LUN administration

Create FlexVol volumes

If you need more storage after you launch your initial Cloud Volumes ONTAP system, you can create new FlexVol volumes for NFS, CIFS, or iSCSI from BlueXP.

BlueXP provides several ways to create a new volume:

- Specify details for a new volume and let BlueXP handle the underlying data aggregates for you. Learn more
- Create a volume on a data aggregate of your choice. Learn more
- Create volume from a template to optimize the volume for the workload requirements for certain applications, such as databases or streaming services. Learn more
- Create a volume on the second node in an HA configuration. Learn more

Before you get started

A few notes about volume provisioning:

- When you create an iSCSI volume, BlueXP automatically creates a LUN for you. We’ve made it simple by creating just one LUN per volume, so there’s no management involved. After you create the volume, use the IQN to connect to the LUN from your hosts.
- You can create additional LUNs from System Manager or the CLI.

Create a volume

The most common way to create a volume is to specify the type of volume that you need and then BlueXP handles the disk allocation for you. But you also have the option to choose the specific aggregate on which you want to create the volume.

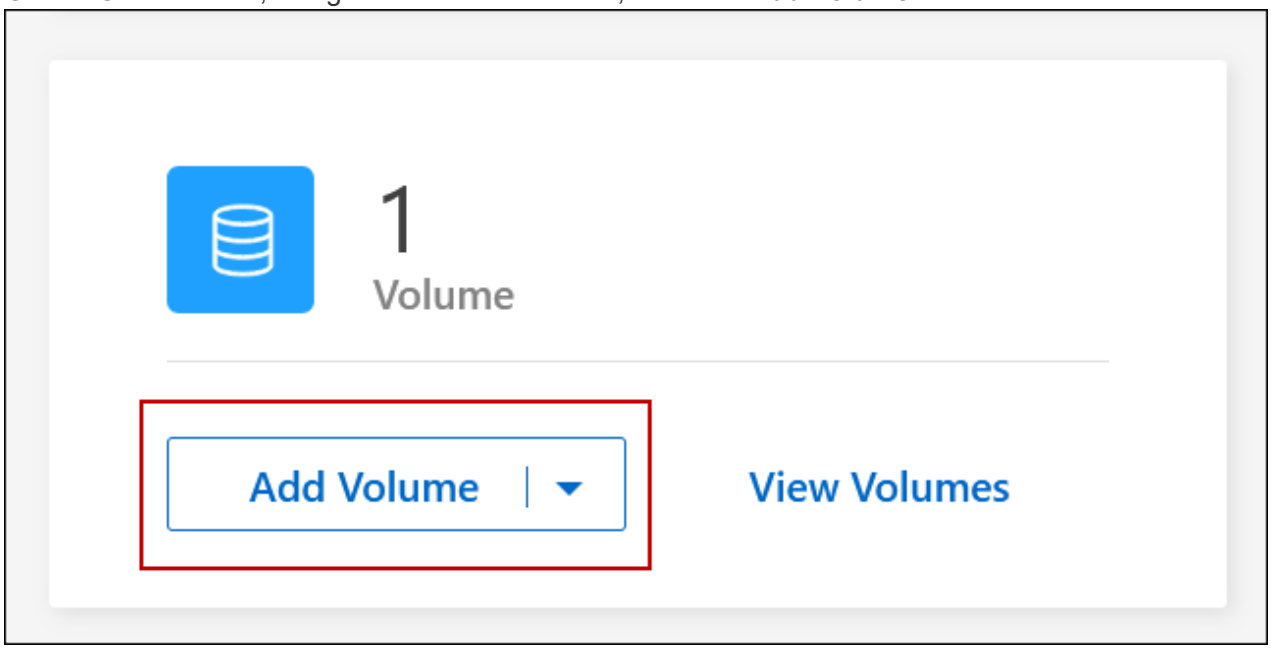
Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the name of the Cloud Volumes ONTAP system on which you want to provision a FlexVol volume.
3. Create a new volume by letting BlueXP handle the disk allocation for you, or choose a specific aggregate for the volume.

   Choosing a specific aggregate is recommended only if you have a good understanding of the data aggregates on your Cloud Volumes ONTAP system.

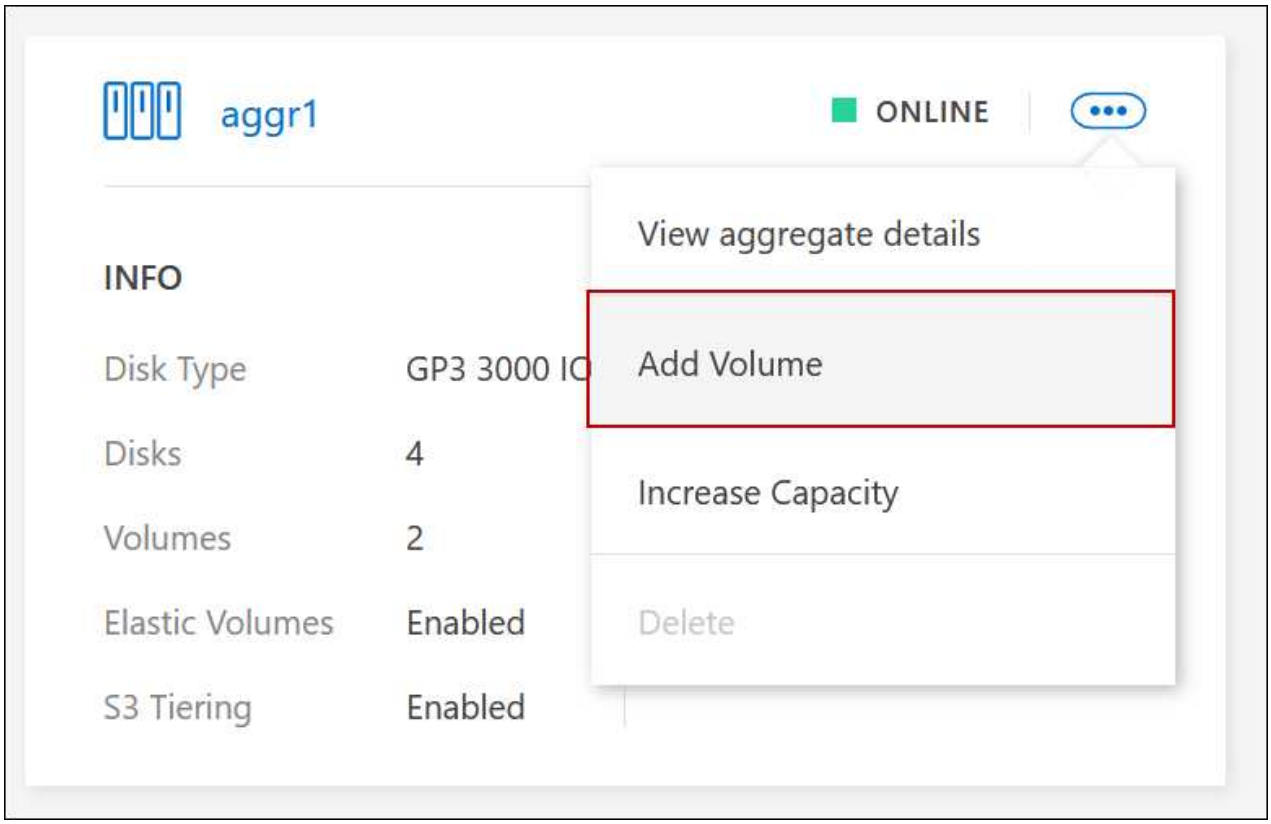
Any aggregate

On the Overview tab, navigate to the Volumes tile, and click **Add Volume**.



Specific aggregate

On the Aggregates tab, navigate to the desired aggregate tile. Click the menu icon, and then click **Add Volume**.



4. Follow the steps in the wizard to create the volume.
   a. **Volumes, Details, Protection, and Tags**: Enter basic details about the volume and select a Snapshot

policy.

Some of the fields on this page are self-explanatory. The following list describes fields for which you might need guidance:

Field	Description
Volume Name	The identifiable name you can enter for the new volume.
Volume Size	The maximum size that you can enter largely depends on whether you enable thin provisioning, which enables you to create a volume that is bigger than the physical storage currently available to it.
Tags	Tags that you add to a volume are associated with the Application Templates service, which can help you organize and simplify the management of your resources.
Storage VM (SVM)	A storage VM is a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an SVM or a vserver. Cloud Volumes ONTAP is configured with one storage VM by default, but some configurations support additional storage VMs. You can specify the Storage VM for the new volume.
Snapshot Policy	A Snapshot copy policy specifies the frequency and number of automatically created NetApp Snapshot copies. A NetApp Snapshot copy is a point-in-time file system image that has no performance impact and requires minimal storage. You can choose the default policy or none. You might choose none for transient data: for example, tempdb for Microsoft SQL Server.

b. **Protocol**: Choose a protocol for the volume (NFS, CIFS, or iSCSI) and then provide the required information.

   If you select CIFS and a server isn't set up, BlueXP prompts you to set up CIFS connectivity after you click **Next**.

   Learn about supported client protocols and versions.

   The following sections describe fields for which you might need guidance. The descriptions are organized by protocol.

NFS

Access control

Choose a custom export policy to make the volume available to clients.

Export policy

Defines the clients in the subnet that can access the volume. By default, BlueXP enters a value that provides access to all instances in the subnet.

CIFS

Permissions and users/groups

Enables you to control the level of access to an SMB share for users and groups (also called access control lists or ACLs). You can specify local or domain Windows users or groups, or UNIX users or groups. If you specify a domain Windows user name, you must include the user's domain using the format domain\username.

DNS Primary and Secondary IP Address

The IP addresses of the DNS servers that provide name resolution for the CIFS server. The listed DNS servers must contain the service location records (SRV) needed to locate the Active Directory LDAP servers and domain controllers for the domain that the CIFS server will join.

Active Directory Domain to join

The FQDN of the Active Directory (AD) domain that you want the CIFS server to join.

Credentials authorized to join the domain

The name and password of a Windows account with sufficient privileges to add computers to the specified Organizational Unit (OU) within the AD domain.

CIFS server NetBIOS name

A CIFS server name that is unique in the AD domain.

Organizational Unit

The organizational unit within the AD domain to associate with the CIFS server. The default is CN=Computers.

- To configure Azure AD Domain Services as the AD server for Cloud Volumes ONTAP, enter **OU=AADDC Computers** or **OU=AADDC Users** in this field.
  Azure Documentation: Create an Organizational Unit (OU) in an Azure AD Domain Services managed domain

DNS Domain

The DNS domain for the Cloud Volumes ONTAP storage virtual machine (SVM). In most cases, the domain is the same as the AD domain.

NTP Server

Select **Use Active Directory Domain** to configure an NTP server using the Active Directory DNS. If you need to configure an NTP server using a different address, then you should use the API. See the BlueXP automation docs for details.

Note that you can configure an NTP server only when creating a CIFS server. It's not configurable after you create the CIFS server.

iSCSI

LUN

iSCSI storage targets are called LUNs (logical units) and are presented to hosts as standard block devices. When you create an iSCSI volume, BlueXP automatically creates a LUN for you. We've made it simple by creating just one LUN per volume, so there's no management involved. After you create the volume, use the IQN to connect to the LUN from your hosts.

Initiator group

Initiator groups (igroups) specify which hosts can access specified LUNs on the storage system

Host initiator (IQN)

iSCSI targets connect to the network through standard Ethernet network adapters (NICs), TCP offload engine (TOE) cards with software initiators, converged network adapters (CNAs) or dedicated host bust adapters (HBAs) and are identified by iSCSI qualified names (IQNs).

c. **Disk Type**: Choose an underlying disk type for the volume based on your performance needs and cost requirements.

   - Sizing your system in Azure

d. **Usage Profile & Tiering Policy**: Choose whether to enable or disable storage efficiency features on the volume and then select a volume tiering policy.

   ONTAP includes several storage efficiency features that can reduce the total amount of storage that you need. NetApp storage efficiency features provide the following benefits:

   Thin provisioning

   Presents more logical storage to hosts or users than you actually have in your physical storage pool. Instead of preallocating storage space, storage space is allocated dynamically to each volume as data is written.

   Deduplication

   Improves efficiency by locating identical blocks of data and replacing them with references to a single shared block. This technique reduces storage capacity requirements by eliminating redundant blocks of data that reside in the same volume.

   Compression

   Reduces the physical capacity required to store data by compressing data within a volume on primary, secondary, and archive storage.

e. **Review**: Review details about the volume and then click **Add**.

Result

BlueXP creates the volume on the Cloud Volumes ONTAP system.

Create a volume from a template

If your organization has created Cloud Volumes ONTAP volume templates so you can deploy volumes that are optimized for the workload requirements for certain applications, follow the steps in this section.

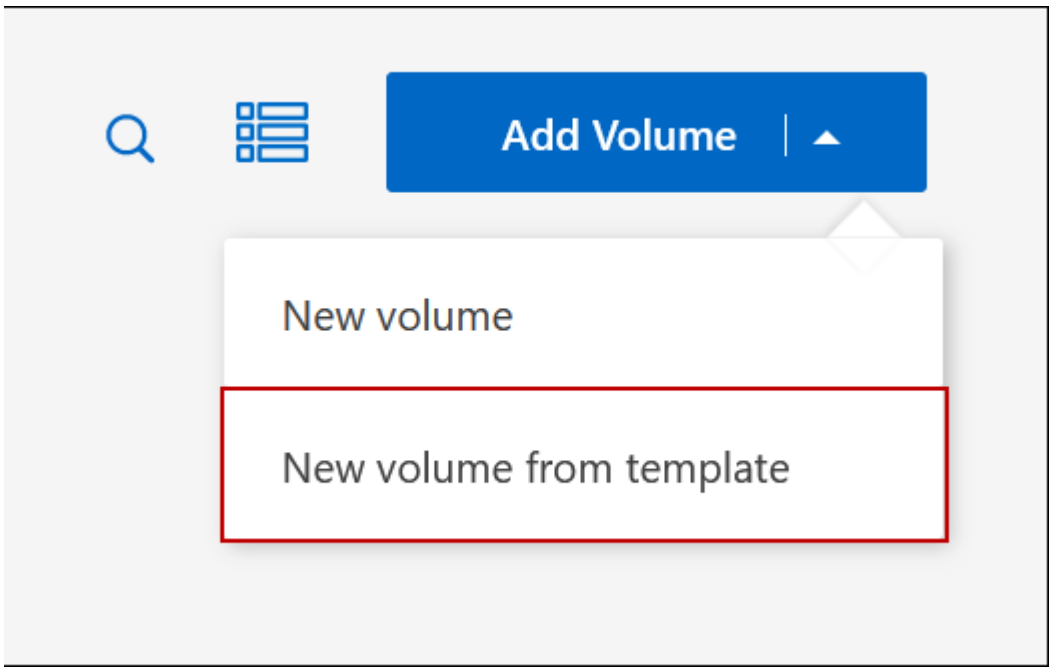
The template should make your job easier because certain volume parameters will already be defined in the template, such as disk type, size, protocol, snapshot policy, cloud provider, and more. When a parameter is

already predefined, you can just skip to the next volume parameter.

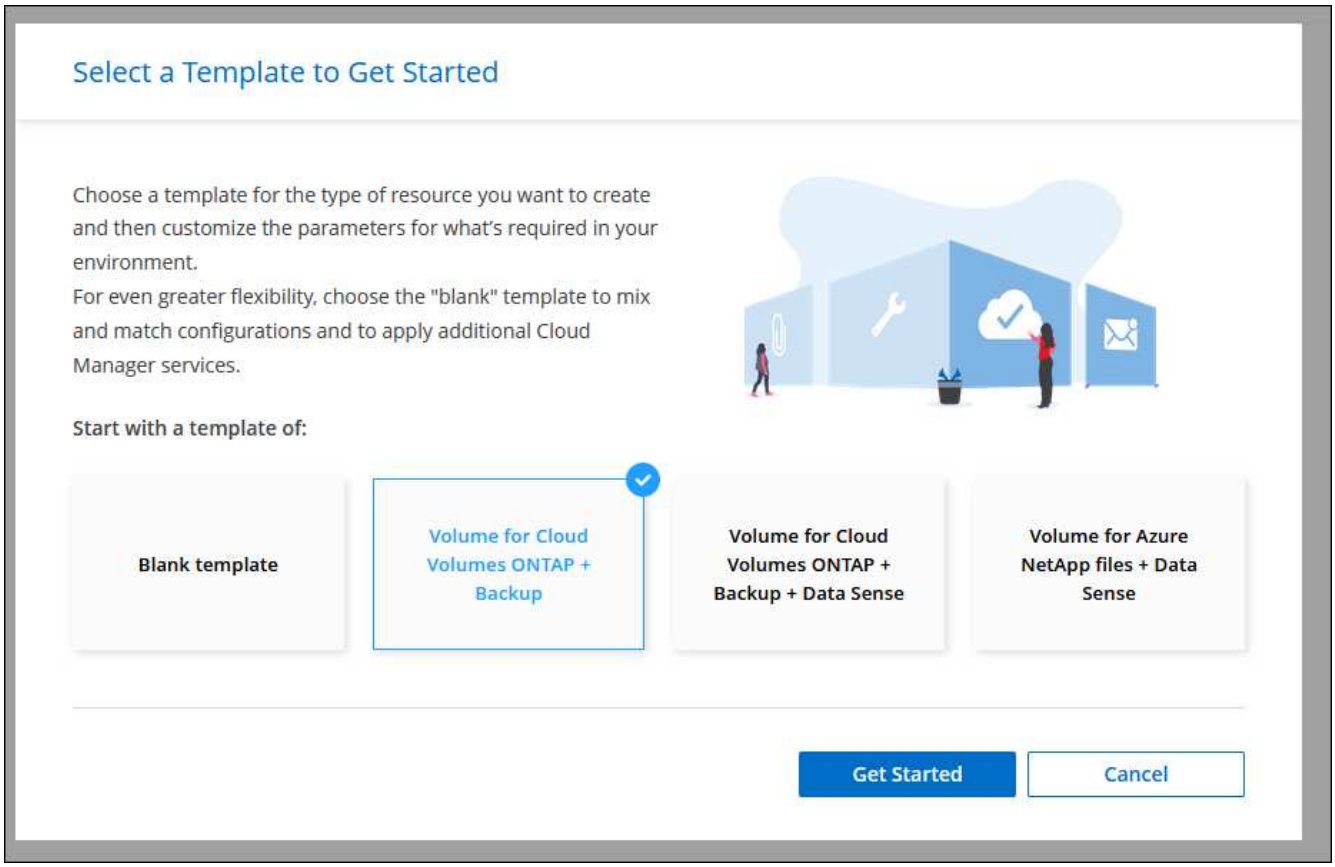
> You can only create NFS or CIFS volumes when using templates.

Steps

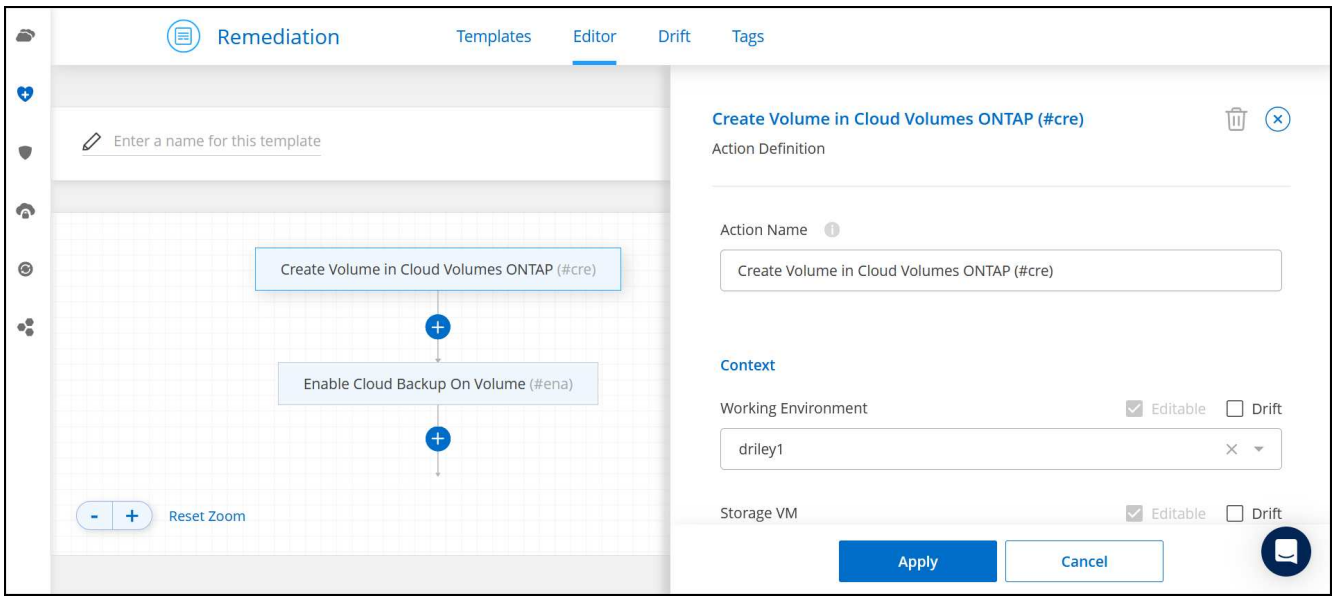
1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, click the name of the Cloud Volumes ONTAP system on which you want to provision a volume.
3. Navigate to the Volumes tab and click **Add Volume** > **New Volume From Template**.



4. In the *Select Template* page, select the template that you want to use to create the volume and click **Next**.



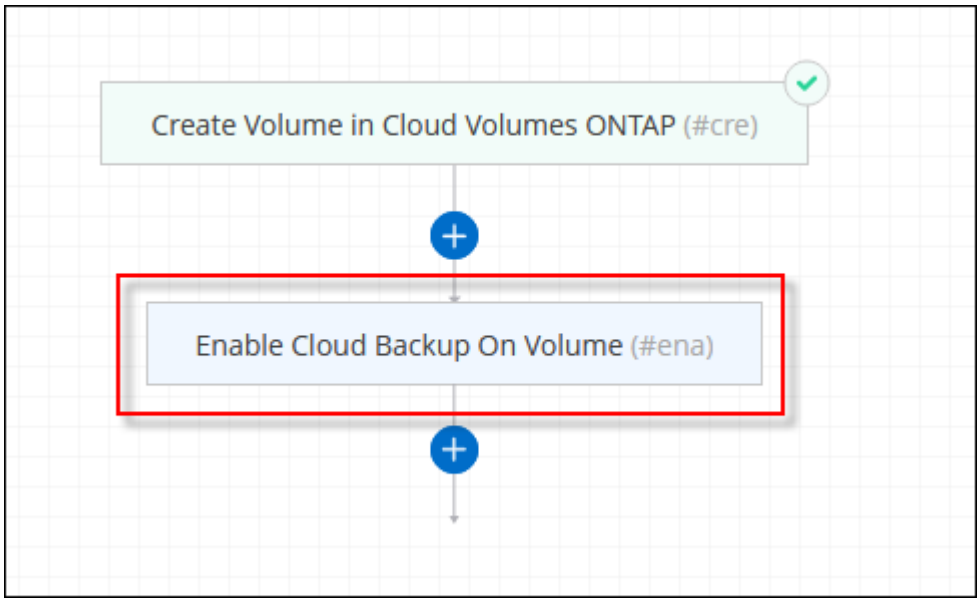
The *Editor* page is displayed.



5. Above the *Action* panel, enter a name for the template.
6. Under *Context*, the Working Environment is filled in with the name of the working environment with which you started. Select the **Storage VM** where the volume will be created.
7. Add values for all of the parameters that are not hard-coded from the template. See Create a volume for details about all the parameters needed to complete the deployment of a Cloud Volumes ONTAP volume.
8. Click **Apply** to save the configured parameters to the selected Action.

9. If there are no other Actions that you need to define (for example, configuring BlueXP backup and recovery), click **Save Template**.

   If there are other actions, click the action in the left pane to display the parameters you need to complete.

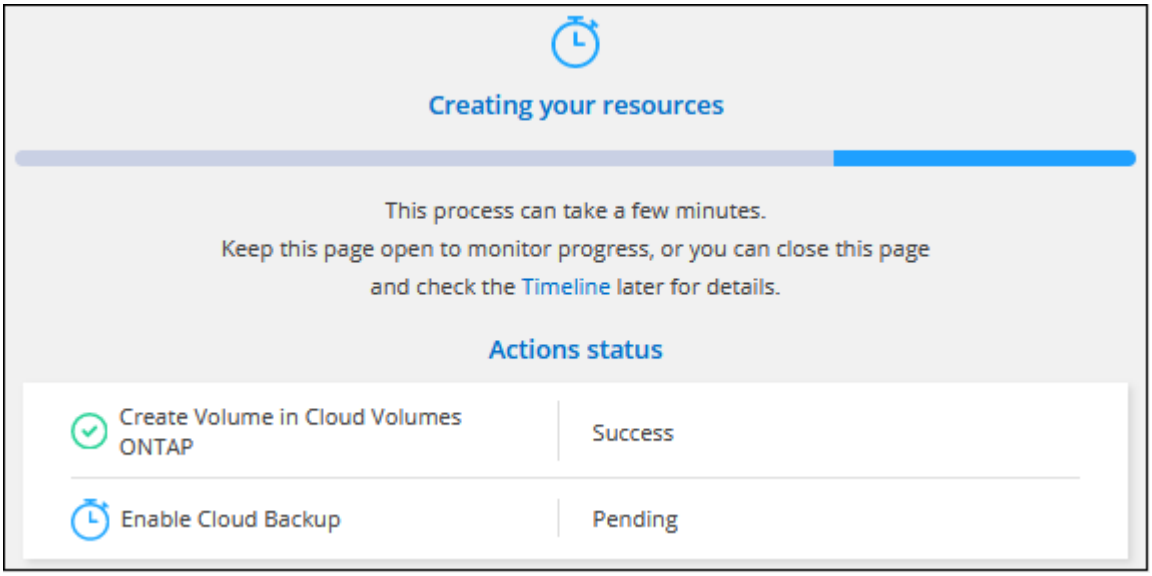


   For example, if the Enable Cloud Backup on Volume action requires that you select a backup policy, you can do that now.

10. Once configuration for the template actions are complete, click **Save Template**.

Result

Cloud Volumes ONTAP provisions the volume and displays a page so that you can see the progress.



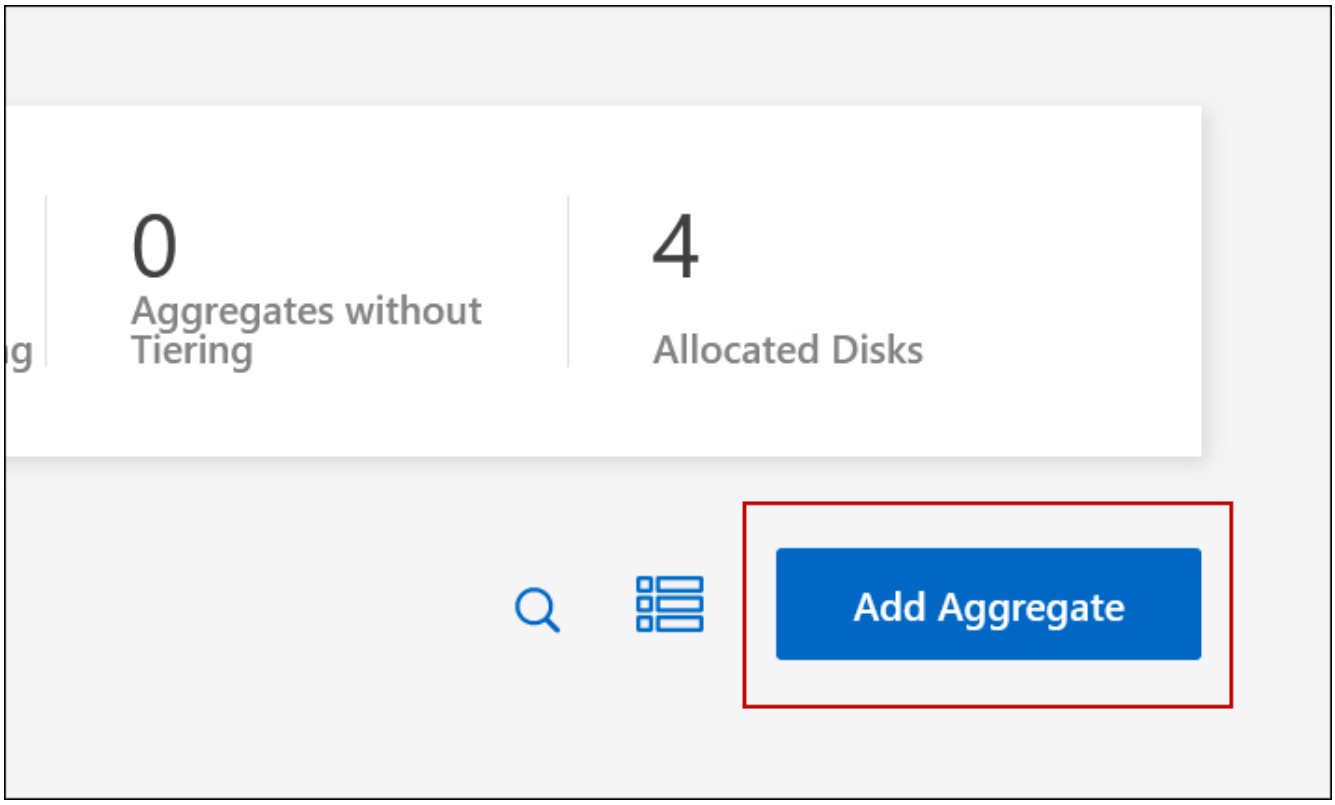
Additionally, if any secondary action is implemented in the template, for example, enabling BlueXP backup and recovery on the volume, that action is also performed.

Create a volume on the second node in an HA configuration

By default, BlueXP creates volumes on the first node in an HA configuration. If you need an active-active configuration, in which both nodes serve data to clients, you must create aggregates and volumes on the second node.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the name of the Cloud Volumes ONTAP working environment on which you want to manage aggregates.
3. On the Aggregates tab, click **Add Aggregate**.
4. From the *Add Aggregate* screen, create the aggregate.



5. For Home Node, choose the second node in the HA pair.
6. After BlueXP creates the aggregate, select it and then click **Create volume**.
7. Enter details for the new volume, and then click **Create**.

Result

BlueXP creates the volume on the second node in the HA pair.

After you create a volume

If you provisioned a CIFS share, give users or groups permissions to the files and folders and verify that those users can access the share and create a file.

If you want to apply quotas to volumes, you must use System Manager or the CLI. Quotas enable you to restrict or track the disk space and number of files used by a user, group, or qtree.

Manage existing volumes

BlueXP enables you to manage volumes and CIFS servers. It also prompts you to move volumes to avoid capacity issues.

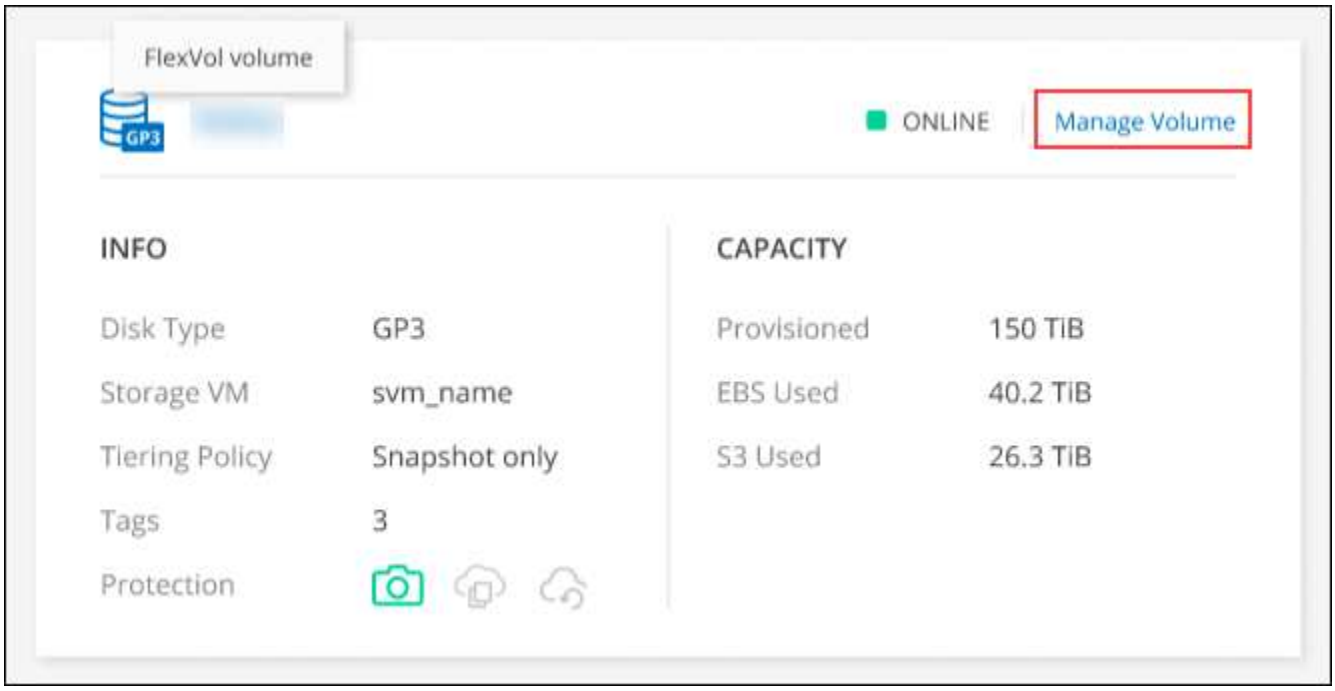
You can manage volumes in BlueXP Standard View or Advanced View. The Standard View provides a limited set of options to modify your volumes. The Advanced View provides advanced level of management, such as cloning, resizing, changing settings for anti-ransomware, analytics, protection, and activity tracking, and moving volumes across tiers. See Administer Cloud Volumes ONTAP using the Advanced View.

Manage volumes

By using the Standard View of BlueXP, you can manage volumes according to your storage needs. You can view, edit, clone, restore, and delete volumes.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the Cloud Volumes ONTAP working environment on which you want to manage volumes.
3. In the working environment, click the **Volumes** tab.



4. On the Volumes tab, navigate to the desired volume title and then click **Manage volume** to access the Manage Volumes right-side panel.

Task	Action
View information about a volume	Under Volume Actions in the Manage volumes panel, click **View volume details**.

Task	Action
Get the NFS mount command	a. Under Volume Actions in the Manage volumes panel, click **Mount Command**. b. Click **Copy**.
Clone a volume	a. Under Volume Actions in the Manage volumes panel, click **Clone the volume**. b. Modify the clone name as needed, and then click **Clone**. This process creates a FlexClone volume. A FlexClone volume is a writable, point-in-time copy that is space-efficient because it uses a small amount of space for metadata, and then only consumes additional space as data is changed or added. To learn more about FlexClone volumes, see the ONTAP 9 Logical Storage Management Guide.
Edit volume tags (read-write volumes only)	a. Under Volume Actions in the Manage volumes panel, click **Edit volume tags** to modify the volume tag assigned to the selected volume. b. Enter the volume tag key and value under the provided fields. c. To include additional tags, click **Add New Tag**. d. Click **Save**.
Edit a volume (read-write volumes only)	a. Under Volume Actions in the Manage volumes panel, click **Edit volume settings** b. Modify the volume's Snapshot policy, NFS protocol version, NFS access control list (export policy), or share permissions, and then click **Apply**. If you need custom Snapshot policies, you can create them by using System Manager.
Delete a volume	a. Under Volume Actions in the Manage volumes panel, click **Delete the volume**. b. Under the Delete Volume window, enter the name of the volume you want to delete. c. Click **Delete** again to confirm.
Create a Snapshot copy on demand	a. Under Protection Actions in the Manage Volumes panel, click **Create a Snapshot copy**. b. Change the name, if needed, and then click **Create**.

Task	Action
Restore data from a Snapshot copy to a new volume	a. Under Protection Actions in the Manage Volumes panel, click **Restore from Snapshot copy**. b. Select a Snapshot copy, enter a name for the new volume, and then click **Restore**.
Change the underlying disk type	a. Under Advanced Actions in the Manage Volumes panel, click **Change Disk Type**. b. Select the disk type, and then click **Change**. BlueXP moves the volume to an existing aggregate that uses the selected disk type or it creates a new aggregate for the volume.
Change the tiering policy	a. Under Advanced Actions in the Manage Volumes panel, click **Change Tiering Policy**. b. Select a different policy and click **Change**. BlueXP moves the volume to an existing aggregate that uses the selected disk type with tiering, or it creates a new aggregate for the volume.
Delete a volume	a. Select a volume, and then click **Delete**. b. Type the name of the volume in the dialog. c. Click **Delete** again to confirm.

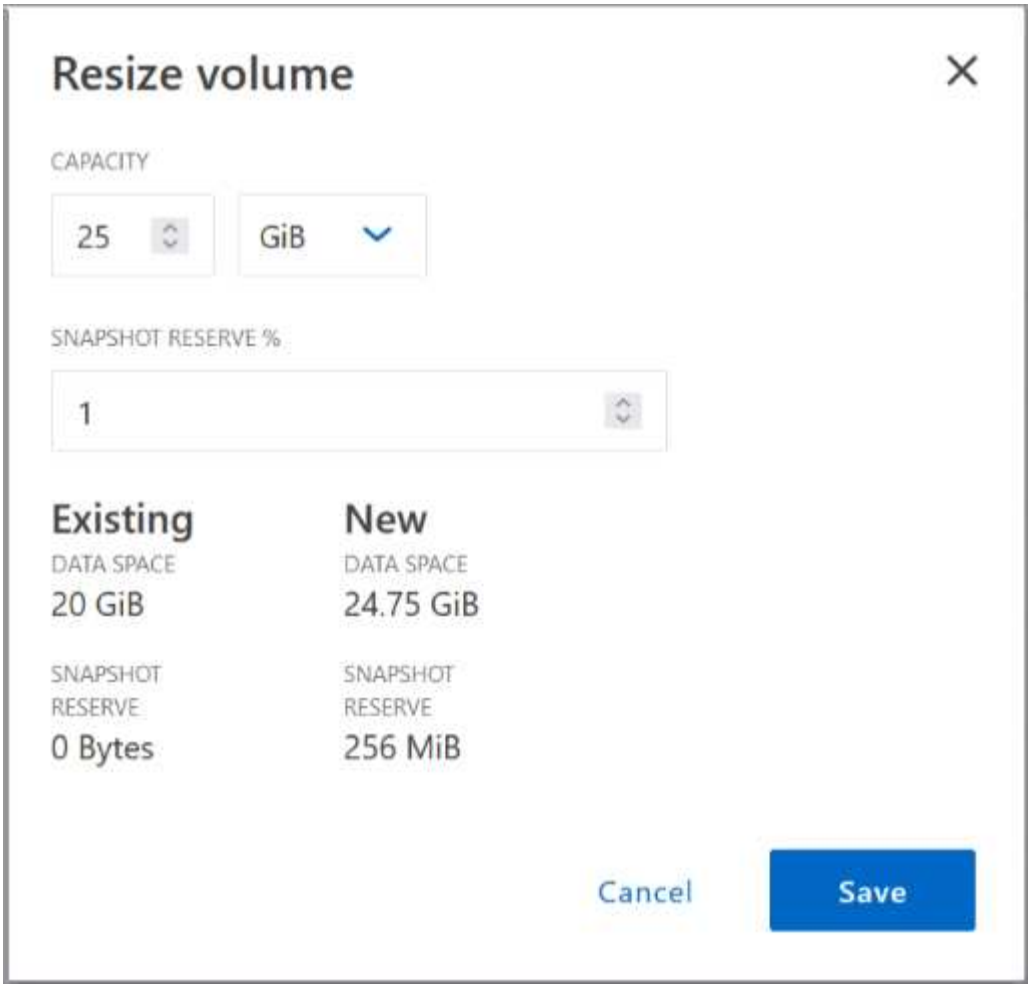
Resize a volume

By default, a volume automatically grows to a maximum size when it's out of space. The default value is 1,000, which means the volume can grow to 11 times its size. This value is configurable in the Connector's settings.

If you need to resize your volume, you can do it from the Advanced View in BlueXP.

Steps

1. Open the Advanced View to resize a volume through System Manager. See How to get started.
2. From the left navigation menu, select **Storage > Volumes**.
3. From the list of volumes, identify the one that you should resize.
4. Click the options icon .
5. Select **Resize**.
6. On the **Resize Volume** screen, edit the capacity and Snapshot reserve percentage as required. You can compare the existing, available space with the modified capacity.
7. Click **Save**.



Be sure to take your system's capacity limits into consideration as you resize volumes. Go to the Cloud Volumes ONTAP Release Notes for more details.

Modify the CIFS server

If you change your DNS servers or Active Directory domain, you need to modify the CIFS server in Cloud Volumes ONTAP so that it can continue to serve storage to clients.

Steps

1. From the Overview tab of the working environment, click the Feature tab under the right-side panel.
2. Under the CIFS Setup field, click the **pencil icon** to display the CIFS Setup window.
3. Specify settings for the CIFS server:

Task	Action
Select Storage VM (SVM)	Selecting the Cloud Volume ONTAP storage virtual machine (SVM) displays it's configured CIFS information.
Active Directory Domain to join	The FQDN of the Active Directory (AD) domain that you want the CIFS server to join.
Credentials authorized to join the domain	The name and password of a Windows account with sufficient privileges to add computers to the specified Organizational Unit (OU) within the AD domain.

Task	Action
DNS Primary and Secondary IP Address	The IP addresses of the DNS servers that provide name resolution for the CIFS server. The listed DNS servers must contain the service location records (SRV) needed to locate the Active Directory LDAP servers and domain controllers for the domain that the CIFS server will join.
DNS Domain	The DNS domain for the Cloud Volumes ONTAP storage virtual machine (SVM). In most cases, the domain is the same as the AD domain.
CIFS server NetBIOS name	A CIFS server name that is unique in the AD domain.
Organizational Unit	The organizational unit within the AD domain to associate with the CIFS server. The default is CN=Computers. • To configure Azure AD Domain Services as the AD server for Cloud Volumes ONTAP, enter **OU=AADDC Computers** or **OU=AADDC Users** in this field. Azure Documentation: Create an Organizational Unit (OU) in an Azure AD Domain Services managed domain

4. Click **Set**.

Result

Cloud Volumes ONTAP updates the CIFS server with the changes.

Move a volume

Move volumes for capacity utilization, improved performance, and to satisfy service-level agreements.

You can move a volume in System Manager by selecting a volume and the destination aggregate, starting the volume move operation, and optionally monitoring the volume move job. When using System Manager, a volume move operation finishes automatically.

Steps

1. Use System Manager or the CLI to move the volumes to the aggregate.

   In most situations, you can use System Manager to move volumes.

   For instructions, see the ONTAP 9 Volume Move Express Guide.

Move a volume when BlueXP displays an Action Required message

BlueXP might display an Action Required message that says moving a volume is necessary to avoid capacity issues, but that you need to correct the issue yourself. If this happens, you need to identify how to correct the issue and then move one or more volumes.

> BlueXP displays these Action Required messages when an aggregate has reached 90% used capacity. If data tiering is enabled, the messages display when an aggregate has reached 80% used capacity. By default, 10% free space is reserved for data tiering. Learn more about the free space ratio for data tiering.

Steps

1. Identify how to correct capacity issues.
2. Based on your analysis, move volumes to avoid capacity issues:
   - Move volumes to another system to avoid capacity issues.
   - Move volumes to another aggregate to avoid capacity issues.

Identify how to correct capacity issues

If BlueXP can't provide recommendations for moving a volume to avoid capacity issues, you must identify the volumes that you need to move and whether you should move them to another aggregate on the same system or to another system.

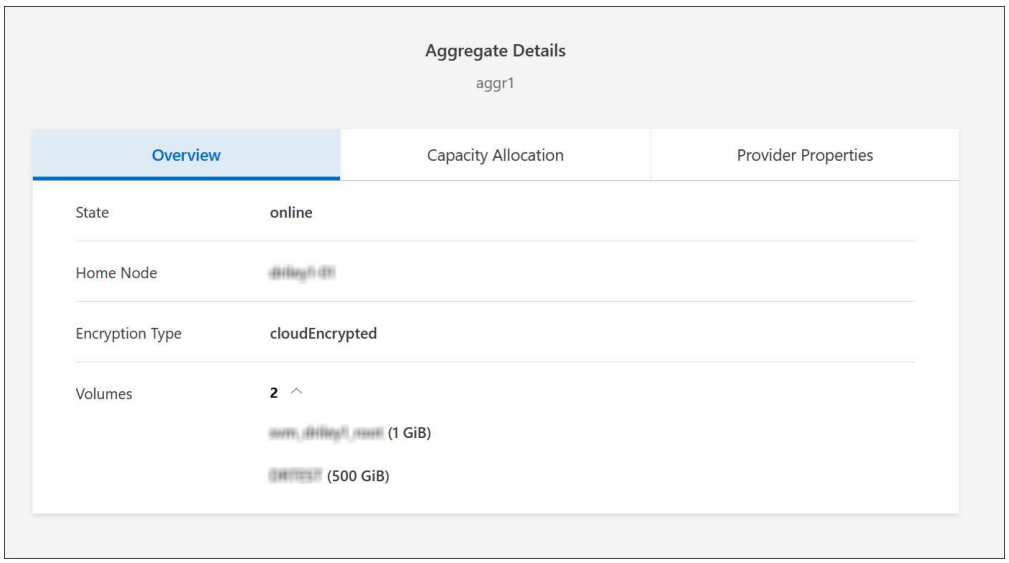
Steps

1. View the advanced information in the Action Required message to identify the aggregate that has reached its capacity limit.

   For example, the advanced information should say something similar to the following: Aggregate aggr1 has reached its capacity limit.

2. Identify one or more volumes to move out of the aggregate:
   a. In the working environment, click the **Aggregates tab**.
   b. Navigate to the desired aggregate tile, and then click the **… (ellipse icon) > View aggregate details**.
   c. Under the Overview tab of the Aggregate Details screen, review the size of each volume and choose one or more volumes to move out of the aggregate.

      You should choose volumes that are large enough to free space in the aggregate so that you avoid additional capacity issues in the future.



3. If the system has not reached the disk limit, you should move the volumes to an existing aggregate or a new aggregate on the same system.

   For details, see Move volumes to another aggregate to avoid capacity issues.

4. If the system has reached the disk limit, do any of the following:

a. Delete any unused volumes.
b. Rearrange volumes to free space on an aggregate.

   For details, see Move volumes to another aggregate to avoid capacity issues.

c. Move two or more volumes to another system that has space.

   For details, see Move volumes to another aggregate to avoid capacity issues.

Move volumes to another system to avoid capacity issues

You can move one or more volumes to another Cloud Volumes ONTAP system to avoid capacity issues. You might need to do this if the system reached its disk limit.

About this task

You can follow the steps in this task to correct the following Action Required message:

> Moving a volume is necessary to avoid capacity issues; however, BlueXP cannot perform this action for you because the system has reached the disk limit.

Steps

1. Identify a Cloud Volumes ONTAP system that has available capacity, or deploy a new system.
2. Drag and drop the source working environment on the target working environment to perform a one-time data replication of the volume.

   For details, see Replicating data between systems.

3. Go to the Replication Status page, and then break the SnapMirror relationship to convert the replicated volume from a data protection volume to a read/write volume.

   For details, see Managing data replication schedules and relationships.

4. Configure the volume for data access.

   For information about configuring a destination volume for data access, see the ONTAP 9 Volume Disaster Recovery Express Guide.

5. Delete the original volume.

   For details, see Manage volumes.

Move volumes to another aggregate to avoid capacity issues

You can move one or more volumes to another aggregate to avoid capacity issues.

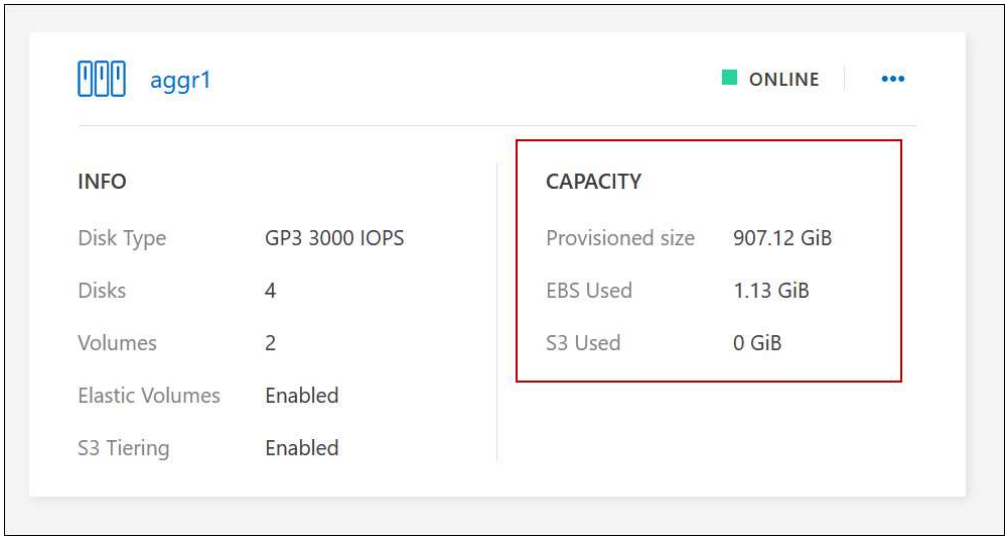
About this task

You can follow the steps in this task to correct the following Action Required message:

Moving two or more volumes is necessary to avoid capacity issues; however, BlueXP cannot perform this action for you.

Steps

1. Verify whether an existing aggregate has available capacity for the volumes that you need to move:
   a. In the working environment, click the **Aggregates tab**.
   b. Navigate to the desired aggregate tile, and then click the **… (ellipse icon) > View aggregate details**.
   c. Under the aggregate tile, view the available capacity (provisioned size minus used aggregate capacity).



2. If needed, add disks to an existing aggregate:
   a. Select the aggregate, then click the **… (ellipse icon) > Add Disks**.
   b. Select the number of disks to add, and then click **Add**.
3. If no aggregates have available capacity, create a new aggregate.

   For details, see Creating aggregates.

4. Use System Manager or the CLI to move the volumes to the aggregate.
5. In most situations, you can use System Manager to move volumes.

   For instructions, see the ONTAP 9 Volume Move Express Guide.

Reasons why a volume move might perform slowly

Moving a volume might take longer than you expect if any of the following conditions are true for Cloud Volumes ONTAP:

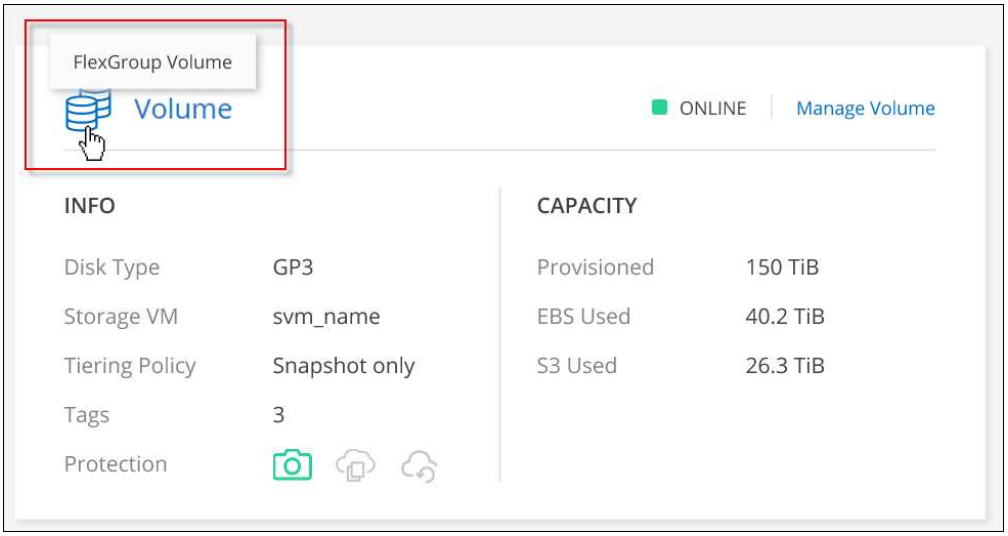
- The volume is a clone.
- The volume is a parent of a clone.
- The source or destination aggregate has a single Throughput Optimized HDD (st1) disk.
- One of the aggregates uses an older naming scheme for objects. Both aggregates have to use the same name format.

An older naming scheme is used if data tiering was enabled on an aggregate in the 9.4 release or earlier.

- The encryption settings don't match on the source and destination aggregates, or a rekey is in progress.
- The *-tiering-policy* option was specified on the volume move to change the tiering policy.
- The *-generate-destination-key* option was specified on the volume move.

View FlexGroup Volumes

You can view FlexGroup volumes created through CLI or System Manager directly through the Volumes tab within BlueXP. Identical to the information provided for FlexVol volumes, BlueXP provides detailed information for created FleGroup volumes through a dedicated Volumes tile. Under the Volumes tile, you can identify each FlexGroup volume group through the icon's hover text. Additionally, you can identify and sort FlexGroup volumes under the volumes list view through the Volume Style column.



Currently, you can only view existing FlexGroup volumes under BlueXP. The ability to create FlexGroup volumes in BlueXP is not available but planned for a future release.

Tiering inactive data to low-cost object storage

You can reduce storage costs for Cloud Volumes ONTAP by combining an SSD or HDD performance tier for hot data with an object storage capacity tier for inactive data. Data tiering is powered by FabricPool technology. For a high-level overview, see Data tiering overview.

To set up data tiering, you need to do the following:

1. **Choose a supported configuration**

Most configurations are supported. If you have a Cloud Volumes ONTAP system running the most recent version, then you should be good to go. Learn more.

2. **Ensure connectivity between Cloud Volumes ONTAP and object storage**

- For Azure, you won't need to do anything as long as BlueXP has the required permissions. Learn more.

3 **Ensure that you have an aggregate with tiering enabled**

Data tiering must be enabled on an aggregate in order to enable data tiering on a volume. You should be aware of the requirements for new volumes and for existing volumes. Learn more.

4 **Choose a tiering policy when creating, modifying, or replicating a volume**

BlueXP prompts you to choose a tiering policy when you create, modify, or replicate a volume.

- Tiering data on read-write volumes
- Tiering data on data protection volumes

> **What's not required for data tiering?**
>
> - You don't need to install a feature license to enable data tiering.
> - You don't need to create an object store for the capacity tier. BlueXP does that for you.
> - You don't need to enable data tiering at the system level.
>
> BlueXP creates an object store for cold data when the system is created, as long as there are no connectivity or permissions issues. After that, you just need to enable data tiering on volumes (and in some cases, on aggregates).

Configurations that support data tiering

You can enable data tiering when using specific configurations and features.

Support in Azure

- Data tiering is supported in Azure as follows:
  - Version 9.4 in with single node systems
  - Version 9.6 in with HA pairs
- The performance tier can be Premium SSD managed disks, Standard SSD managed disks, or Standard HDD managed disks.

Feature interoperability

- Data tiering is supported with encryption technologies.
- Thin provisioning must be enabled on volumes.

Requirements

Depending on your cloud provider, certain connections and permissions must be set up so that Cloud Volumes ONTAP can tier cold data to object storage.

Requirements to tier cold data to Azure Blob storage

You don't need to set up a connection between the performance tier and the capacity tier as long as BlueXP has the required permissions. BlueXP enables a VNet service endpoint for you if the custom role for the Connector has these permissions:

```
"Microsoft.Network/virtualNetworks/subnets/write",
"Microsoft.Network/routeTables/join/action",
```

The permissions are included in the custom role by default. View Azure permission for the Connector

Enabling data tiering after implementing the requirements

BlueXP creates an object store for cold data when the system is created, as long as there are no connectivity or permissions issues. If you didn't implement the requirements listed above until after you created the system, then you'll need to manually enable tiering through the API or System Manager, which creates the object store.

> The ability to enable tiering through the BlueXP user interface will be available in a future Cloud Volumes ONTAP release.

Ensuring that tiering is enabled on aggregates

Data tiering must be enabled on an aggregate in order to enable data tiering on a volume. You should be aware of the requirements for new volumes and for existing volumes.

- **New volumes**

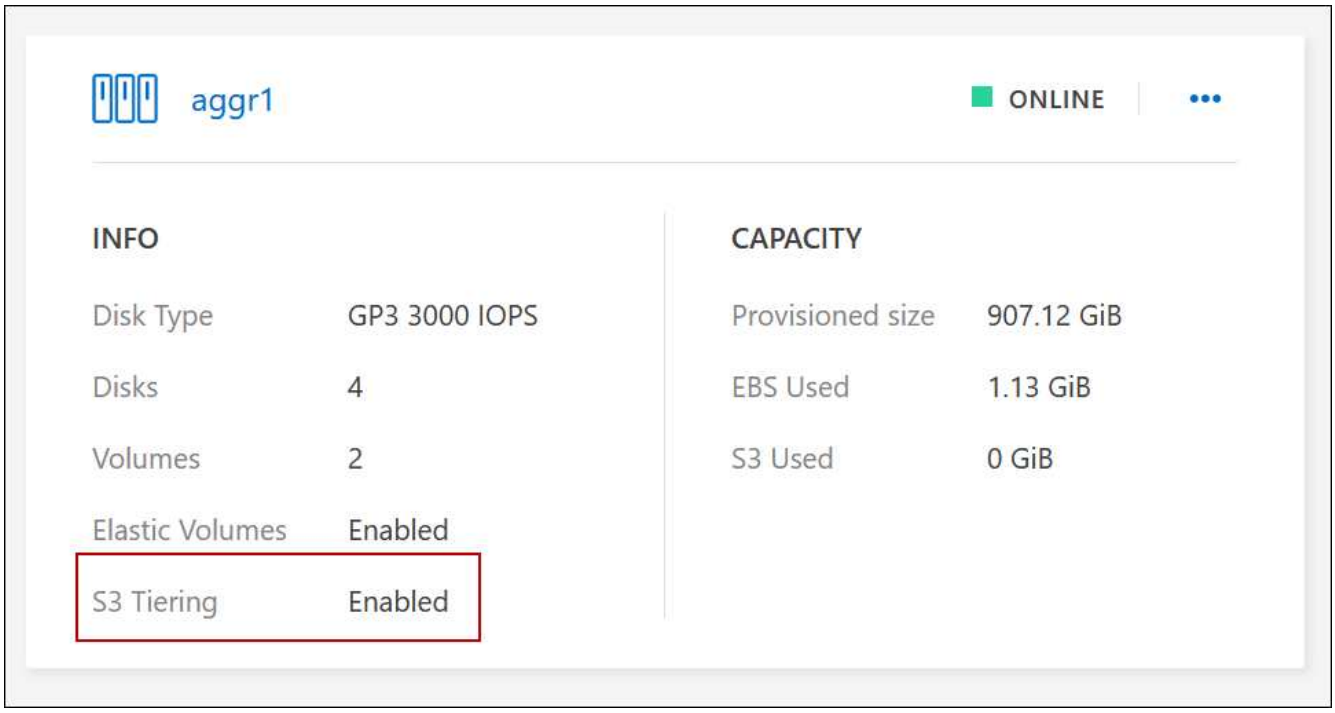
  If you're enabling data tiering on a new volume, then you don't need to worry about enabling data tiering on an aggregate. BlueXP creates the volume on an existing aggregate that has tiering enabled, or it creates a new aggregate for the volume if a data tiering-enabled aggregate doesn't already exist.

- **Existing volumes**

  If you want to enable data tiering on an existing volume, then you'll need to ensure that data tiering is enabled on the underlying aggregate. If data tiering isn't enabled on the existing aggregate, then you'll need to use System Manager to attach an existing aggregate to the object store.

Steps to confirm whether tiering is enabled on an aggregate

1. Open the working environment in BlueXP.
2. Click the Aggregates tab.
3. Navigate to the desired tile and verify whether tiering is enabled or disabled on the aggregate.



Steps to enable tiering on an aggregate

1. In System Manager, click **Storage > Tiers**.
2. Click the action menu for the aggregate and select **Attach Cloud Tiers**.
3. Select the cloud tier to attach and click **Save**.

What's next?

You can now enable data tiering on new and existing volumes, as explained in the next section.

Tiering data from read-write volumes

Cloud Volumes ONTAP can tier inactive data on read-write volumes to cost-effective object storage, freeing up the performance tier for hot data.

Steps

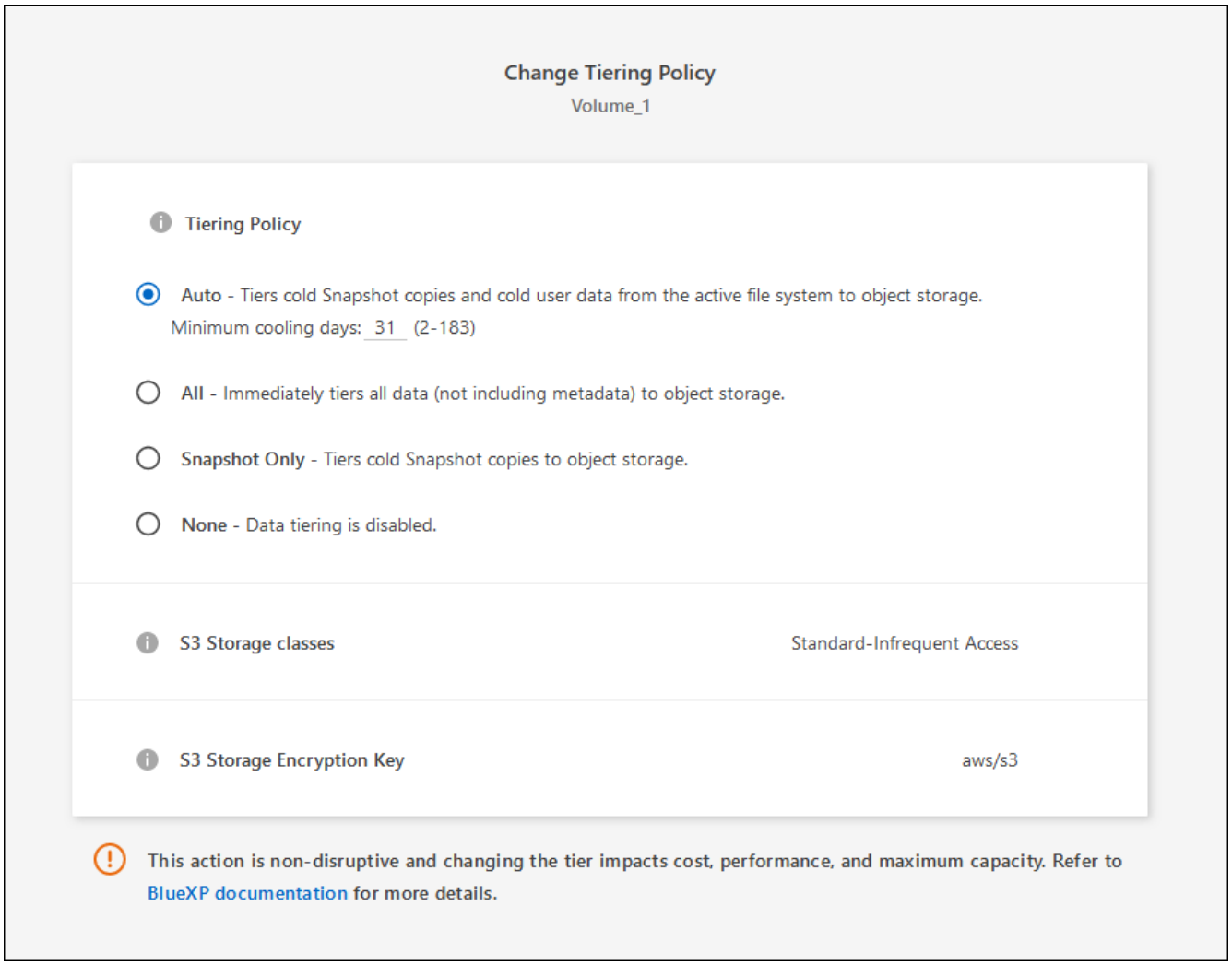
1. In Volumes tab under the working environment, create a new volume or change the tier of an existing volume:

Task	Action
Create a new volume	Click **Add New Volume**.
Modify an existing volume	Select the desired volume tile, click **Manage volume** to access the Manage Volumes right-side panel, and then click **Advanced actions** and **Change tiering policy** under the right panel.

2. Select a tiering policy.

   For a description of these policies, see Data tiering overview.

   Example



BlueXP creates a new aggregate for the volume if a data tiering-enabled aggregate does not already exist.

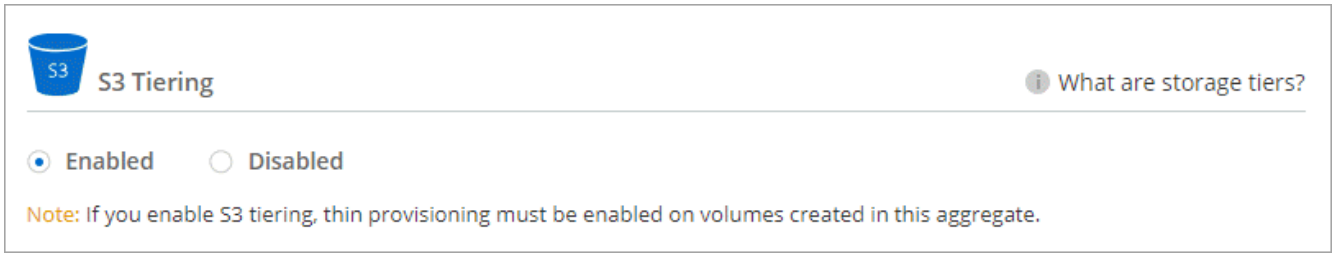
Tiering data from data protection volumes

Cloud Volumes ONTAP can tier data from a data protection volume to a capacity tier. If you activate the destination volume, the data gradually moves to the performance tier as it is read.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, select the working environment that contains the source volume, and then drag it to the working environment to which you want to replicate the volume.
3. Follow the prompts until you reach the tiering page and enable data tiering to object storage.

   Example



For help with replicating data, see Replicating data to and from the cloud.

Changing the storage class for tiered data

After you deploy Cloud Volumes ONTAP, you can reduce your storage costs by changing the storage class for inactive data that hasn't been accessed for 30 days. The access costs are higher if you do access the data, so you must take that into consideration before you change the storage class.

The storage class for tiered data is system wide—it's not per volume.

For information about supported storage classes, see Data tiering overview.

Steps

1. From the working environment, click the menu icon and then click **Storage Classes** or **Blob Storage Tiering**.
2. Choose a storage class and then click **Save**.

Changing the free space ratio for data tiering

The free space ratio for data tiering defines how much free space is required on Cloud Volumes ONTAP SSDs/HDDs when tiering data to object storage. The default setting is 10% free space, but you can tweak the setting based on your requirements.

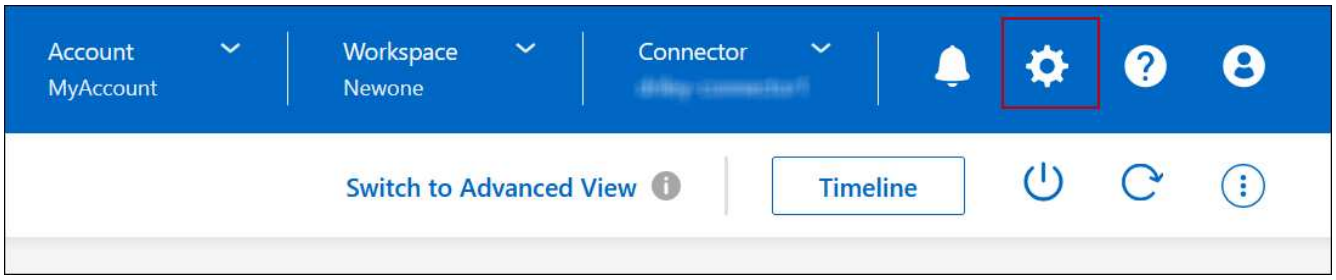
For example, you might choose less than 10% free space to ensure that you are utilizing the purchased capacity. BlueXP can then purchase additional disks for you when additional capacity is required (up until you reach the disk limit for the aggregate).

> If there isn't sufficient space, then Cloud Volumes ONTAP can't move the data and you might experience performance degradation. Any change should be done with caution. If you're unsure, reach out to NetApp support for guidance.

The ratio is important for disaster recovery scenarios because as data is read from the object store, Cloud Volumes ONTAP moves the data to SSDs/HDDs to provide better performance. If there isn't sufficient space, then Cloud Volumes ONTAP can't move the data. Take this into consideration when changing the ratio so that you can meet your business requirements.

Steps

1. In the upper right of the BlueXP console, click the **Settings** icon, and select **Cloud Volumes ONTAP Settings**.



2. Under **Capacity**, click **Aggregate Capacity Thresholds - Free Space Ratio for Data Tiering**.
3. Change the free space ratio based on your requirements and click **Save**.

Changing the cooling period for the auto tiering policy

If you enabled data tiering on a Cloud Volumes ONTAP volume using the *auto* tiering policy, you can adjust the default cooling period based on your business needs. This action is supported using the API and CLI only.

The cooling period is the number of days that user data in a volume must remain inactive before it is considered "cold" and moved to object storage.

The default cooling period for the auto tiering policy is 31 days. You can change the cooling period as follows:

- 9.8 or later: 2 days to 183 days
- 9.7 or earlier: 2 days to 63 days

Step

1. Use the *minimumCoolingDays* parameter with your API request when creating a volume or modifying an existing volume.

Connect a LUN to a host

When you create an iSCSI volume, BlueXP automatically creates a LUN for you. We’ve made it simple by creating just one LUN per volume, so there’s no management involved. After you create the volume, use the IQN to connect to the LUN from your hosts.

Note the following:

- BlueXP’s automatic capacity management doesn’t apply to LUNs. When BlueXP creates a LUN, it disables the autogrow feature.
- You can create additional LUNs from System Manager or the CLI.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the Cloud Volumes ONTAP working environment on which you want to manage volumes.
3. In the working environment, click the **Volumes** tab.
4. On the Volumes tab, navigate to the desired volume title and then click **Manage volume** to access the Manage Volumes right-side panel.
5. Click **Target iQN**.
6. Click **Copy** to copy the iQN name.

7. Set up an iSCSI connection from the host to the LUN.
    - ONTAP 9 iSCSI express configuration for Red Hat Enterprise Linux: Starting the iSCSI sessions with the target
    - ONTAP 9 iSCSI express configuration for Windows: Starting iSCSI sessions with the target
    - ONTAP SAN host configuration

Accelerate data access with FlexCache volumes

A FlexCache volume is a storage volume that caches SMB and NFS read data from an origin (or source) volume. Subsequent reads to the cached data result in faster access to that data.

You can use FlexCache volumes to speed up access to data or to offload traffic from heavily accessed volumes. FlexCache volumes help improve performance, especially when clients need to access the same data repeatedly, because the data can be served directly without having to access the origin volume. FlexCache volumes work well for system workloads that are read-intensive.

BlueXP provides management of FlexCache volumes with the BlueXP volume caching service.

You can also use the ONTAP CLI or ONTAP System Manager to create and manage FlexCache volumes:

- FlexCache Volumes for Faster Data Access Power Guide
- Creating FlexCache volumes in System Manager

BlueXP generates a FlexCache license for all new Cloud Volumes ONTAP systems. The license includes a 500 GiB usage limit.

Aggregate administration

Create aggregates

You can create aggregates yourself or let BlueXP do it for you when it creates volumes. The benefit of creating aggregates yourself is that you can choose the underlying disk size, which enables you to size your aggregate for the capacity or the performance that you need.

> All disks and aggregates must be created and deleted directly from BlueXP. You should not perform these actions from another management tool. Doing so can impact system stability, hamper the ability to add disks in the future, and potentially generate redundant cloud provider fees.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the name of the Cloud Volumes ONTAP instance on which you want to manage aggregates.
3. On the Aggregates tab, click **Add Aggregate** and then specify details for the aggregate.

   > **Azure**
   >
   > For help with disk type and disk size, refer to Plan your Cloud Volumes ONTAP configuration in Azure.

4. Click **Go**, and then click **Approve and Purchase**.

Manage aggregates

Manage aggregates yourself by adding disks, viewing information about the aggregates, and by deleting them.

> All disks and aggregates must be created and deleted directly from BlueXP. You should not perform these actions from another management tool. Doing so can impact system stability, hamper the ability to add disks in the future, and potentially generate redundant cloud provider fees.

Before you begin

If you want to delete an aggregate, you must have first deleted the volumes in the aggregate.

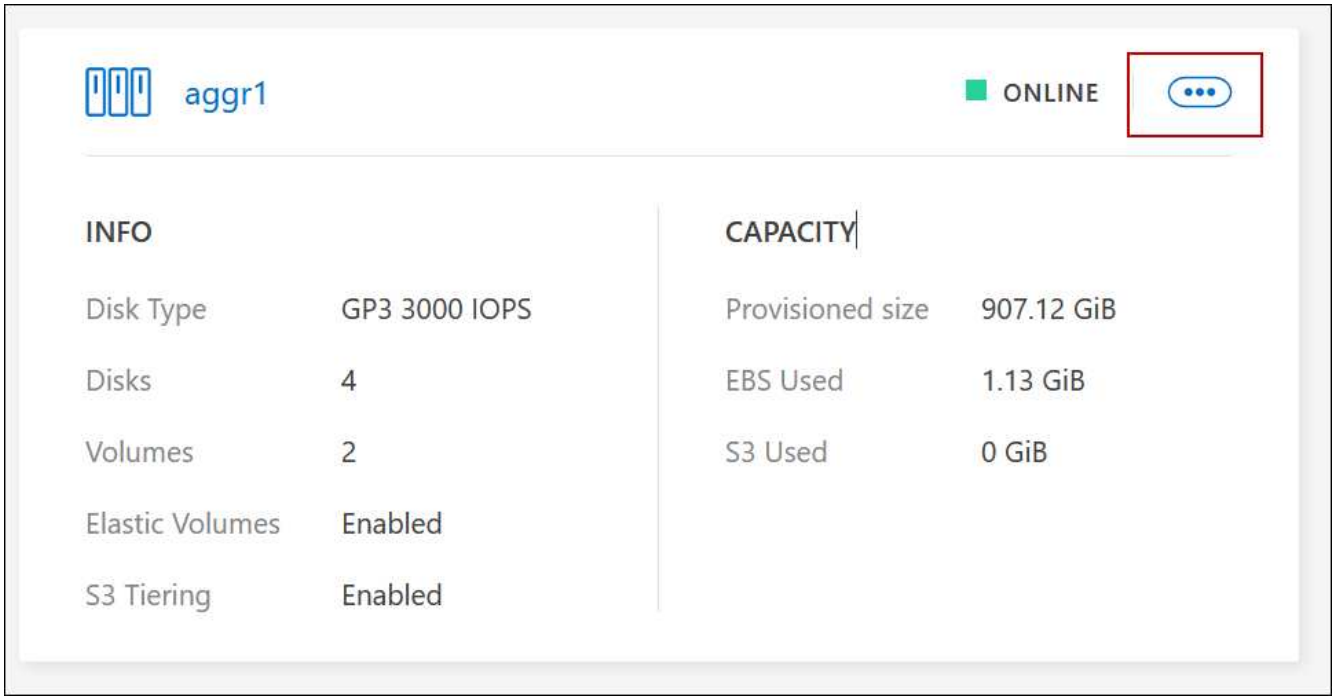
About this task

If an aggregate is running out of space, you can move volumes to another aggregate by using System Manager.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the Cloud Volumes ONTAP working environment on which you want to manage aggregates.

3. In the working environment, click the **Aggregates** tab.
4. On the Aggregates tab, navigate to the desired title and then click the **… (ellipse icon)**.



5. Manage your aggregates:

Task	Action
View information about an aggregate	Under the … (ellipse icon) menu, click **View aggregate details**.
Create a volume on a specific aggregate	Under the … (ellipse icon) menu, click **Add volume**.
Add disks to an aggregate	a. Under the … (ellipse icon) menu, click **Add disks**. b. Select the number of disks that you want to add and click **Add**. All disks in an aggregate must be the same size.
Delete an aggregate	a. Select an aggregate tile that does not contain any volumes click the **… (ellipse icon) > Delete**. b. Click **Delete** again to confirm.

Manage capacity settings on a Connector

Each Connector has settings that determines how it manages aggregate capacity for Cloud Volumes ONTAP.

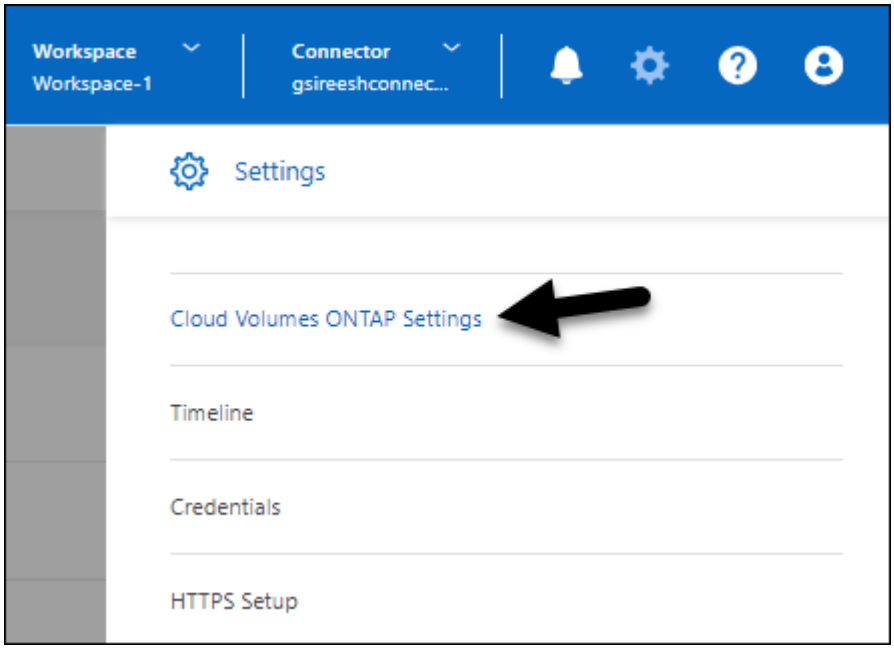
These settings affect all Cloud Volumes ONTAP systems managed by a Connector. If you have another Connector, it can be configured differently.

Required permissions

Account Admin privileges are required to modify Cloud Volumes ONTAP Settings.

Steps

1. In the upper right of the BlueXP console, click the Settings icon, and select **Cloud Volumes ONTAP Settings**.



2. Under **Capacity**, modify any of the following settings:

Capacity Management Mode

Choose whether BlueXP notifies you of storage capacity decisions or whether BlueXP automatically manages capacity requirements for you.

Learn how Capacity Management Mode works.

Aggregate Capacity Threshold - Free Space Ratio

This ratio is a key parameter in capacity management decisions, and understanding its impact is essential regardless of whether you are in an automatic or manual mode of capacity management. It is recommended to set this threshold with consideration of your specific storage needs and anticipated growth to maintain a balance between resource utilization and cost.

In the manual mode, if the free space ratio on an aggregate drops below the specified threshold, it triggers a notification, alerting you that you should take actions to address the low free space ratio. It is important to monitor these notifications and manually manage the aggregate capacity to avoid service disruption and ensure optimal performance.

The free space ratio is calculated as follows:
(aggregate capacity - total used capacity on the aggregate) / aggregate capacity

See Automatic capacity management to learn now capacity is automatically managed in Cloud Volumes ONTAP.

Aggregate Capacity Thresholds - Free Space Ratio for Data Tiering

Defines how much free space is required on the performance tier (disks) when tiering data to a capacity tier (object storage).

The ratio is important for disaster recovery scenarios. As data is read from the capacity tier, Cloud Volumes ONTAP moves data to the performance tier to provide better performance. If there isn't sufficient space, then Cloud Volumes ONTAP can't move the data.

3. Click **Save**.

Storage VM administration

Manage storage VMs in BlueXP

A storage VM is a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an *SVM* or a *vserver*. Cloud Volumes ONTAP is configured with one storage VM by default, but some configurations support additional storage VMs.

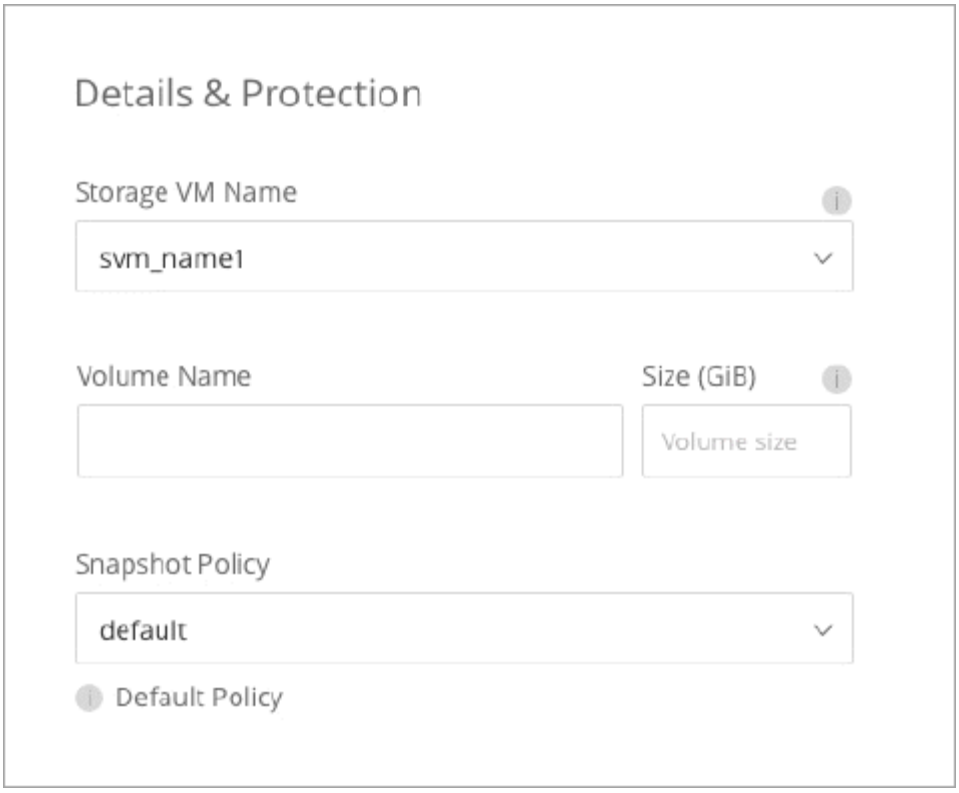
Supported number of storage VMs

Multiple storage VMs are supported with certain configurations. Go to the Cloud Volumes ONTAP Release Notes to verify the supported number of storage VMs for your version of Cloud Volumes ONTAP.

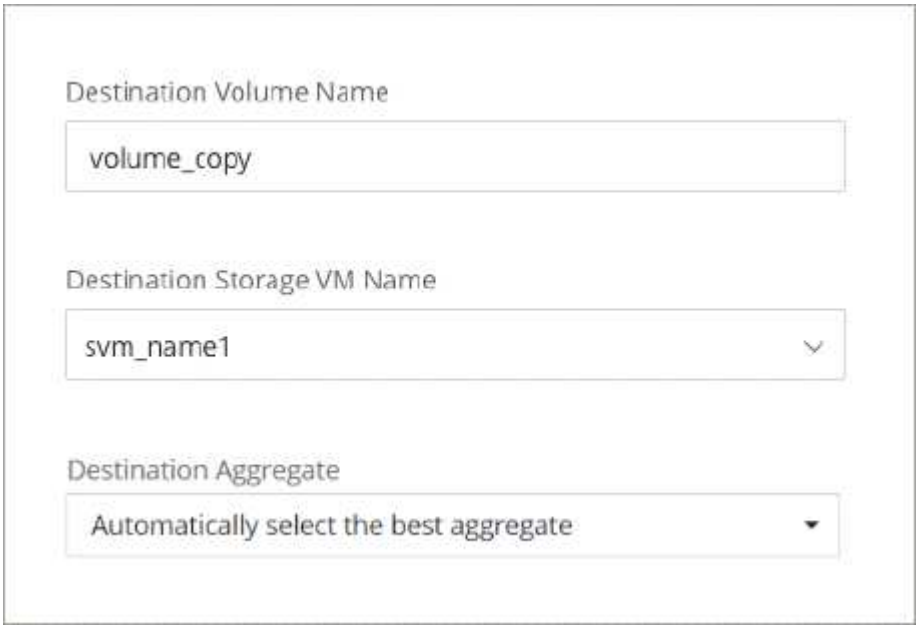
Work with multiple storage VMs

BlueXP supports any additional storage VMs that you create from System Manager or the CLI.

For example, the following image shows how you can choose a storage VM when you create a volume.



And the following image shows how you can choose a storage VM when replicating a volume to another system.



Modify the name of the default storage VM

BlueXP automatically names the single storage VM that it creates for Cloud Volumes ONTAP. From System Manager, CLI, or API, you can modify the name of the storage VM if you have strict naming standards. For example, you might want the name to match how you name the storage VMs for your ONTAP clusters.

Create data-serving storage VMs for Cloud Volumes ONTAP in Azure

A storage VM is a virtual machine running within ONTAP that provides storage and data services to your clients. You might know this as an *SVM* or a *vserver*. Cloud Volumes ONTAP is configured with one storage VM by default, but additional storage VMs are supported when running Cloud Volumes ONTAP in Azure.

To create additional data-serving storage VMs, you need to allocate IP addresses in Azure and then run ONTAP commands to create the storage VM and data LIFs.

> To perform additional NIC-related tasks, you can assign a network contributor role or custom role with appropriate permissions in Azure. For more information on these NIC-related permissions, see the Microsoft Azure documentation.

Supported number of storage VMs

Multiple storage VMs are supported with specific Cloud Volumes ONTAP configurations starting with the 9.9.0 release. Go to the Cloud Volumes ONTAP Release Notes to verify the supported number of storage VMs for your version of Cloud Volumes ONTAP.

All other Cloud Volumes ONTAP configurations support one data-serving storage VM and one destination storage VM used for disaster recovery. You can activate the destination storage VM for data access if there's an outage on the source storage VM.

Allocate IP addresses in Azure

You need to allocate IP addresses in Azure before you create a storage VM and allocate LIFs.

Single node system

IP addresses must be assigned to nic0 in Azure before you create a storage VM and allocate LIFs.

You'll need to create an IP address for data LIF access and another optional IP address for a storage VM (SVM) management LIF. This management LIF provides a connection to management tools like SnapCenter.

Steps

1. Log in to the Azure portal and open the **Virtual machine** service.
2. Click the name of the Cloud Volumes ONTAP VM.
3. Click **Networking**.
4. Click the name of the network interface for nic0.
5. Under **Settings**, click **IP configurations**.
6. Click **Add**.
7. Enter a name for the IP configuration, select **Dynamic**, and then click **OK**.
8. Click the name of the IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

   It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

If you want to create an SVM management LIF, repeat these steps to create an additional IP address.

After you finish

Copy the private IP addresses that you just created. You'll need to specify those IP addresses when you create LIFs for the new storage VM.

HA pair

How you allocate IP addresses for an HA pair depends on the storage protocol that you're using.

iSCSI

iSCSI IP addresses must be assigned to nic0 in Azure before you create a storage VM and allocate LIFs. IPs for iSCSI are assigned to nic0 and not the load balancer because iSCSI uses ALUA for failover.

You'll need to create the following IP addresses:

- One IP address for iSCSI data LIF access from node 1
- One IP address for iSCSI data LIF access from node 2
- An optional IP address for a storage VM (SVM) management LIF

  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. Log in to the Azure portal and open the **Virtual machine** service.
2. Click the name of the Cloud Volumes ONTAP VM for node 1.
3. Click **Networking**.
4. Click the name of the network interface for nic0.
5. Under **Settings**, click **IP configurations**.
6. Click **Add**.
7. Enter a name for the IP configuration, select **Dynamic**, and then click **OK**.
8. Click the name of the IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

   It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

9. Repeat these steps on node 2.
10. If you want to create an SVM management LIF, repeat these steps on node 1.

NFS

IP addresses that you use for NFS are allocated in the load balancer so that the IP addresses can migrate to the other node in case failover events occur.

You'll need to create the following IP addresses:

- One IP address for NAS data LIF access from node 1
- One IP address for NAS data LIF access from node 2
- An optional IP address for a storage VM (SVM) management LIF

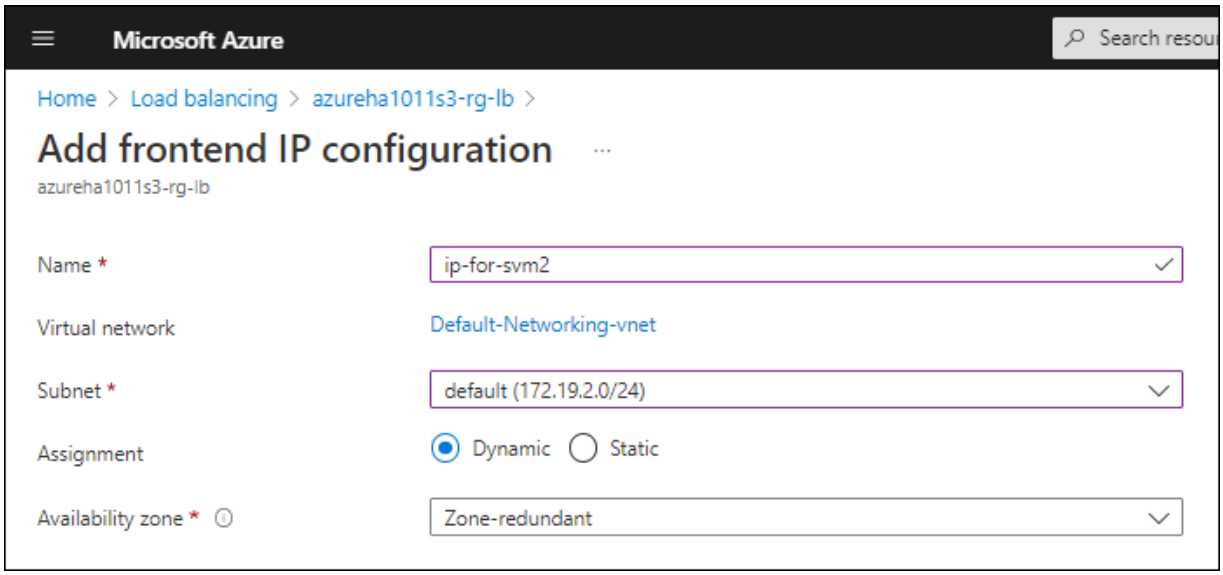
  The iSCSI LIFs are required for DNS communication. An iSCSI LIF is used for this purpose because it doesn't migrate on failover.

  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. In the Azure portal, open the **Load balancers** service.

2. Click the name of the load balancer for the HA pair.
3. Create one frontend IP configuration for data LIF access from node 1, another for data LIF access from node 2, and another optional frontend IP for a storage VM (SVM) management LIF.
   a. Under **Settings**, click **Frontend IP configuration**.
   b. Click **Add**.
   c. Enter a name for the frontend IP, select the subnet for the Cloud Volumes ONTAP HA pair, leave **Dynamic** selected, and in regions with Availability Zones, leave **Zone-redundant** selected to ensure that the IP address remains available if a zone fails.

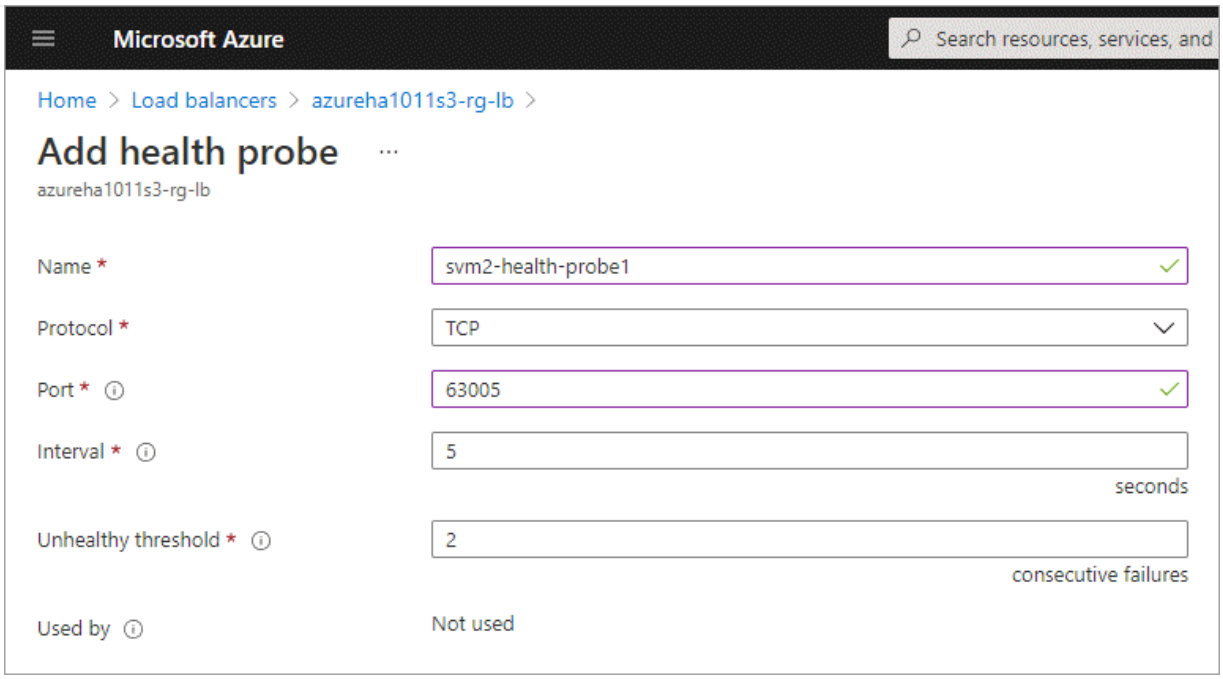


   d. Click the name of the frontend IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

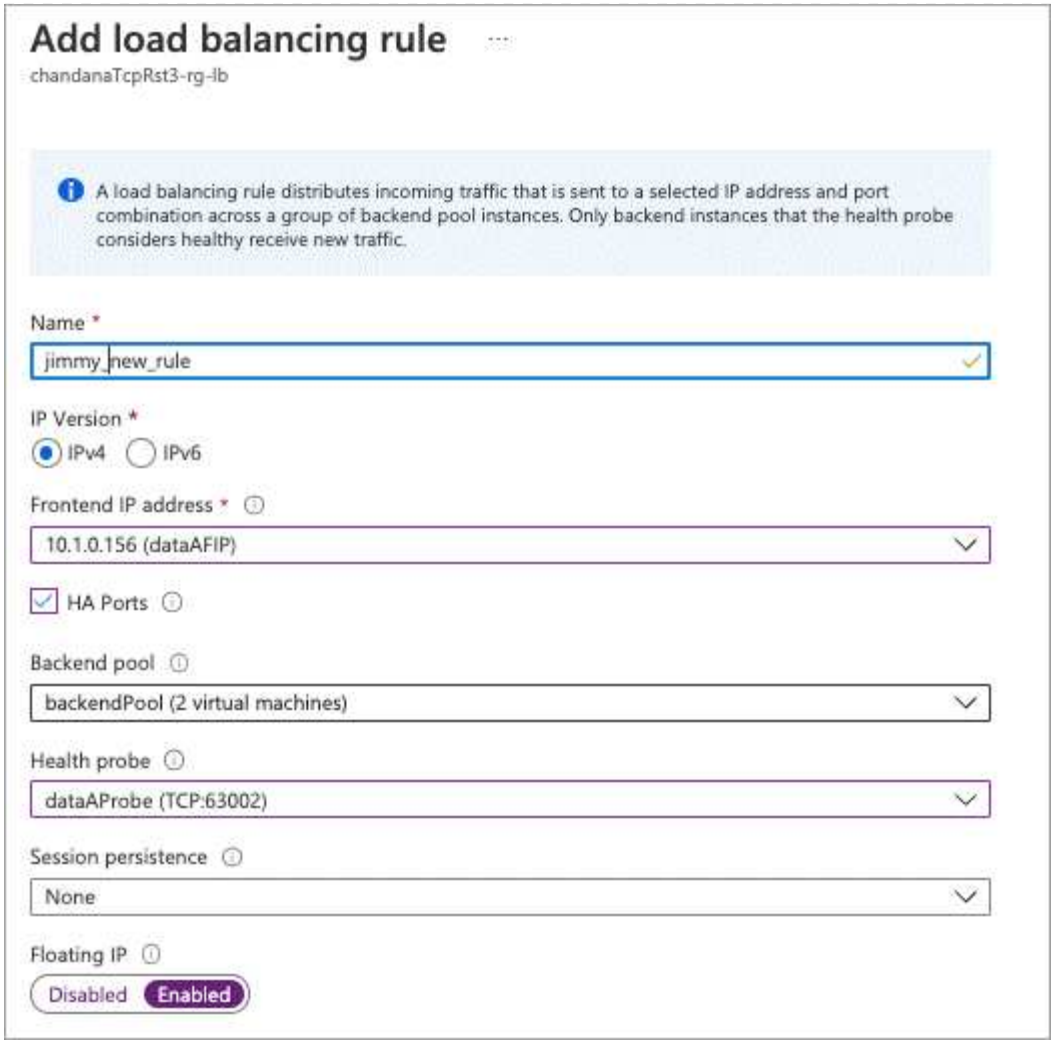
      It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

4. Add a health probe for each frontend IP that you just created.
   a. Under the load balancer's **Settings**, click **Health probes**.
   b. Click **Add**.
   c. Enter a name for the health probe and enter a port number that's between 63005 and 65000. Keep the default values for the other fields.

      It's important that the port number is between 63005 and 65000. For example, if you are creating three health probes, you could enter probes that use the port numbers 63005, 63006, and 63007.



5. Create new load balancing rules for each frontend IP.
   a. Under the load balancer's **Settings**, click **Load balancing rules**.
   b. Click **Add** and enter the required information:
      - **Name**: Enter a name for the rule.
      - **IP Version**: Select **IPv4**.
      - **Frontend IP address**: Select one of the frontend IP addresses that you just created.
      - **HA Ports**: Enable this option.
      - **Backend pool**: Keep the default Backend pool that was already selected.
      - **Health probe**: Select the health probe that you created for the selected frontend IP.
      - **Session persistence**: Select **None**.
      - **Floating IP**: Select **Enabled**.



6. Ensure that the network security group rules for Cloud Volumes ONTAP allows the load balancer to send TCP probes for the health probes that were created in step 4 above. Note that this is allowed by default.

SMB

IP addresses that you use for SMB data are allocated in the load balancer so that the IP addresses can migrate to the other node in case failover events occur.

You'll need to create the following IP addresses in the load balancer:

- One IP address for NAS data LIF access from node 1
- One IP address for NAS data LIF access from node 2
- One IP address for an iSCSI LIF on node 1 in each VM's respective NIC0
- One IP address for an iSCSI LIF on node 2

  The iSCSI LIFs are required for DNS and SMB communication. An iSCSI LIF is used for this purpose because it doesn't migrate on failover.

- An optional IP address for a storage VM (SVM) management LIF

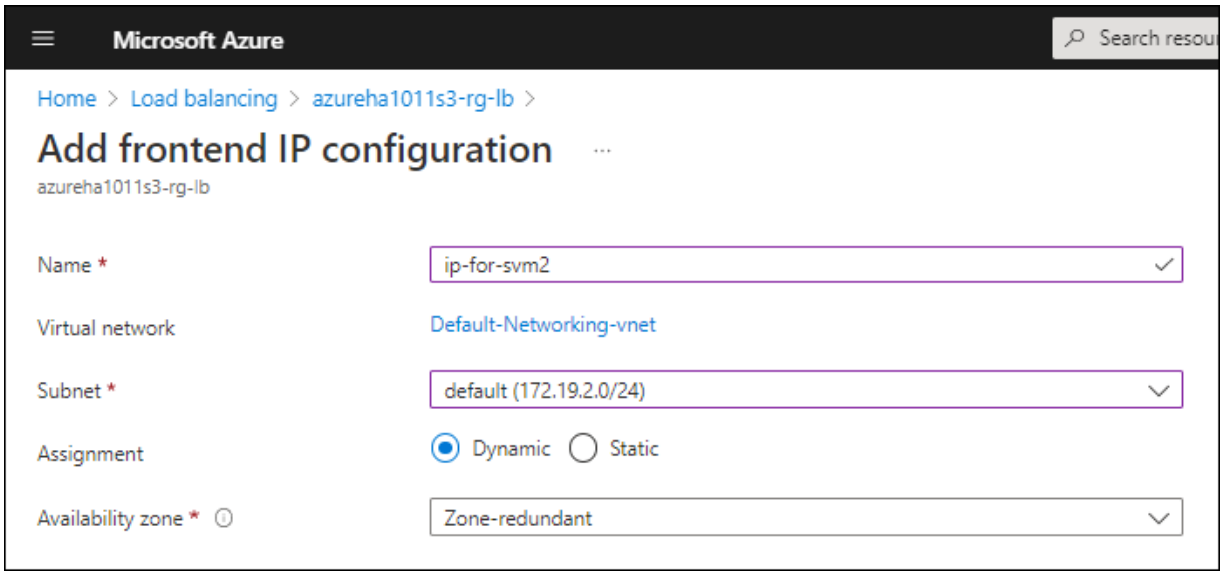
  This management LIF provides a connection to management tools like SnapCenter.

Steps

1. In the Azure portal, open the **Load balancers** service.
2. Click the name of the load balancer for the HA pair.
3. Create the required number of frontend IP configurations for the data and SVM LIFs only:

> A frontend IP should only be created under the NIC0 for each corresponding SVM. For more information on how to add the IP address to the SVM NIC0, see "Step 7 [hyperlink]"

   a. Under **Settings**, click **Frontend IP configuration**.
   b. Click **Add**.
   c. Enter a name for the frontend IP, select the subnet for the Cloud Volumes ONTAP HA pair, leave **Dynamic** selected, and in regions with Availability Zones, leave **Zone-redundant** selected to ensure that the IP address remains available if a zone fails.

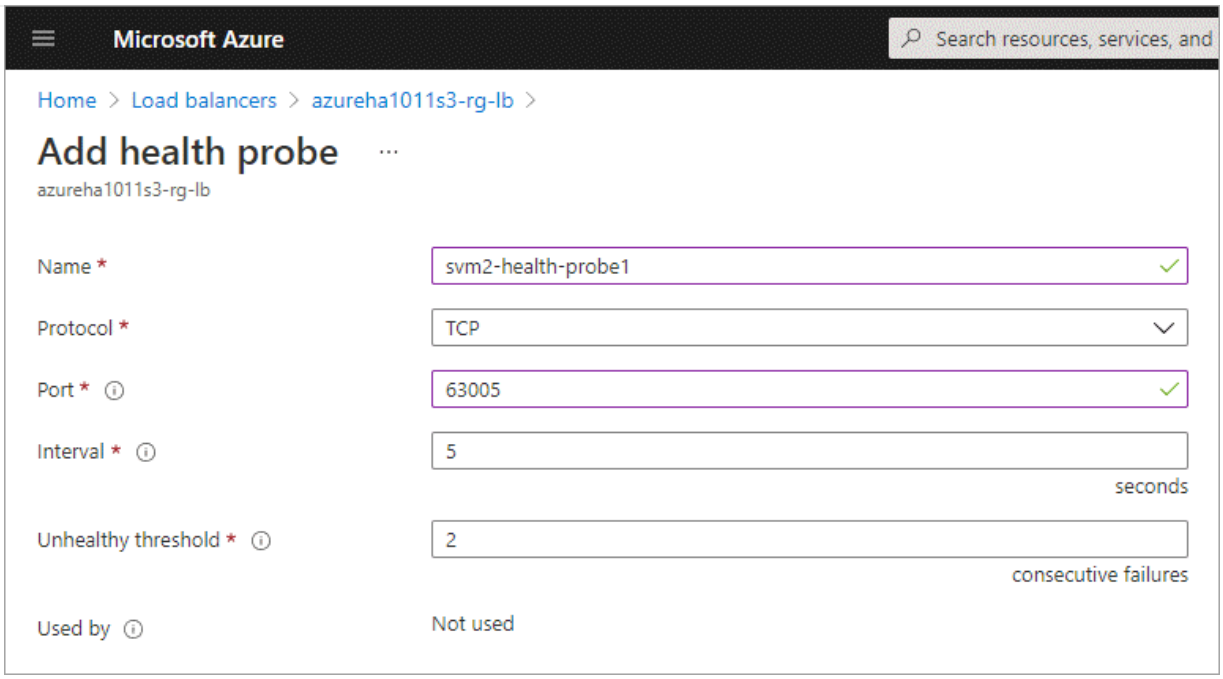


   d. Click the name of the frontend IP configuration that you just created, change the **Assignment** to **Static**, and click **Save**.

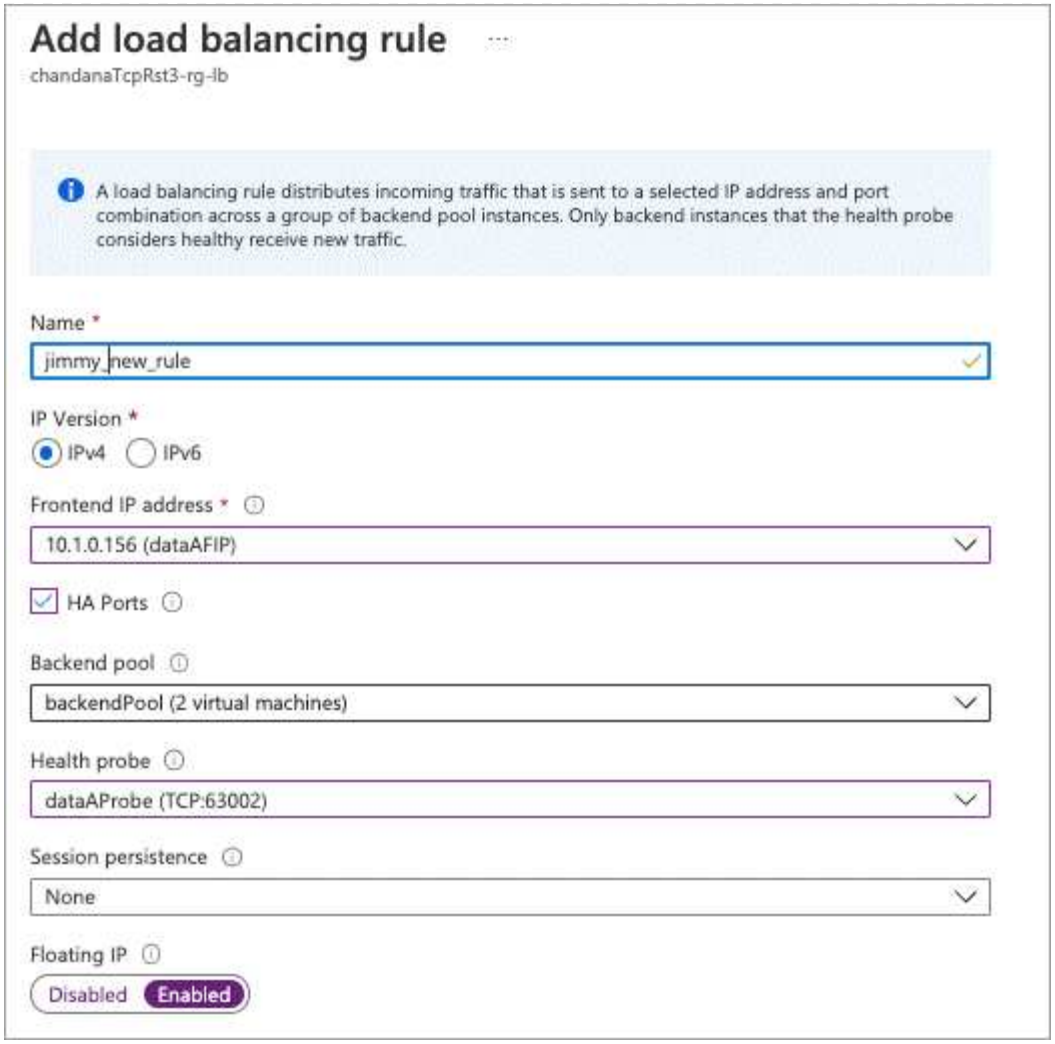
      It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

4. Add a health probe for each frontend IP that you just created.
   a. Under the load balancer's **Settings**, click **Health probes**.
   b. Click **Add**.
   c. Enter a name for the health probe and enter a port number that's between 63005 and 65000. Keep the default values for the other fields.

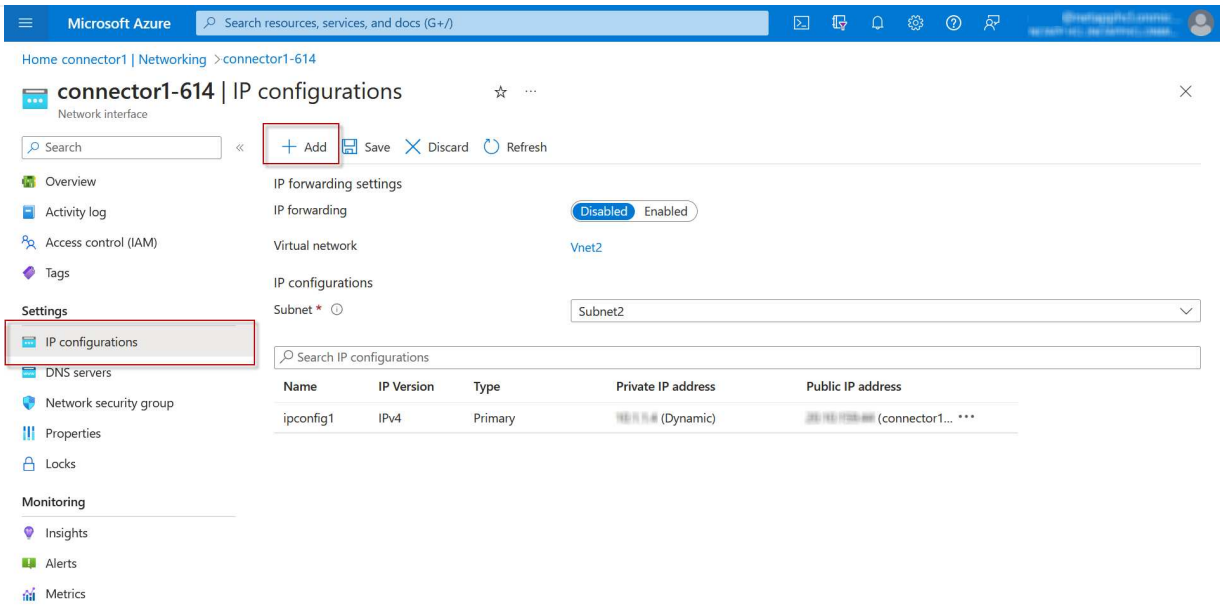
      It's important that the port number is between 63005 and 65000. For example, if you are creating three health probes, you could enter probes that use the port numbers 63005, 63006, and 63007.



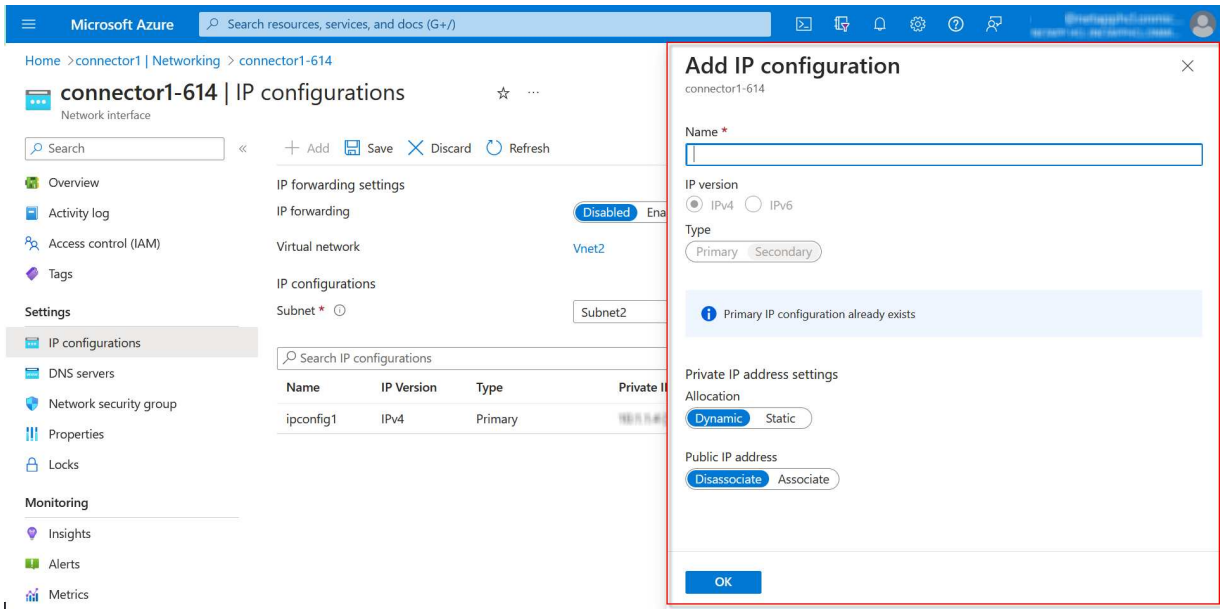
5. Create new load balancing rules for each frontend IP.
   a. Under the load balancer's **Settings**, click **Load balancing rules**.
   b. Click **Add** and enter the required information:
      - **Name**: Enter a name for the rule.
      - **IP Version**: Select **IPv4**.
      - **Frontend IP address**: Select one of the frontend IP addresses that you just created.
      - **HA Ports**: Enable this option.
      - **Backend pool**: Keep the default Backend pool that was already selected.
      - **Health probe**: Select the health probe that you created for the selected frontend IP.
      - **Session persistence**: Select **None**.
      - **Floating IP**: Select **Enabled**.



6. Ensure that the network security group rules for Cloud Volumes ONTAP allows the load balancer to send TCP probes for the health probes that were created in step 4 above. Note that this is allowed by default.
7. For iSCSI LIFs, add the IP address for NIC0.
   a. Click the name of the Cloud Volumes ONTAP VM.
   b. Click **Networking**.
   c. Click the name of the network interface for nic0.
   d. Under Settings, click **IP configurations**.
   e. Click **Add**.



f. Enter a name for the IP configuration, select Dynamic, and then click **OK**.



g. Click the name of the IP configuration that you just created, change the Assignment to Static, and click **Save**.

> It's best to use a static IP address because a static IP ensures that the IP address won't change, which can help to prevent unnecessary outages to your application.

After you finish

Copy the private IP addresses that you just created. You'll need to specify those IP addresses when you create LIFs for the new storage VM.

Create a storage VM and LIFs

After you allocate IP addresses in Azure, you can create a new storage VM on a single node system or on an HA pair.

Single node system

How you create a storage VM and LIFs on a single node system depends on the storage protocol that you're using.

iSCSI

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -home-port e0a -address
<iscsi-ip-address> -netmask-length <# of mask bits> -lif <lif-name>
-home-node <name-of-node1> -data-protocol iscsi
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

NFS

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol cifs,nfs -address <nas-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy disabled -firewall-policy data -home-port e0a -auto
-revert true -failover-group Default
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

SMB

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create a data LIF:

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol cifs,nfs -address <nas-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy disabled -firewall-policy data -home-port e0a -auto
-revert true -failover-group Default
```

3. Optional: Create a storage VM management LIF.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

HA pair

How you create a storage VM and LIFs on an HA pair depends on the storage protocol that you’re using.

iSCSI

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create data LIFs:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

3. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default
```

   This management LIF provides a connection to management tools like SnapCenter.

4. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you

can create volumes on the storage VM.

5. If you're running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.

   a. Enter the following command to access advanced mode.

   ```
   ::> set adv -con off
   ```

   Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

   ```
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service data-fpolicy-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-ad-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-dns-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-ldap-client
   network interface service-policy remove-service -vserver <svm-
   name> -policy default-data-files -service management-nis-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service data-fpolicy-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-ad-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-dns-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-ldap-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-blocks -service management-nis-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service data-fpolicy-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-ad-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-dns-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-ldap-client
   network interface service-policy add-service -vserver <svm-name>
   -policy default-data-iscsi -service management-nis-client
   ```

NFS

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create data LIFs:
   a. Use the following command to create a NAS LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node1> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe1>
```

   b. Use the following command to create a NAS LIF on node 2.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node2> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe2>
```

3. Create iSCSI LIFs to provide DNS communication:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

4. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

   This management LIF provides a connection to management tools like SnapCenter.

5. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

   This management LIF provides a connection to management tools like SnapCenter.

6. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

   This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

7. If you’re running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.
   a. Enter the following command to access advanced mode.

```
::> set adv -con off
```

      Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

```
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service data-fpolicy-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ad-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-dns-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ldap-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-nis-client
```

SMB

Follow these steps to create a new storage VM, along with the required LIFs.

Steps

1. Create the storage VM and a route to the storage VM.

```
vserver create -vserver <svm-name> -subtype default -rootvolume
<root-volume-name> -rootvolume-security-style unix
```

```
network route create -destination 0.0.0.0/0 -vserver <svm-name>
-gateway <ip-of-gateway-server>
```

2. Create NAS data LIFs:
   a. Use the following command to create a NAS LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node1> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe1>
```

   b. Use the following command to create a NAS LIF on node 2.

```
network interface create -vserver <svm-name> -lif <lif-name>
-role data -data-protocol cifs,nfs -address <nfs-cifs-ip-address>
-netmask-length <length> -home-node <name-of-node2> -status-admin
up -failover-policy system-defined -firewall-policy data -home
-port e0a -auto-revert true -failover-group Default -probe-port
<port-number-for-azure-health-probe2>
```

3. Create iSCSI LIFs to provide DNS communication:
   a. Use the following command to create an iSCSI LIF on node 1.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node1> -data-protocol iscsi
```

   b. Use the following command to create an iSCSI LIF on node 2.

```
network interface create -vserver <svm-name> -home-port e0a
-address <iscsi-ip-address> -netmask-length <# of mask bits> -lif
<lif-name> -home-node <name-of-node2> -data-protocol iscsi
```

4. Optional: Create a storage VM management LIF on node 1.

```
network interface create -vserver <svm-name> -lif <lif-name> -role
data -data-protocol none -address <svm-mgmt-ip-address> -netmask
-length <length> -home-node <name-of-node1> -status-admin up
-failover-policy system-defined -firewall-policy mgmt -home-port e0a
-auto-revert false -failover-group Default -probe-port <port-number-
for-azure-health-probe3>
```

This management LIF provides a connection to management tools like SnapCenter.

5. Assign one or more aggregates to the storage VM.

```
vserver add-aggregates -vserver svm_2 -aggregates aggr1,aggr2
```

This step is required because the new storage VM needs access to at least one aggregate before you can create volumes on the storage VM.

6. If you’re running Cloud Volumes ONTAP 9.11.1 or later, modify the network service policies for the storage VM.

   a. Enter the following command to access advanced mode.

```
::> set adv -con off
```

   Modifying the services is required because it ensures that Cloud Volumes ONTAP can use the iSCSI LIF for outbound management connections.

```
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service data-fpolicy-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ad-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-dns-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-ldap-client
network interface service-policy remove-service -vserver <svm-
name> -policy default-data-files -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-blocks -service management-nis-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service data-fpolicy-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ad-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-dns-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-ldap-client
network interface service-policy add-service -vserver <svm-name>
-policy default-data-iscsi -service management-nis-client
```

What's next?

After you create a storage VM on an HA pair, it's best to wait 12 hours before you provision storage on that SVM. Starting with the Cloud Volumes ONTAP 9.10.1 release, BlueXP scans the settings for an HA pair's load balancer at a 12-hour interval. If there are new SVMs, BlueXP will enable a setting that provides shorter unplanned failover.

Set up SVM disaster recovery

BlueXP doesn't provide any setup or orchestration support for storage VM (SVM) disaster recovery. You must use System Manager or the CLI.

If you set up SnapMirror SVM replication between two Cloud Volumes ONTAP systems, the replication must be between two HA pair systems or two single node systems. You can't set up SnapMirror SVM replication between an HA pair and a single node system.

Refer to the following documents for CLI instructions.

- SVM Disaster Recovery Preparation Express Guide
- SVM Disaster Recovery Express Guide

Security and data encryption

Encrypting volumes with NetApp encryption solutions

Cloud Volumes ONTAP supports NetApp Volume Encryption (NVE) and NetApp Aggregate Encryption (NAE). NVE and NAE are software-based solutions that enable FIPS 140-2–compliant data-at-rest encryption of volumes. Learn more about these encryption solutions.

Both NVE and NAE are supported with an external key manager.

If you use NVE, you have the option to use your cloud provider's key vault to protect ONTAP encryption keys:

- Azure Key Vault (AKV)

New aggregates will have NAE enabled by default after you set up an external key manager. New volumes that aren't part of an NAE aggregate will have NVE enabled by default (for example, if you have existing aggregates that were created before setting up an external key manager).

Cloud Volumes ONTAP doesn't support onboard key management.

What you'll need

Your Cloud Volumes ONTAP system should be registered with NetApp support. A NetApp Volume Encryption license is automatically installed on each Cloud Volumes ONTAP system that is registered with NetApp Support.

- Adding NetApp Support Site accounts to BlueXP
- Registering pay-as-you-go systems

BlueXP doesn't install the NVE license on systems that reside in the China region.

Steps

1. Review the list of supported key managers in the NetApp Interoperability Matrix Tool.

   Search for the **Key Managers** solution.

2. Connect to the Cloud Volumes ONTAP CLI.
3. Configure external key management.
   - Azure: Azure Key Vault (AKV)

Manage keys with Azure Key Vault

You can use Azure Key Vault (AKV) to protect your ONTAP encryption keys in an Azure-deployed application.

AKV can be used to protect NetApp Volume Encryption (NVE) keys only for data SVMs.

Key management with AKV can be enabled with the CLI or the ONTAP REST API.

When using AKV, be aware that by default a data SVM LIF is used to communicate with the cloud key management endpoint. A node management network is used to communicate with the cloud provider's authentication services (login.microsoftonline.com). If the cluster network is not configured correctly, the cluster will not properly utilize the key management service.

Before you begin

- Cloud Volumes ONTAP must be running version 9.10.1 or later
- Volume Encryption (VE) license installed (NetApp Volume Encryption license is automatically installed on each Cloud Volumes ONTAP system that is registered with NetApp Support)
- You must have a Multi-tenant Encryption Key Management (MT_EK_MGMT) license
- You must be a cluster or SVM administrator
- An Active Azure subscription

Limitations

- AKV can only be configured on a data SVM
- NAE can not be used using AKV. NAE requires an external-supported KMIP server.

Configuration process

The outlined steps capture how to register your Cloud Volumes ONTAP configuration with Azure and how to create an Azure Key Vault and keys. If you have already completed these steps, ensure you have the correct configuration settings, particularly in Create an Azure Key Vault, and then proceed to Cloud Volumes ONTAP configuration.

- Azure Application Registration
- Create Azure client secret
- Create an Azure Key Vault
- Create encryption key
- Create an Azure Active Directory Endpoint (HA only)
- Cloud Volumes ONTAP configuration

Azure Application Registration

1. You must first register your application in the Azure subscription that you want the Cloud Volumes ONTAP to use for access the Azure Key Vault. Within the Azure portal, select **App registrations**.
2. Select **New registration**.
3. Provide a name for your application and select a supported application type. The default single tenant suffices for Azure Key Vault usage. Select **Register**.
4. In the Azure Overview window, select the application you have registered. Copy the **application (client) ID** and the **directory (tenant) ID** to a secure location. They will be required later in the registration process.

Create Azure client secret

1. In the Azure portal for your Azure Key Vault app registration, select the **Certificates & secrets** pane.
2. Select **New client secret**. Enter a meaningful name for your client secret. NetApp recommends a 24-

month expiration period; however, your specific cloud governance policies may require a different setting.

3. Click **Add** to create the client secret. Copy the secret string listed in the **Value** column and store it in a secure location for use later in Cloud Volumes ONTAP configuration. The secret value will not be displayed again after you navigate away from the page.

Create an Azure Key Vault

1. If you have an existing Azure Key Vault, you can connect it to your Cloud Volumes ONTAP configuration; however, you must adapt the access policies to the settings in this process.
2. In the Azure portal, navigate to the **Key Vaults** section.
3. Click **+Create** and enter the required information including resource group, region, and pricing tier. In addition, enter the number of days to retain deleted vaults and select **Enable purge protection** on the key vault.
4. Select **Next** to choose an access policy.
5. Select the following options:
   a. Under **Access configuration**, select the **Vault access policy**.
   b. Under **Resource access**, select **Azure Disk Encryption for volume encryption**.
6. Select **+Create** to add an access policy.
7. Under **Configure from a template**, click the drop-down menu and then select the **Key, Secret, and Certificate Management** template.
8. Choose each of the drop-down permissions menus (key, secret, certificate) and then **Select all** at the top of the menu list to select all the permissions available. You should have:
   - **Key permissions**: 20 selected
   - **Secret permissions**: 8 selected
   - **Certificate permissions**: 16 selected

Create an access policy

1 Permissions 2 Principal 3 Application (optional) 4 Review + create

Configure from a template

Key, Secret, & Certificate Management

Key permissions

Key Management Operations

- [x] Select all
- [x] Get
- [x] List
- [x] Update
- [x] Create
- [x] Import
- [x] Delete
- [x] Recover
- [x] Backup
- [x] Restore

Cryptographic Operations

- [x] Select all
- [x] Decrypt
- [x] Encrypt
- [x] Unwrap Key
- [x] Wrap Key
- [x] Verify
- [x] Sign

Privileged Key Operations

- [x] Select all
- [x] Purge
- [x] Release

Rotation Policy Operations

- [x] Select all
- [x] Rotate
- [x] Get Rotation Policy
- [x] Set Rotation Policy

Secret permissions

Secret Management Operations

- [x] Select all
- [x] Get
- [x] List
- [x] Set
- [x] Delete
- [x] Recover
- [x] Backup
- [x] Restore

Privileged Secret Operations

- [x] Select all
- [x] Purge

Certificate permissions

Certificate Management Operations

- [x] Select all
- [x] Get
- [x] List
- [x] Update
- [x] Create
- [x] Import
- [x] Delete
- [x] Recover
- [x] Backup
- [x] Restore
- [x] Manage Contacts
- [x] Manage Certificate Authorities
- [x] Get Certificate Authorities
- [x] List Certificate Authorities
- [x] Set Certificate Authorities
- [x] Delete Certificate Authorities

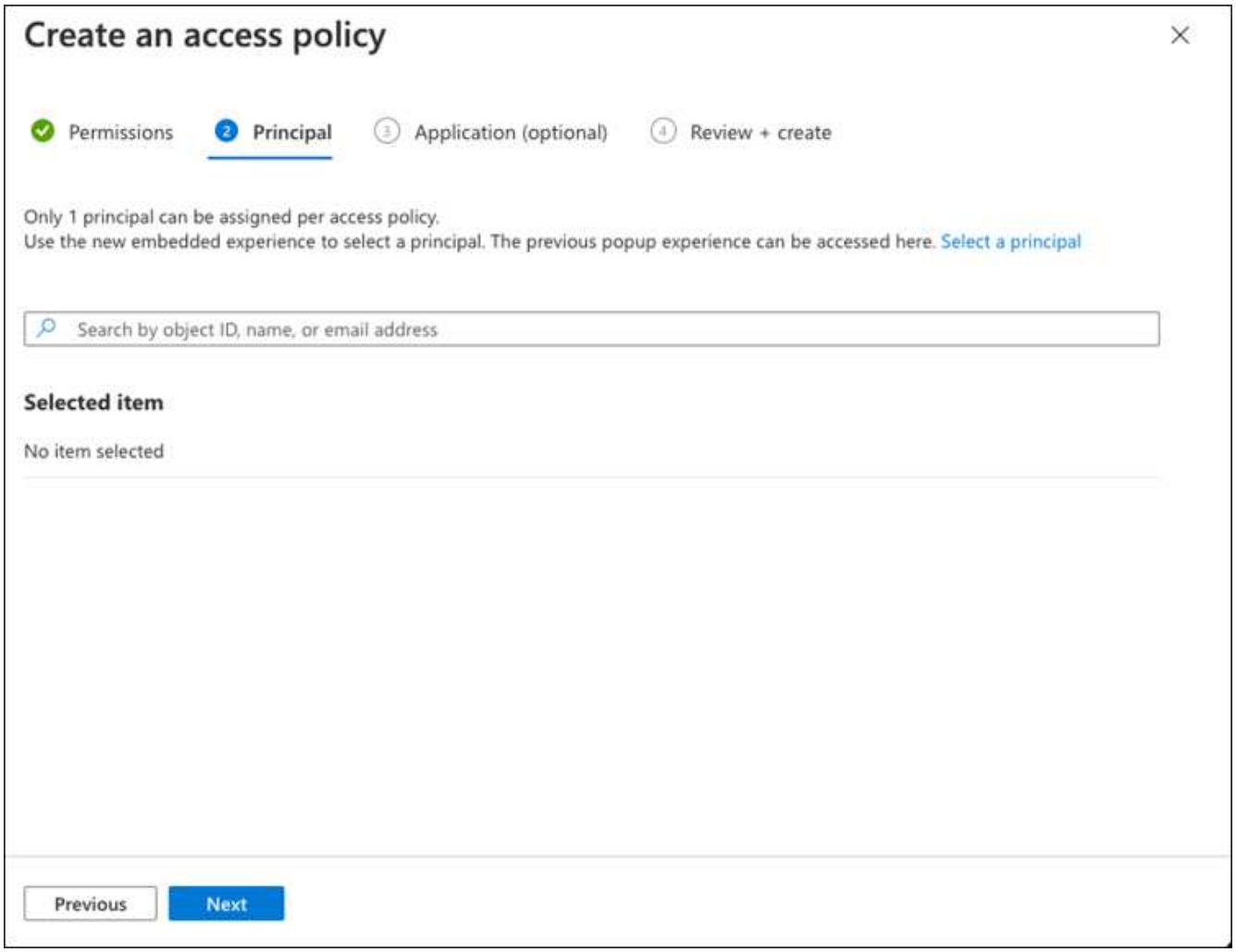
Privileged Certificate Operations

- [x] Select all
- [x] Purge

Previous Next

9. Click **Next** to select the **Principal** Azure registered application you created in Azure Application Registration. Select **Next**.

> Only one principal can be assigned per policy.



10. Click **Next** two times until you arrive at **Review and create**. Then, click **Create**.
11. Select **Next** to advance to **Networking** options.
12. Choose the appropriate network access method or select **All networks** and **Review + Create** to create the key vault. (Network access method may be prescribed by a governance policy or your corporate cloud security team.)
13. Record the Key Vault URI: In the key vault you created, navigate to the Overview menu and copy the **Vault URI** from the right-hand column. You need this for a later step.

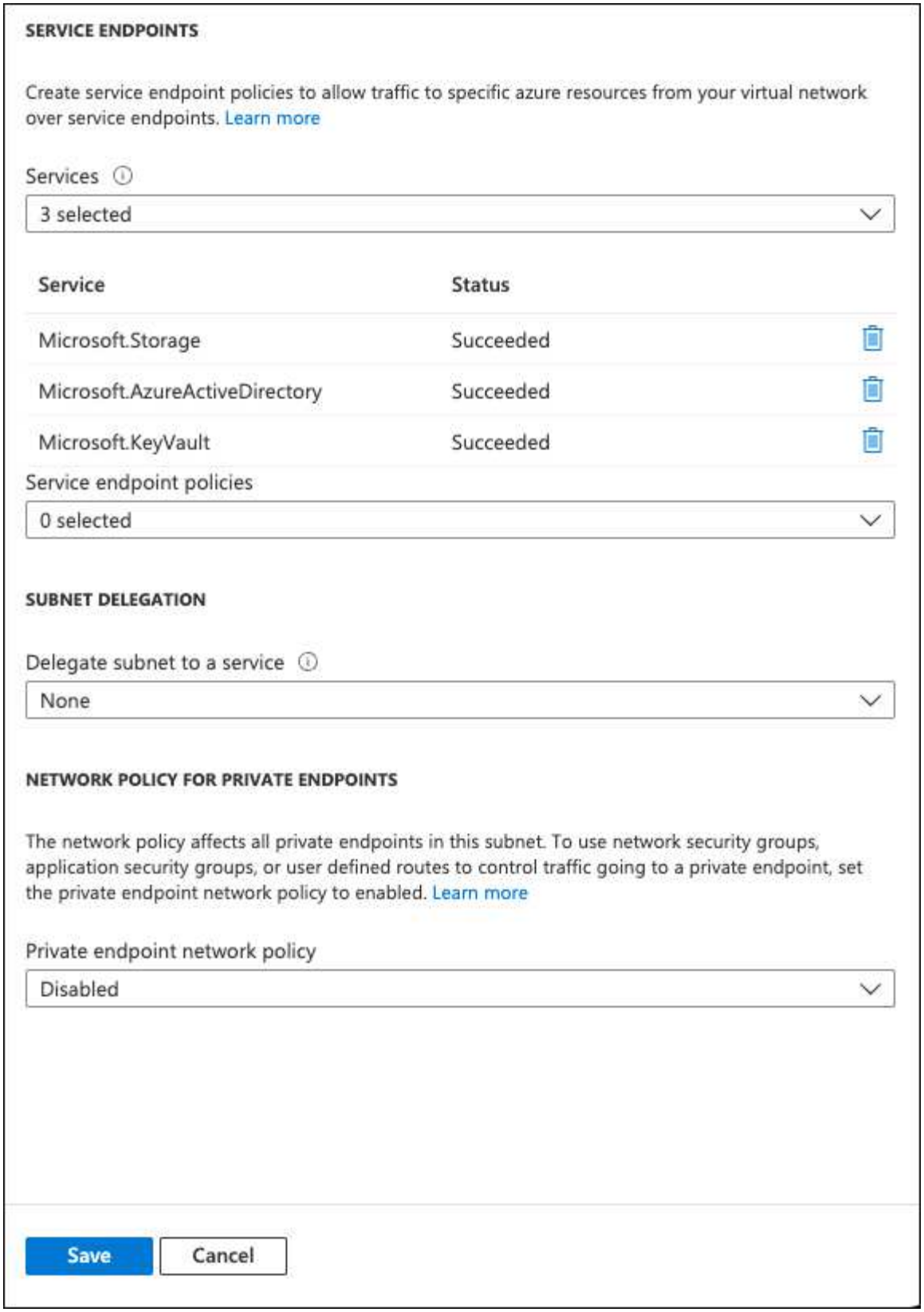
Create encryption key

1. In the menu for the Key Vault you have created for Cloud Volumes ONTAP, navigate to the **Keys** option.
2. Select **Generate/import** to create a new key.
3. Leave the default option set to **Generate**.
4. Provide the following information:
    - Encryption key name

    - Key type: RSA
    - RSA key size: 2048
    - Enabled: Yes
5. Select **Create** to create the encryption key.
6. Return to the **Keys** menu and select the key you just created.
7. Select the key ID under **Current version** to view the key properties.
8. Locate the **Key Identifier** field. Copy the URI up to but not including the hexadecimal string.

Create an Azure Active Directory Endpoint (HA only)

1. This process is only required if you are configuring Azure Key Vault for an HA Cloud Volumes ONTAP Working Environment.
2. In the Azure portal navigate to **Virtual Networks**.
3. Select the Virtual Network where you deployed the Cloud Volumes ONTAP working environment and select the **Subnets** menu on the left side of the page.
4. Select the subnet name for you Cloud Volumes ONTAP deployment from the list.
5. Navigate to the **Service Endpoints** heading. In the drop-down menu, select the following:
    - **Microsoft.AzureActiveDirectory**
    - **Microsoft.KeyVault**
    - **Microsoft.Storage** (optional)



6. Select **Save** to capture your settings.

Cloud Volumes ONTAP configuration

1. Connect to the cluster management LIF with your preferred SSH client.
2. Enter the advanced privilege mode in ONTAP:

```
set advanced -con off
```

3. Identify the desired data SVM and verify its DNS configuration:
   `vserver services name-service dns show`
   a. If a DNS entry for the desired data SVM exists and it contains an entry for the Azure DNS, then no action is required. If it does not, add a DNS server entry for the data SVM that points to the Azure DNS, private DNS, or on-premise server. This should match the entry for the cluster admin SVM:
      `vserver services name-service dns create -vserver SVM_name -domains domain -name-servers IP_address`
   b. Verify the DNS service has been created for the data SVM:
      `vserver services name-service dns show`
4. Enable Azure Key Vault using the client ID and tenant ID saved after the application registration:
   `security key-manager external azure enable -vserver SVM_name -client-id Azure_client_ID -tenant-id Azure_tenant_ID -name key_vault_URI -key-id full_key_URI`

   > The `_full_key_URI` value must utilize the `<https:// <key vault host name>/keys/<key label>` format.

5. Upon successful enablement of the Azure Key Vault, enter the `client secret value` when prompted.
6. Check the status of the key manager:
   `security key-manager external azure check`
   The output will look like:

```
::*> security key-manager external azure check

Vserver: data_svm_name
Node: akvlab01-01

Category: service_reachability
    Status: OK

Category: ekmip_server
    Status: OK

Category: kms_wrapped_key_status
    Status: UNKNOWN
    Details: No volumes created yet for the vserver. Wrapped KEK status
will be available after creating encrypted volumes.

3 entries were displayed.
```

If the `service_reachability` status is not `OK`, the SVM cannot reach the Azure Key Vault service with all the required connectivity and permissions. Ensure that your Azure network policies and routing don't block your private vNet from reaching the Azure KeyVault Public endpoint. If they do, consider using an Azure Private endpoint to access the Key vault from within the vNet. You may also need to add a static hosts entry on your SVM to resolve the private IP address for your endpoint.

The `kms_wrapped_key_status` will report `UNKNOWN` at initial configuration. Its status will change to `OK` after the first volume is encrypted.

7. OPTIONAL: Create a test volume to verify the functionality of NVE.

   ```
   vol create -vserver SVM_name -volume volume_name -aggregate aggr -size size
   -state online -policy default
   ```

   If configured correctly, Cloud Volumes ONTAP will automatically create the volume and enable volume encryption.

8. Confirm the volume was created and encrypted correctly. If it is, the `-is-encrypted` parameter will display as `true`.
   ```
   vol show -vserver SVM_name -fields is-encrypted
   ```

Improving protection against ransomware

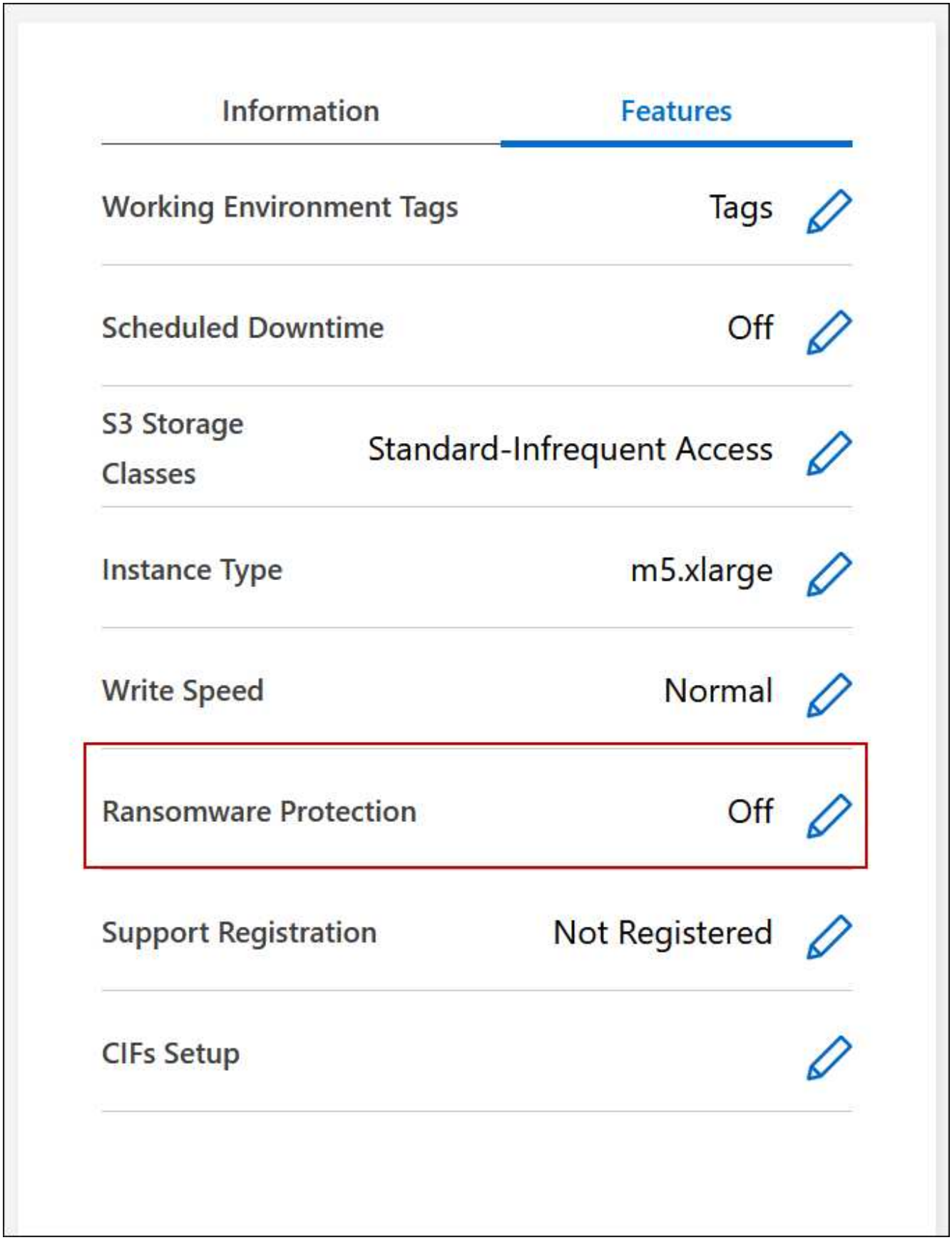
Ransomware attacks can cost a business time, resources, and reputation. BlueXP enables you to implement two NetApp solutions for ransomware: Protection from common ransomware file extensions and Autonomous Ransomware Protection (ARP). These solutions provide effective tools for visibility, detection, and remediation.

Protection from common ransomware file extensions

Available through BlueXP, the Ransomware Protection setting allows you to utilize the ONTAP FPolicy functionality to guard against common ransomware file extension types.

Steps

1. On the Canvas page, double-click the name of the system you configure to ransomware protection.
2. On the Overview tab, click the Features panel and then click the pencil icon next to **Ransomware Protection**.



3. Implement the NetApp solution for ransomware:
   a. Click **Activate Snapshot Policy**, if you have volumes that do not have a Snapshot policy enabled.

NetApp Snapshot technology provides the industry's best solution for ransomware remediation. The key to a successful recovery is restoring from uninfected backups. Snapshot copies are read-only, which prevents ransomware corruption. They can also provide the granularity to create images of a single file copy or a complete disaster recovery solution.

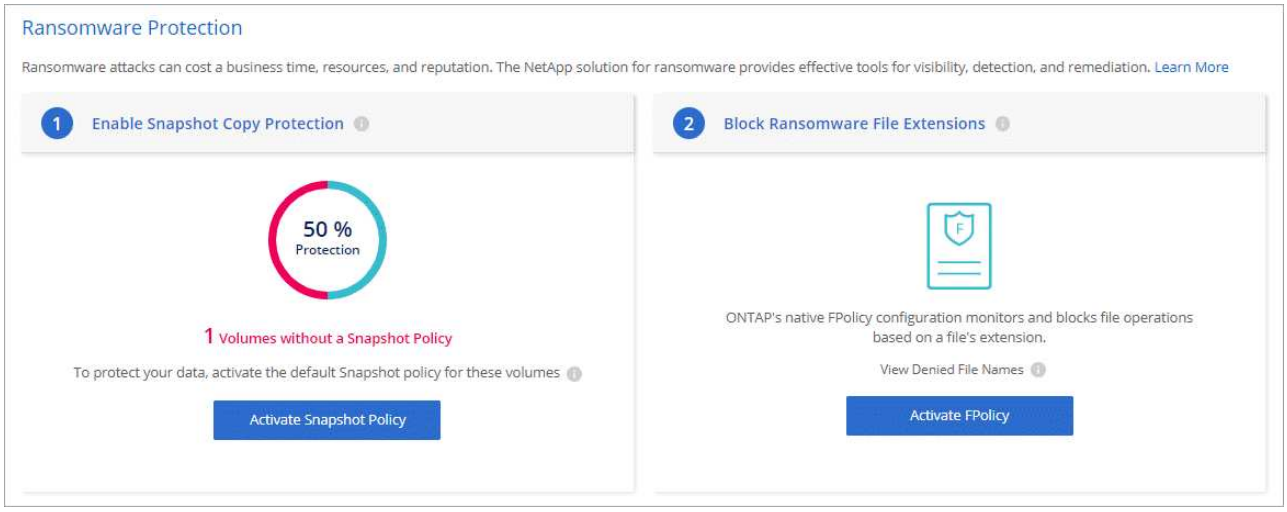
b. Click **Activate FPolicy** to enable ONTAP's FPolicy solution, which can block file operations based on a file's extension.

   This preventative solution improves protection from ransomware attacks by blocking common ransomware file types.

   The default FPolicy scope blocks files that have the following extensions:

   micro, encrypted, locked, crypto, crypt, crinf, r5a, XRNT, XTBL, R16M01D05, pzdc, good, LOL!, OMG!, RDM, RRK, encryptedRS, crjoker, EnCiPhErEd, LeChiffre

   > BlueXP creates this scope when you activate FPolicy on Cloud Volumes ONTAP. The list is based on common ransomware file types. You can customize the blocked file extensions by using the *vserver fpolicy policy scope* commands from the Cloud Volumes ONTAP CLI.



Autonomous Ransomware Protection

Cloud Volumes ONTAP supports the Autonomous Ransomware Protection (ARP) feature, which performs analyses on workloads to proactively detect and warn about abnormal activity that might indicate a ransomware attack.

Separate from the file extension protections provided through the ransomware protection setting, the ARP feature uses workload analysis to alert the user on potential attacks based on detected "abnormal activity". Both the ransomware protection setting and the ARP feature can be used in conjunction for comprehensive ransomware protection.

The ARP feature is available for use with BYOL licenses only (1 to 36 month terms) on both node-based and capacity-based licensing models. You must contact your NetApp sales representative to purchase a new, separate, add-on license for use with the ARP feature in Cloud Volumes ONTAP.

The ARP license is considered a "floating" license, which means it is not bound to a single Cloud Volumes ONTAP instance and can be applied to multiple Cloud Volumes ONTAP environments.

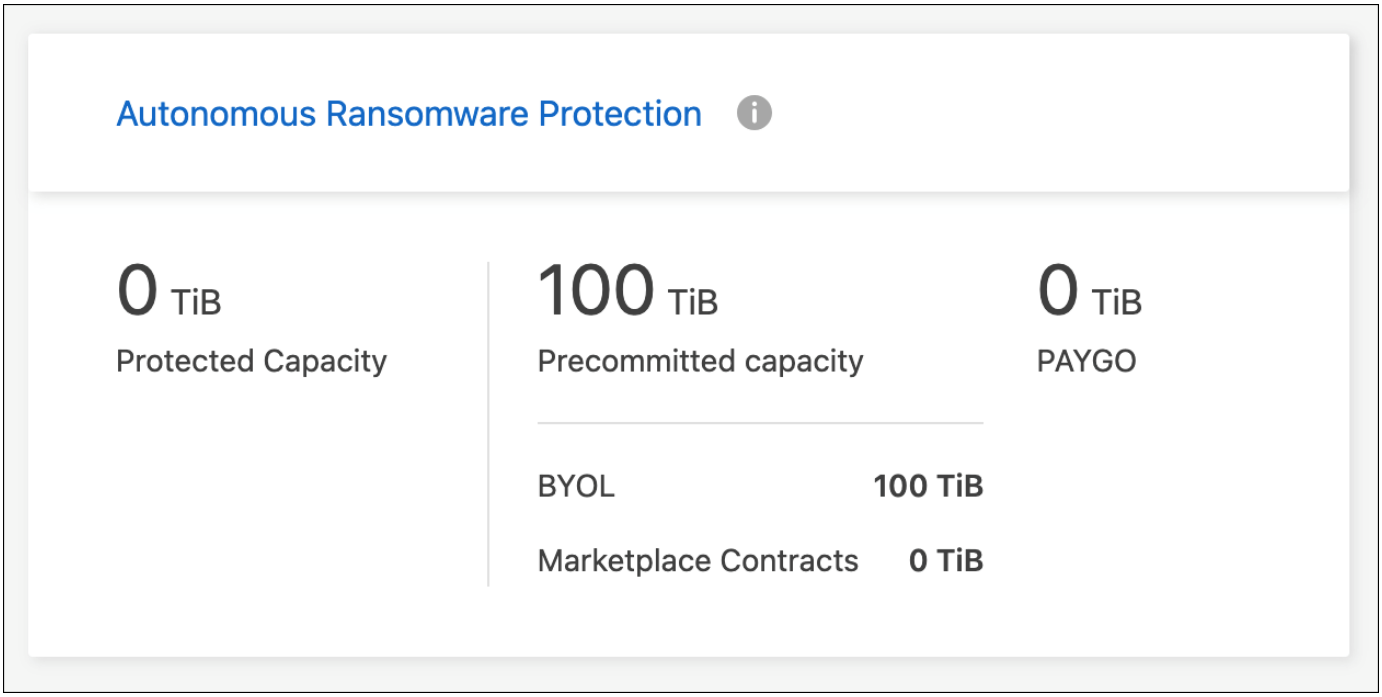
The usage of the ARP feature with node-based Cloud Volumes ONTAP licenses is not currently reflected in Digital Wallet. The ability to view node-based ARP usage will be available under Digital Wallet in a future release.

Upon purchase of an add-on license and adding it to the Digital Wallet, you can enable ARP on a per volume basis with Cloud Volumes ONTAP. Charging for ARP is metered at a volume level, according to the total provisioned capacity of volumes with the ARP feature enabled. The minimum license capacity is 1TB. However, there is no minimum capacity charging for the ARP feature.

ARP enabled volumes have a designated state of "Learning mode" or "Active". Any volume with an ARP state of "Disabled" is excluded from charging. For example, a Cloud Volumes ONTAP environment with 30 TiB of provisioned capacity can elect to have only a subset of 15 TiB volumes with ARP enabled.

Configuration of ARP for volumes is performed through ONTAP System Manager and ONTAP CLI.

For more information on how to enable ARP with ONTAP System Manager and CLI, see Enable Autonomous Ransomware Protection.



Support is not available for the use of licensed features without a license.

System administration

Upgrade Cloud Volumes ONTAP software

Upgrade Cloud Volumes ONTAP from BlueXP to gain access to the latest new features and enhancements. You should prepare Cloud Volumes ONTAP systems before you upgrade the software.

Upgrade overview

You should be aware of the following before you start the Cloud Volumes ONTAP upgrade process.

Upgrade from BlueXP only

Upgrades of Cloud Volumes ONTAP must be completed from BlueXP. You should not upgrade Cloud Volumes ONTAP by using System Manager or the CLI. Doing so can impact system stability.

How to upgrade

BlueXP provides two ways to upgrade Cloud Volumes ONTAP:

- By following upgrade notifications that appear in the working environment
- By placing the upgrade image at an HTTPS location and then providing BlueXP with the URL

Supported upgrade paths

The version of Cloud Volumes ONTAP that you can upgrade to depends on the version of Cloud Volumes ONTAP that you're currently running.

Current version	Versions that you can directly upgrade to
9.14.0	9.14.1
9.13.1	9.14.1
	9.14.0
9.13.0	9.13.1
9.12.1	9.13.1
	9.13.0
9.12.0	9.12.1
9.11.1	9.12.1
	9.12.0
9.11.0	9.11.1
9.10.1	9.11.1
	9.11.0
9.10.0	9.10.1
9.9.1	9.10.1
	9.10.0
9.9.0	9.9.1
9.8	9.9.1
9.7	9.8
9.6	9.7

Current version	Versions that you can directly upgrade to
9.5	9.6
9.4	9.5
9.3	9.4
9.2	9.3
9.1	9.2
9.0	9.1
8.3	9.0

Note the following:

- The supported upgrade paths for Cloud Volumes ONTAP are different than they are for an on-premises ONTAP cluster.
- If you upgrade by following the upgrade notifications that appear in a working environment, BlueXP will prompt you to upgrade to a release that follows these supported upgrade paths.
- If you upgrade by placing an upgrade image at an HTTPS location, be sure to follow these supported upgrade paths.
- In some cases, you might need to upgrade a few times to reach your target release.

  For example, if you're running version 9.8 and you want to upgrade to 9.10.1, you first need to upgrade to version 9.9.1 and then to 9.10.1.

Patch releases

Starting in January 2024, patch upgrades are only available in BlueXP if they are a patch release for the three latest versions of Cloud Volumes ONTAP. We use the latest GA release to determine the three latest versions to display in BlueXP. For example, if the current GA release is 9.13.1, patches for 9.11.1-9.13.1 appear in BlueXP. If you want to upgrade to a patch release for versions 9.11.1 or below, you will need to use the manual upgrade procedure by downloading the ONTAP image.

As a general rule for patch (P) releases, you can upgrade from one version release to any P-release of the current version you're running or the next version.

Here are a couple examples:

- 9.13.0 > 9.13.1P15
- 9.12.1 > 9.13.1P2

Reverting or downgrading

Reverting or downgrading Cloud Volumes ONTAP to a previous release is not supported.

Support registration

Cloud Volumes ONTAP must be registered with NetApp support in order to upgrade the software using any of the methods described on this page. This applies to both PAYGO and BYOL. You'll need to manually register PAYGO systems, while BYOL systems are registered by default.

A system that isn't registered for support will still receive the software update notifications that appear in BlueXP when a new version is available. But you will need to register the system before you can upgrade the software.

Upgrades of the HA mediator

BlueXP also updates the mediator instance as needed during the Cloud Volumes ONTAP upgrade process.

Upgrades in AWS with c4, m4, and r4 EC2 instance types

Cloud Volumes ONTAP no longer supports the c4, m4, and r4 EC2 instance types. You can upgrade existing deployments to Cloud Volumes ONTAP versions 9.8-9.12.1 with these instance types. Before you upgrade we recommend that you change the instance type. If you can't change the instance type, you need to enable enhanced networking before you upgrade. Read the following sections to learn more about changing the instance type and enabling enhanced networking.

In Cloud Volumes ONTAP running versions 9.13.0 and above, you cannot upgrade with c4, m4, and r4 EC2 instance types. In this case, you need to reduce the number of disks and then change the instance type or deploy a new HA-pair configuration with the c5, m5, and r5 EC2 instance types and migrate the data.

Change the instance type

c4, m4, and r4 EC2 instance types allow for more disks per node than the c5, m5, and r5 EC2 instance types. If the disk count per node for the c4, m4, or r4 EC2 instance you're running is below the max disk allowance per node for c5, m5, and r5 instances, you can change the EC2 instance type to c5, m5, or r5.

Check disk and tiering limits by EC2 instance
Change the EC2 instance type for Cloud Volumes ONTAP

If you can't change the instance type, follow the steps in Enable enhanced networking.

Enable enhanced networking

To upgrade to Cloud Volumes ONTAP versions 9.8 and later, you must enable *enhanced networking* on the cluster running the c4, m4, or r4 instance type. To enable ENA, refer to the Knowledge Base article "How to enable Enhanced networking like SR-IOV or ENA on AWS Cloud Volumes ONTAP instances".

Prepare to upgrade

Before performing an upgrade, you must verify that your systems are ready and make any required configuration changes.

- Plan for downtime
- Verify that automatic giveback is still enabled
- Suspend SnapMirror transfers
- Verify that aggregates are online
- Verify that all LIFs are on home ports

Plan for downtime

When you upgrade a single-node system, the upgrade process takes the system offline for up to 25 minutes, during which I/O is interrupted.

In many cases, upgrading an HA pair is nondisruptive and I/O is uninterrupted. During this nondisruptive upgrade process, each node is upgraded in tandem to continue serving I/O to clients.

Session-oriented protocols might cause adverse effects on clients and applications in certain areas during upgrades. For details, refer to ONTAP documentation

Verify that automatic giveback is still enabled

Automatic giveback must be enabled on a Cloud Volumes ONTAP HA pair (this is the default setting). If it isn't, then the operation will fail.

ONTAP 9 Documentation: Commands for configuring automatic giveback

Suspend SnapMirror transfers

If a Cloud Volumes ONTAP system has active SnapMirror relationships, it is best to suspend transfers before you update the Cloud Volumes ONTAP software. Suspending the transfers prevents SnapMirror failures. You must suspend the transfers from the destination system.

> Even though BlueXP backup and recovery uses an implementation of SnapMirror to create backup files (called SnapMirror Cloud), backups do not need to be suspended when a system is upgraded.

About this task

These steps describe how to use System Manager for version 9.3 and later.

Steps

1. Log in to System Manager from the destination system.

   You can log in to System Manager by pointing your web browser to the IP address of the cluster management LIF. You can find the IP address in the Cloud Volumes ONTAP working environment.

   > The computer from which you are accessing BlueXP must have a network connection to Cloud Volumes ONTAP. For example, you might need to log in to BlueXP from a jump host that's in your cloud provider network.

2. Click **Protection > Relationships**.
3. Select the relationship and click **Operations > Quiesce**.

Verify that aggregates are online

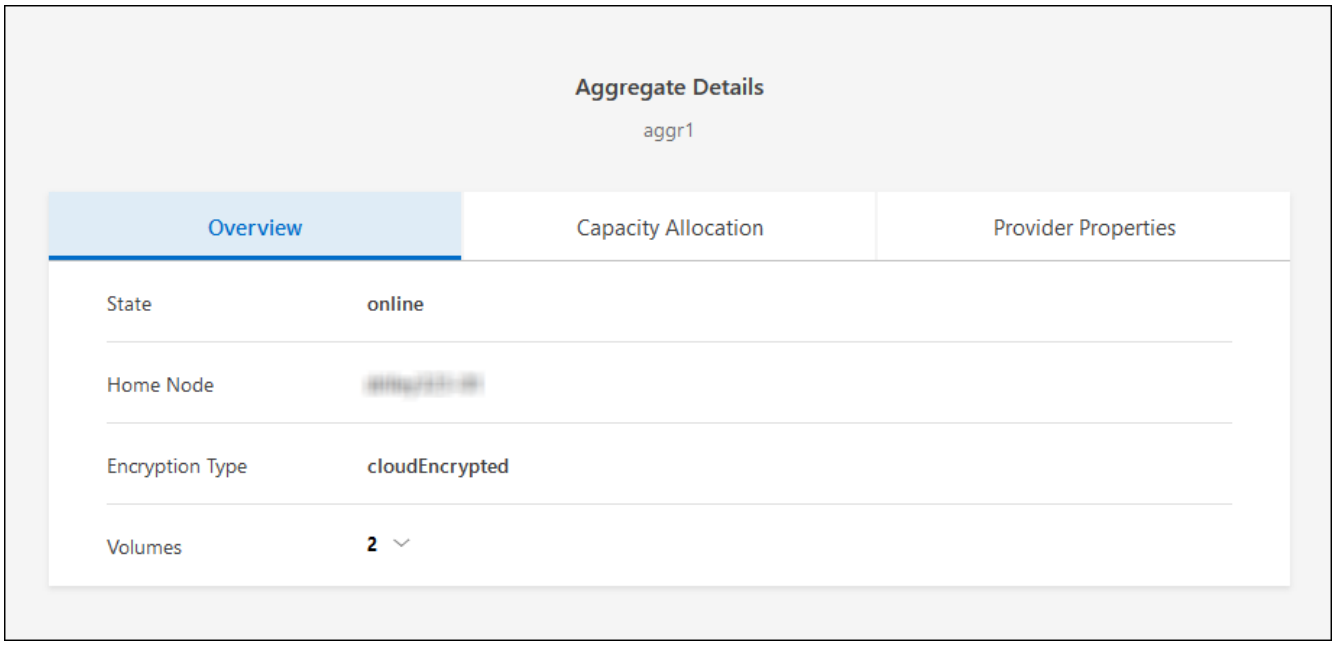
Aggregates for Cloud Volumes ONTAP must be online before you update the software. Aggregates should be online in most configurations, but if they are not, then you should bring them online.

About this task

These steps describe how to use System Manager for version 9.3 and later.

Steps

1. In the working environment, click the **Aggregates** tab.
2. Under the aggregate title, click the ellipse button, and then select **View Aggregate details**.



3. If the aggregate is offline, use System Manager to bring the aggregate online:
   a. Click **Storage > Aggregates & Disks > Aggregates**.
   b. Select the aggregate, and then click **More Actions > Status > Online**.

Verify that all LIFs are on home ports

Before you upgrade, all LIFs must be on home ports. Refer to ONTAP documentation to verify that all LIFs are on home ports.

If an upgrade failure error occurs, refer to the Knowledge Base article "Cloud Volumes ONTAP upgrade fails".

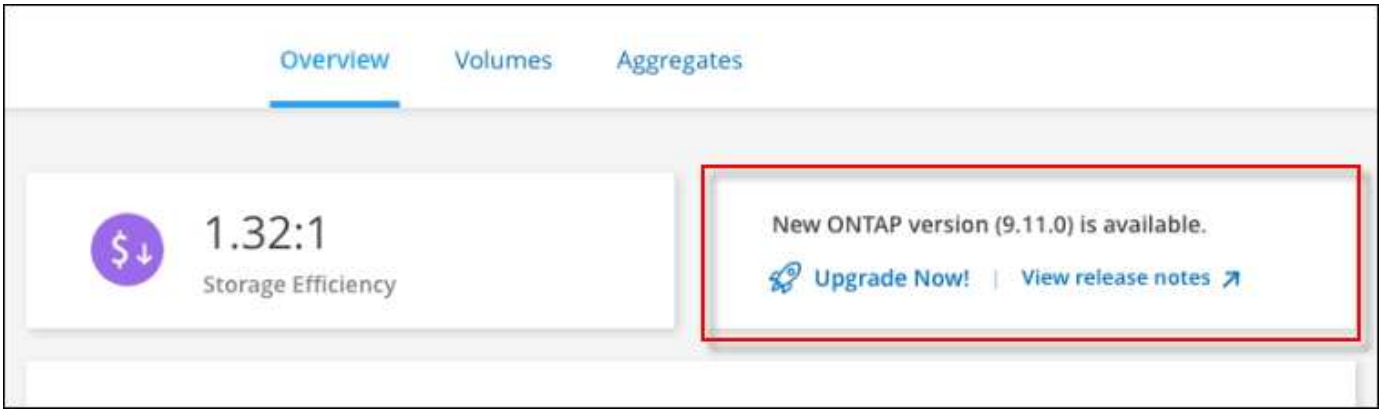
Upgrade Cloud Volumes ONTAP

BlueXP notifies you when a new version is available for upgrade. You can start the upgrade process from this notification. For details, see Upgrade from BlueXP notifications.

Another way to perform software upgrades by using an image on an external URL. This option is helpful if BlueXP can’t access the S3 bucket to upgrade the software or if you were provided with a patch. For details, see Upgrade from an image available at a URL.

Upgrade from BlueXP notifications

BlueXP displays a notification in Cloud Volumes ONTAP working environments when a new version of Cloud Volumes ONTAP is available:



You can start the upgrade process from this notification, which automates the process by obtaining the software image from an S3 bucket, installing the image, and then restarting the system.

Before you begin

BlueXP operations such as volume or aggregate creation must not be in progress on the Cloud Volumes ONTAP system.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. Select a working environment.

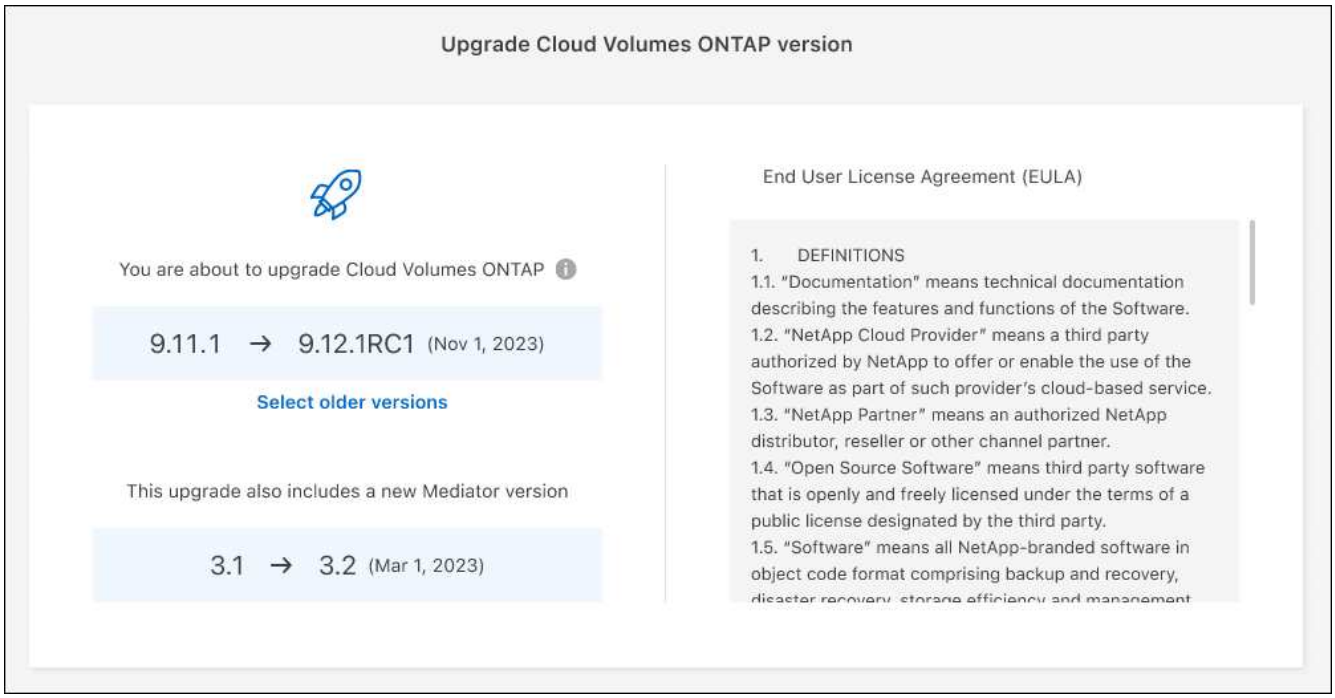
   A notification appears in the Overview tab if a new version is available:

3. If a new version is available, click **Upgrade Now!**

   Before you can upgrade Cloud Volumes ONTAP through the BlueXP notification, you must have a NetApp Support Site account.

4. In the Upgrade Cloud Volumes ONTAP page, read the EULA, and then select **I read and approve the EULA**.
5. Click **Upgrade**.

   The Upgrade Cloud Volumes ONTAP page selects the latest available Cloud Volumes ONTAP version for upgrade by default. If available, older versions of Cloud Volumes ONTAP can instead be selected for your upgrade by clicking **Select older versions**.
   Refer to the Supported upgrade paths list for the appropriate upgrade path based on your current Cloud Volumes ONTAP version.



6. To check the status of the upgrade, click the Settings icon and select **Timeline**.

Result

BlueXP starts the software upgrade. You can perform actions on the working environment when the software update is complete.

After you finish

If you suspended SnapMirror transfers, use System Manager to resume the transfers.

Upgrade from an image available at a URL

You can place the Cloud Volumes ONTAP software image on the Connector or on an HTTP server and then initiate the software upgrade from BlueXP. You might use this option if BlueXP can't access the S3 bucket to upgrade the software.

Before you begin

- BlueXP operations such as volume or aggregate creation must not be in progress on the Cloud Volumes ONTAP system.
- If you use HTTPS to host ONTAP images, the upgrade can fail due to SSL authentication issues, which are caused by missing certificates. The workaround is to generate and install a CA-signed certificate to be used for authentication between ONTAP and BlueXP.

  Go to the NetApp Knowledge Base to view step-by-step instructions:

  NetApp KB: How to configure BlueXP as an HTTPS server to host upgrade images

Steps

1. Optional: Set up an HTTP server that can host the Cloud Volumes ONTAP software image.

   If you have a VPN connection to the virtual network, you can place the Cloud Volumes ONTAP software image on an HTTP server in your own network. Otherwise, you must place the file on an HTTP server in the cloud.

2. If you use your own security group for Cloud Volumes ONTAP, ensure that the outbound rules allow HTTP connections so Cloud Volumes ONTAP can access the software image.

   The predefined Cloud Volumes ONTAP security group allows outbound HTTP connections by default.

3. Obtain the software image from the NetApp Support Site.
4. Copy the software image to a directory on the Connector or on an HTTP server from which the file will be served.

   Two paths are available. The correct path depends on your Connector version.

   - `/opt/application/netapp/cloudmanager/docker_occm/data/ontap/images/`
   - `/opt/application/netapp/cloudmanager/ontap/images/`

5. From the working environment in BlueXP, click the **… (ellipse icon)**, and then click **Update Cloud Volumes ONTAP**.
6. On the Update Cloud Volumes ONTAP version page, enter the URL, and then click **Change Image**.

   If you copied the software image to the Connector in the path shown above, you would enter the following URL:

   http://<Connector-private-IP-address>/ontap/images/<image-file-name>

   In the URL, **image-file-name** must follow the format "cot.image.9.13.1P2.tgz".

7. Click **Proceed** to confirm.

Result

BlueXP starts the software update. You can perform actions on the working environment once the software update is complete.

After you finish

If you suspended SnapMirror transfers, use System Manager to resume the transfers.

Registering pay-as-you-go systems

Support from NetApp is included with Cloud Volumes ONTAP PAYGO systems, but you must first activate support by registering the systems with NetApp.

Registering a PAYGO system with NetApp is required to upgrade ONTAP software using any of the methods described on this page.

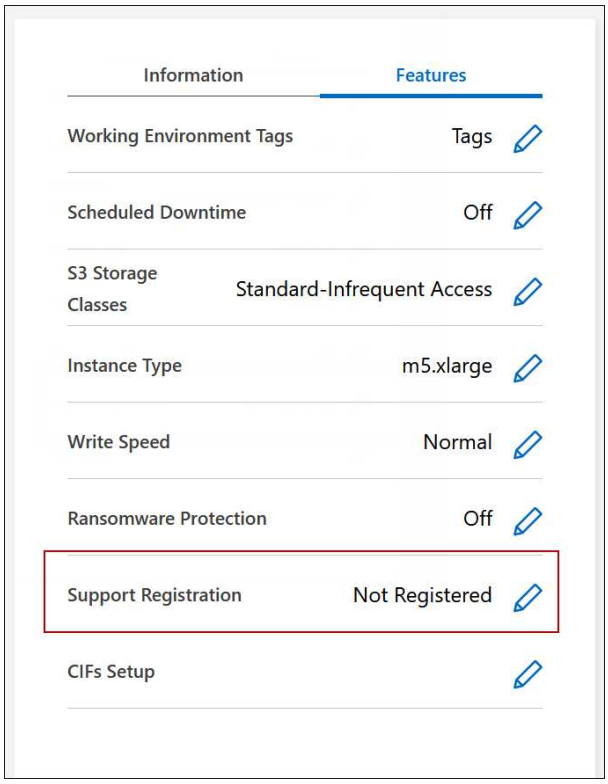
A system that isn't registered for support will still receive the software update notifications that appear in BlueXP when a new version is available. But you will need to register the system before you can upgrade the software.

Steps

1. If you have not yet added your NetApp Support Site account to BlueXP, go to **Account Settings** and add it now.

Learn how to add NetApp Support Site accounts.

2. On the Canvas page, double-click the name of the system you want to register..
3. On the Overview tab, click the Features panel and then click the pencil icon next to **Support Registration**.



4. Select a NetApp Support Site account and click **Register**.

Result

BlueXP registers the system with NetApp.

Managing the state of Cloud Volumes ONTAP

You can stop and start Cloud Volumes ONTAP from BlueXP to manage your cloud compute costs.

Scheduling automatic shutdowns of Cloud Volumes ONTAP

You might want to shut down Cloud Volumes ONTAP during specific time intervals to lower your compute costs. Rather than do this manually, you can configure BlueXP to automatically shut down and then restart systems at specific times.

About this task

- When you schedule an automatic shutdown of your Cloud Volumes ONTAP system, BlueXP postpones the shutdown if an active data transfer is in progress.

  BlueXP shuts down the system after the transfer is complete.

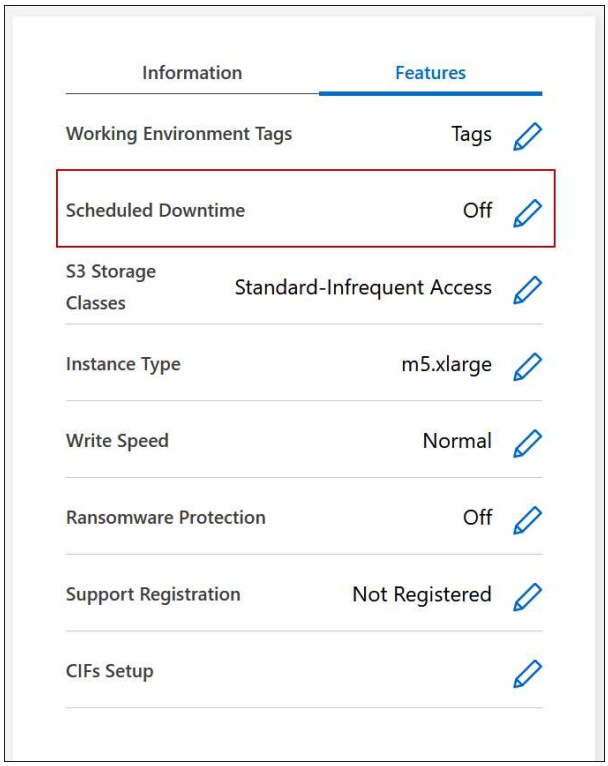
- This task schedules automatic shutdowns of both nodes in an HA pair.
- Snapshots of boot and root disks are not created when turning off Cloud Volumes ONTAP through

scheduled shutdowns.

Snapshots are automatically created only when performing a manual shutdown, as described in the next section.

Steps

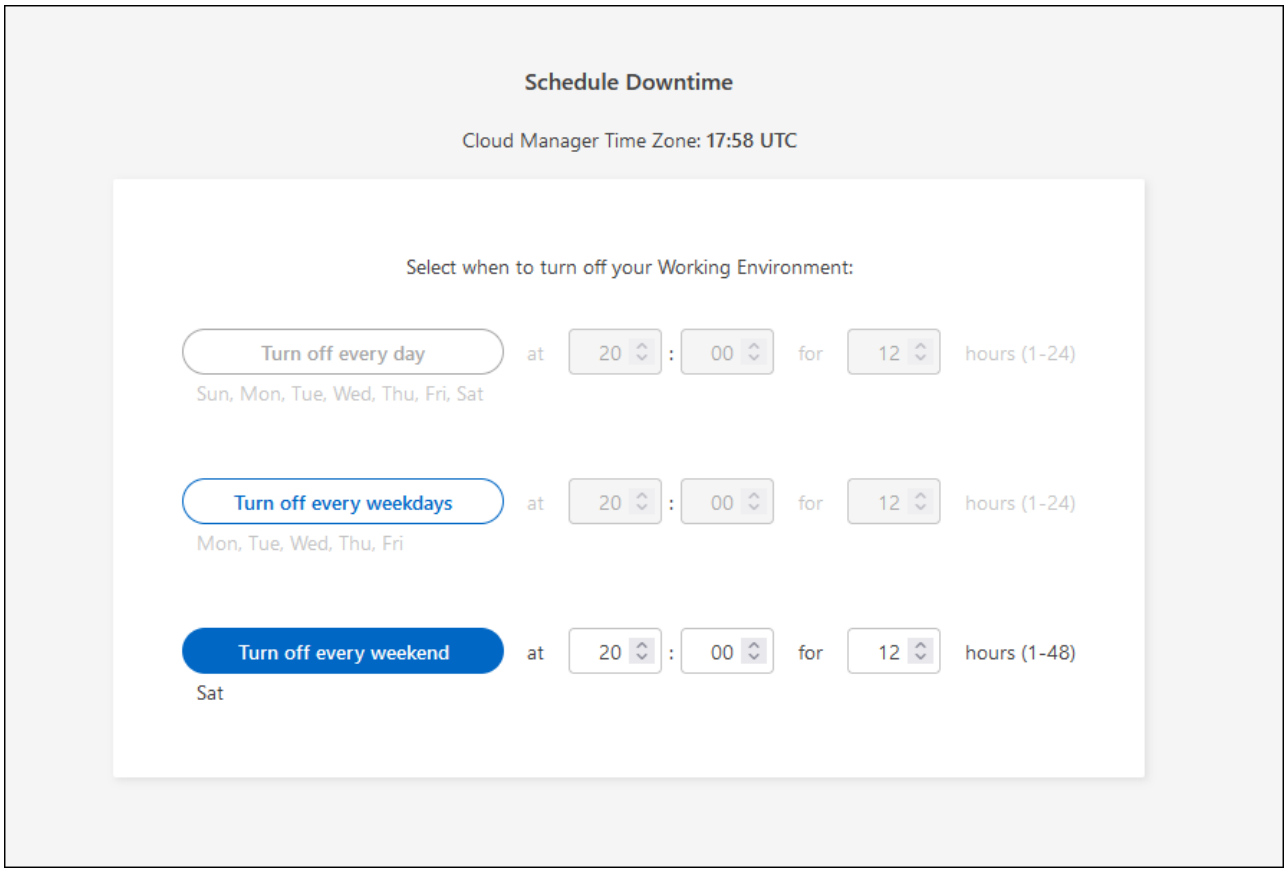
1. On the Canvas page, double-click the desired working environment.
2. On the Overview tab, click the Features panel and then click the pencil icon next to **Scheduled Downtime**.



3. Specify the shutdown schedule:
   a. Choose whether you want to shut down the system every day, every weekday, every weekend, or any combination of the three options.
   b. Specify when you want to turn off the system and for how long you want it turned off.

      Example

      The following image shows a schedule that instructs BlueXP to shut down the system every Saturday at 20:00 P.M. (8:00 PM) for 12 hours. BlueXP restarts the system every Monday at 12:00 a.m.



4. Click **Save**.

Result

BlueXP saves the schedule. The corresponding Scheduled Downtime line item under the Features panel displays 'On'.

Stopping Cloud Volumes ONTAP

Stopping Cloud Volumes ONTAP saves you from accruing compute costs and creates snapshots of the root and boot disks, which can be helpful for troubleshooting.

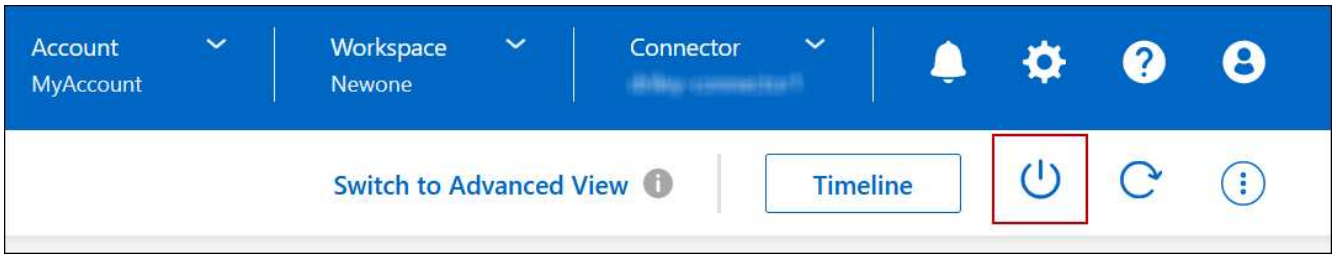
> To reduce costs, BlueXP periodically deletes older snapshots of root and boot disks. Only the two most recent snapshots are retained for both the root and boot disks.

About this task

When you stop an HA pair, BlueXP shuts down both nodes.

Steps

1. From the working environment, click the **Turn off** icon.



2. Keep the option to create snapshots enabled because the snapshots can enable system recovery.
3. Click **Turn Off**.

   It can take up to a few minutes to stop the system. You can restart systems at a later time from the working environment page.

   Snapshots are created automatically upon reboot.

Synchronize the system time using NTP

Specifying an NTP server synchronizes the time between the systems in your network, which can help prevent issues due to time differences.

Specify an NTP server using the BlueXP API or from the user interface when you create a CIFS server.

Modify system write speed

BlueXP enables you to choose a normal or high write speed for Cloud Volumes ONTAP. The default write speed is normal. You can change to high write speed if fast write performance is required for your workload.

High write speed is supported with all types of single node systems and some HA pair configurations. View supported configurations in the Cloud Volumes ONTAP Release Notes

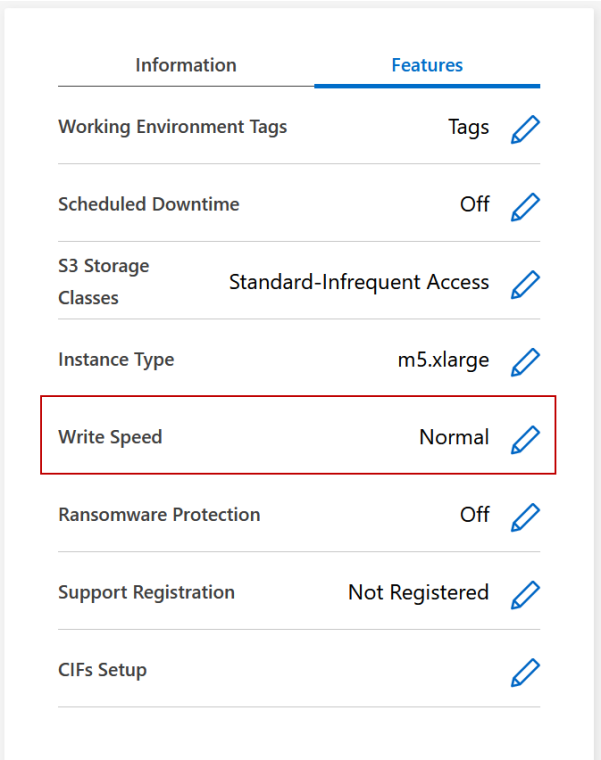
Before you change the write speed, you should understand the differences between the normal and high settings.

About this task

- Ensure that operations such as volume or aggregate creation are not in progress.
- Be aware that this change restarts the Cloud Volumes ONTAP system. This is disruptive process that requires downtime for the entire system.

Steps

1. On the Canvas page, double-click the name of the system you configure to the write speed.
2. On the Overview tab, click the Features panel and then click the pencil icon next to **Write Speed**.



3. Select **Normal** or **High**.

   If you choose High, then you'll need to read the "I understand…" statement and confirm by checking the box.

   > The **High** write speed option is supported with Cloud Volumes ONTAP HA pairs in Google Cloud starting with version 9.13.0.

4. Click **Save**, review the confirmation message, and then click **Approve**.

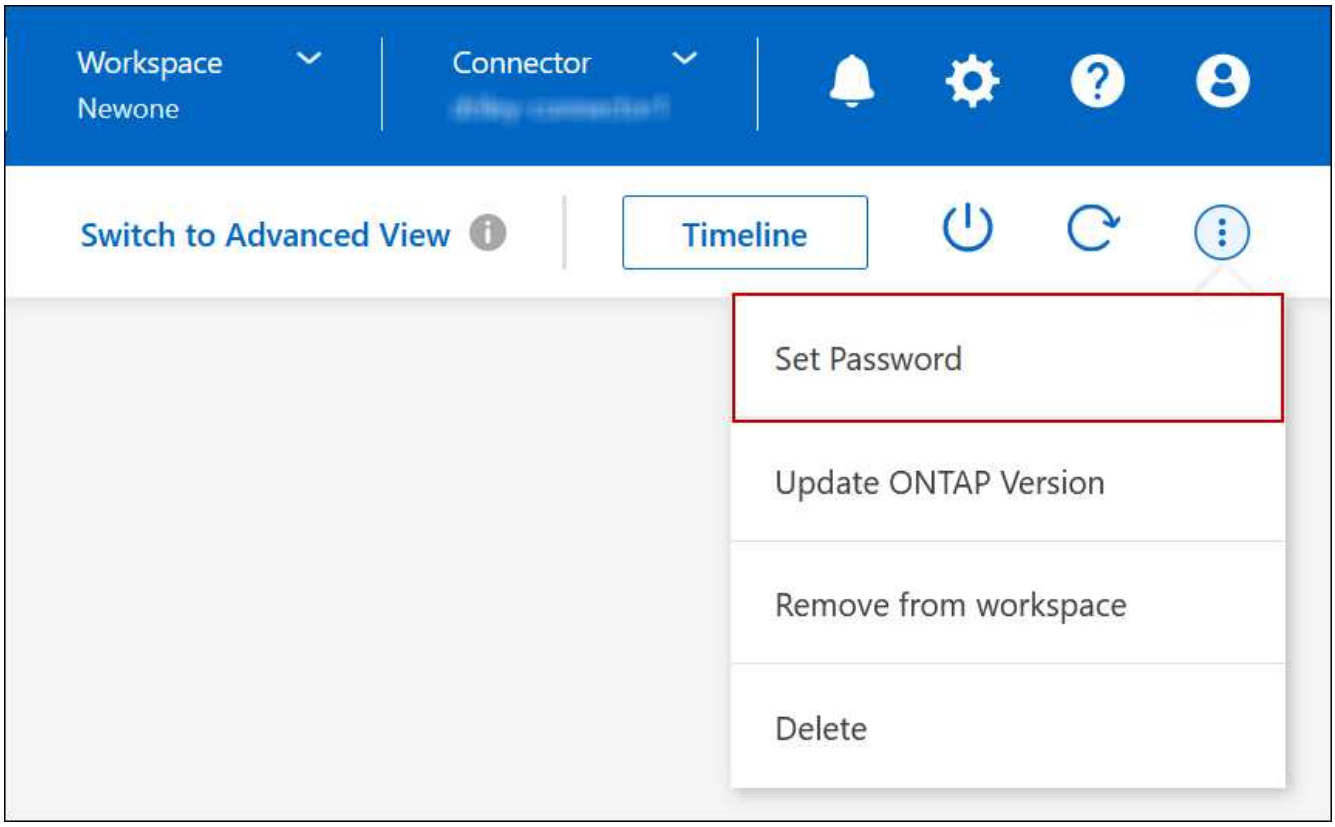
Change the password for Cloud Volumes ONTAP

Cloud Volumes ONTAP includes a cluster admin account. You can change the password for this account from BlueXP, if needed.

> You should not change the password for the admin account through System Manager or the CLI. The password will not be reflected in BlueXP. As a result, BlueXP cannot monitor the instance properly.

Steps

1. On the Canvas page, double-click the name of the Cloud Volumes ONTAP working environment.
2. On the upper right of the BlueXP console, click the ellipse icon, and select **Set password**.



The new password must be different than one of the last six passwords that you used.

Add, remove, or delete systems

Adding existing Cloud Volumes ONTAP systems to BlueXP

You can discover and add existing Cloud Volumes ONTAP systems to BlueXP. You might do this if you deployed a new BlueXP system.

Before you begin

You must know the password for the Cloud Volumes ONTAP admin user account.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, click **Add Working Environment**.
3. Select the cloud provider in which the system resides.
4. Choose the type of Cloud Volumes ONTAP system.
5. Click the link to discover an existing system.
6. On the Region page, choose the region where the instances are running, and then select the instances.
7. On the Credentials page, enter the password for the Cloud Volumes ONTAP admin user, and then click **Go**.

Result

BlueXP adds the Cloud Volumes ONTAP instances to the workspace.

Removing Cloud Volumes ONTAP working environments

The Account Admin can remove a Cloud Volumes ONTAP working environment to move it to another system or to troubleshoot discovery issues.

About this task

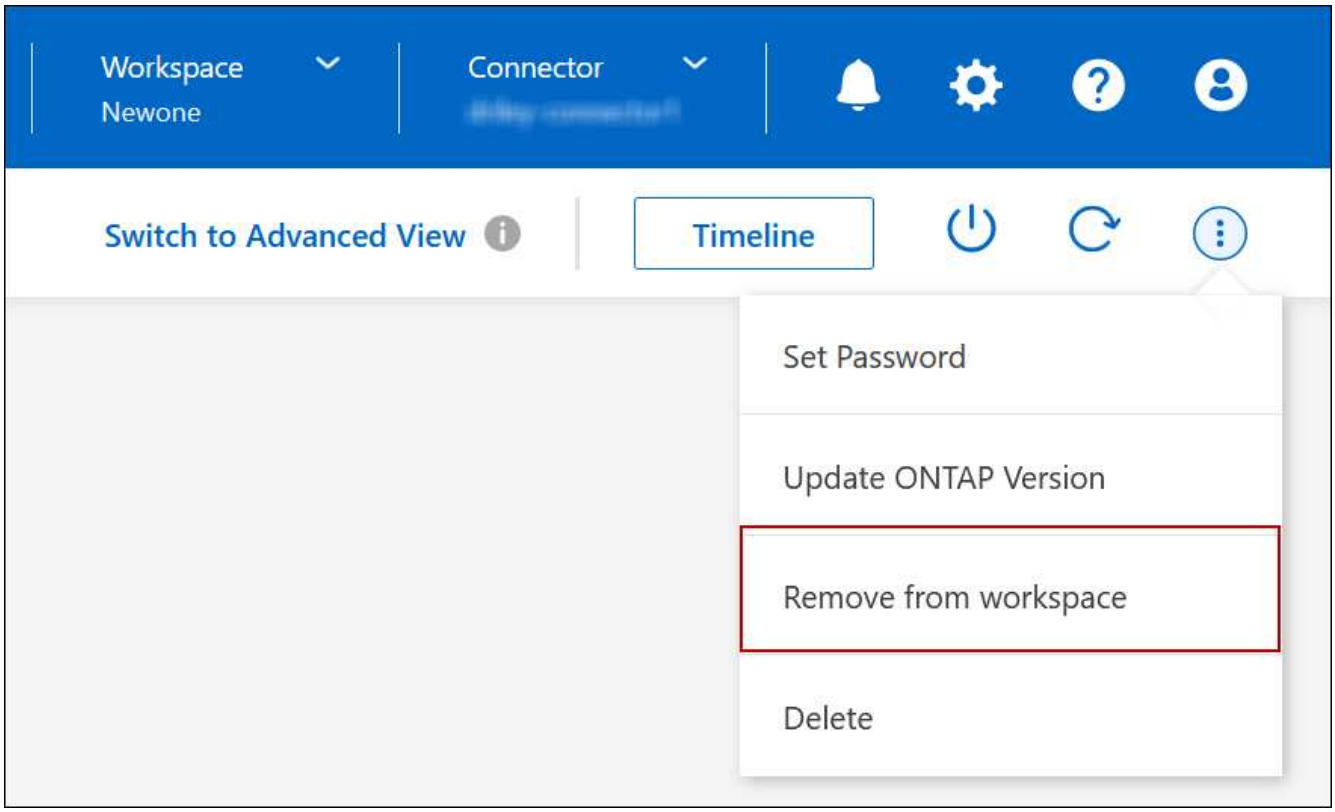
Removing a Cloud Volumes ONTAP working environment removes it from BlueXP. It does not delete the Cloud Volumes ONTAP system. You can later rediscover the working environment.

Removing a working environment from BlueXP enables you to do the following:

- Rediscover it in another workspace
- Rediscover it from another BlueXP system
- Rediscover it if you had problems during the initial discovery

Steps

1. On the Canvas page, double-click on the working environment you want to remove.
2. On the upper right of the BlueXP console, click the ellipse icon, and select **Remove from workspace**.



3. In the Review from Workspace window, click **Remove**.

Result

BlueXP removes the working environment. Users can rediscover this working environment from the Canvas page at any time.

Deleting a Cloud Volumes ONTAP system

You should always delete Cloud Volumes ONTAP systems from BlueXP, rather than from

your cloud provider's console. For example, if you terminate a licensed Cloud Volumes ONTAP instance from your cloud provider, then you can't use the license key for another instance. You must delete the working environment from BlueXP to release the license.

When you delete a working environment, BlueXP terminates Cloud Volumes ONTAP instances and deletes disks and snapshots.

Resources managed by other services like backups for BlueXP backup and recovery and instances for BlueXP classification are not deleted when you delete a working environment. You'll need to manually delete them yourself. If you don't, then you'll continue to receive charges for these resources.

> When BlueXP deploys Cloud Volumes ONTAP in your cloud provider, it enables termination protection on the instances. This option helps prevent accidental termination.

Steps

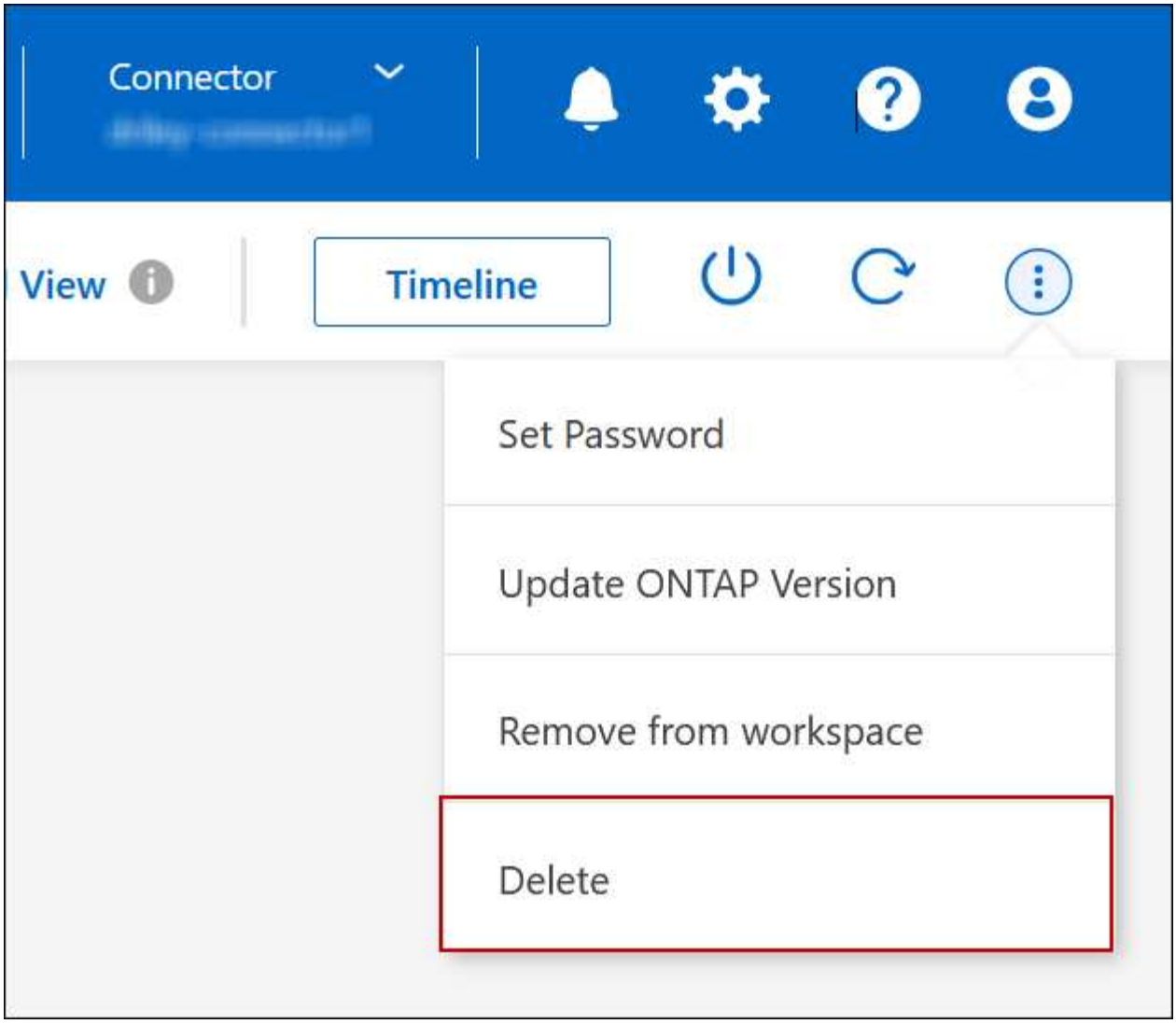
1. If you enabled BlueXP backup and recovery on the working environment, determine whether the backed up data is still required and then delete the backups, if necessary.

   BlueXP backup and recovery is independent from Cloud Volumes ONTAP by design. BlueXP backup and recovery doesn't automatically delete backups when you delete a Cloud Volumes ONTAP system, and there is no current support in the UI to delete the backups after the system has been deleted.

2. If you enabled BlueXP classification on this working environment and no other working environments use this service, then you'll need to delete the instance for the service.

   Learn more about the BlueXP classification instance.

3. Delete the Cloud Volumes ONTAP working environment.
   a. On the Canvas page, double-click the name of the Cloud Volumes ONTAP working environment that you want to delete.
   b. On the upper right of the BlueXP console, click the ellipse icon, and select **Delete**.



c. Under the Delete Working Environment window, type the name of the working environment and then click **Delete**.

   It can take up to 5 minutes to delete the working environment.

Azure administration

Change the Azure VM type for Cloud Volumes ONTAP

You can choose from several VM types when you launch Cloud Volumes ONTAP in Microsoft Azure. You can change the VM type at any time if you determine that it is undersized or oversized for your needs.

About this task

- Automatic giveback must be enabled on a Cloud Volumes ONTAP HA pair (this is the default setting). If it isn't, then the operation will fail.

  ONTAP 9 Documentation: Commands for configuring automatic giveback

- Changing the VM type can affect Microsoft Azure service charges.

- The operation restarts Cloud Volumes ONTAP.

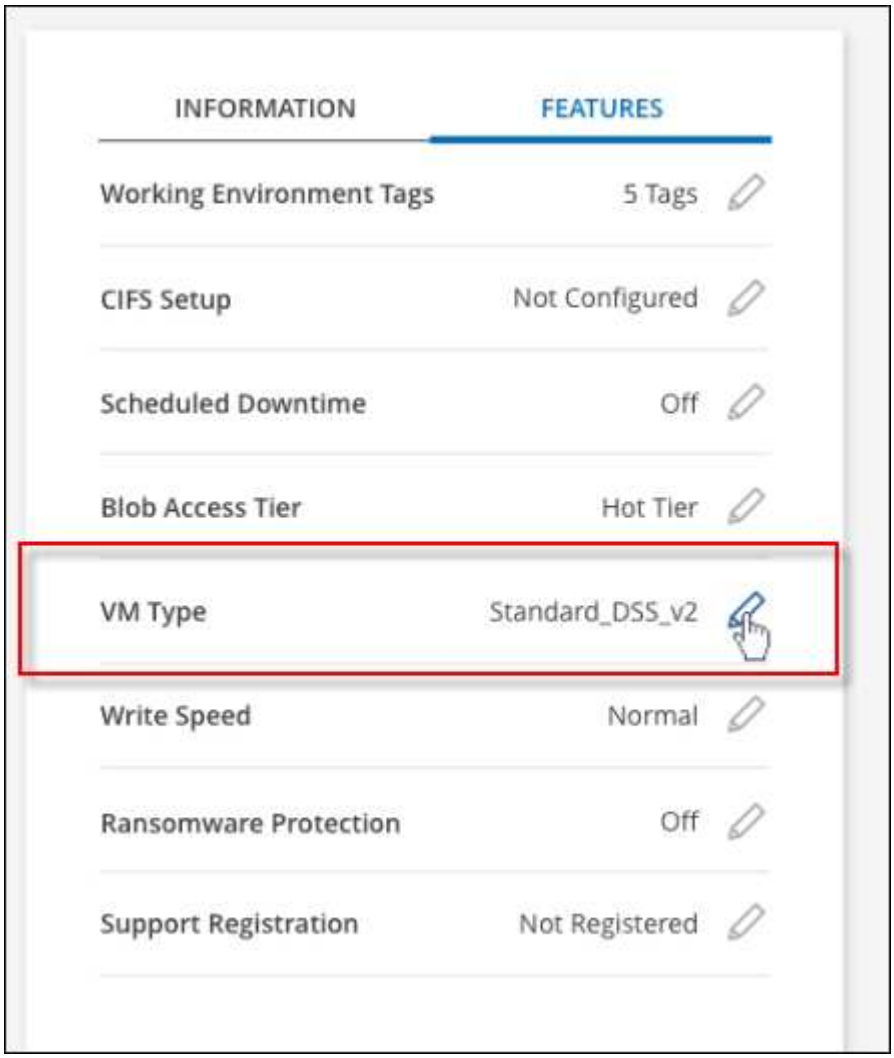
  For single node systems, I/O is interrupted.

  For HA pairs, the change is nondisruptive. HA pairs continue to serve data.

> BlueXP gracefully changes one node at a time by initiating takeover and waiting for give back. NetApp's QA team tested both writing and reading files during this process and didn't see any issues on the client side. As connections changed, we did see retries on the I/O level, but the application layer overcame these short "re-wire" of NFS/CIFS connections.

Steps

1. On the Canvas page, select the working environment.
2. On the Overview tab, click the Features panel and then click the pencil icon next to **VM type**.



   a. If you are using a node-based PAYGO license, you can optionally choose a different license and VM type by clicking the pencil icon next to **License type**.

3. Select a VM type, select the check box to confirm that you understand the implications of the change, and then click **Change**.

Result

Cloud Volumes ONTAP reboots with the new configuration.

Overriding CIFS locks for Cloud Volumes ONTAP HA pairs in Azure

The Account Admin can enable a setting in BlueXP that prevents issues with Cloud Volumes ONTAP storage giveback during Azure maintenance events. When you enable this setting, Cloud Volumes ONTAP vetoes CIFS locks and resets active CIFS sessions.

About this task

Microsoft Azure schedules periodic maintenance events on its virtual machines. When a maintenance event occurs on a Cloud Volumes ONTAP HA pair, the HA pair initiates storage takeover. If there are active CIFS sessions during this maintenance event, the locks on CIFS files can prevent storage giveback.

If you enable this setting, Cloud Volumes ONTAP will veto the locks and reset the active CIFS sessions. As a result, the HA pair can complete storage giveback during these maintenance events.

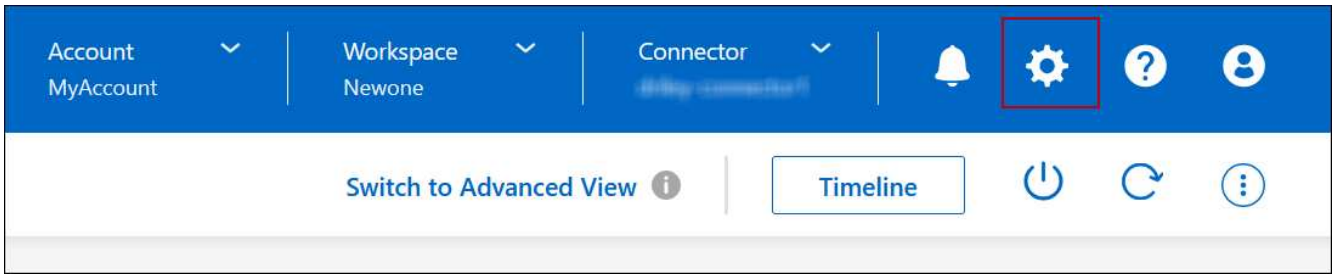
> This process might be disruptive to CIFS clients. Data that is not committed from CIFS clients could be lost.

What you'll need

You need to create a Connector before you can change BlueXP settings. Learn how.

Steps

1. In the upper right of the BlueXP console, click the Settings icon, and select **Cloud Volumes ONTAP Settings**.



2. Under **Azure**, click **Azure CIFS locks for Azure HA working environments**.
3. Click the checkbox to enable the feature and then click **Save**.

Use an Azure Private Link or service endpoints

Cloud Volumes ONTAP uses an Azure Private Link for connections to its associated storage accounts. If needed, you can disable Azure Private Links and use service endpoints instead.

Overview

By default, BlueXP enables an Azure Private Link for connections between Cloud Volumes ONTAP and its associated storage accounts. An Azure Private Link secures connections between endpoints in Azure and provides performance benefits.

If required, you can configure Cloud Volumes ONTAP to use service endpoints instead of an Azure Private Link.

With either configuration, BlueXP always limits network access for connections between Cloud Volumes ONTAP and storage accounts. Network access is limited to the VNet where Cloud Volumes ONTAP is deployed and the VNet where the Connector is deployed.

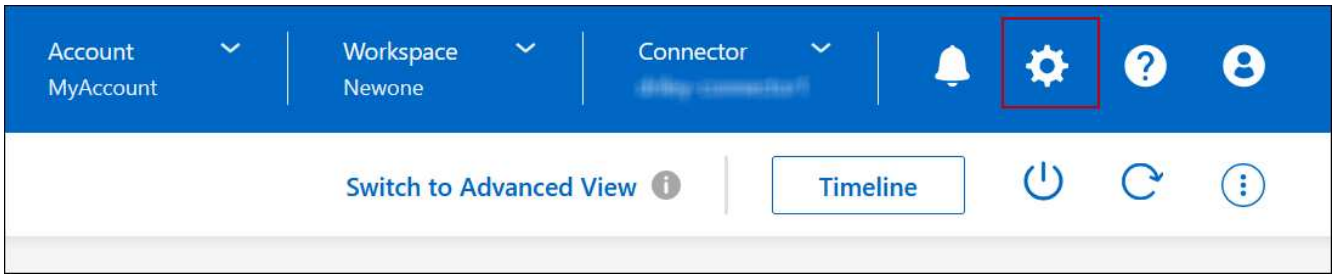
Disable Azure Private Links and use service endpoints instead

If required by your business, you can change a setting in BlueXP so that it configures Cloud Volumes ONTAP to use service endpoints instead of an Azure Private Link. Changing this setting applies to new Cloud Volumes ONTAP systems that you create. Service endpoints are only supported in Azure region pairs between the Connector and Cloud Volumes ONTAP VNets.

The Connector should be deployed in the same Azure region as the Cloud Volumes ONTAP systems that it manages, or in the Azure region pair for the Cloud Volumes ONTAP systems.

Steps

1. In the upper right of the BlueXP console, click the Settings icon, and select **Cloud Volumes ONTAP Settings**.



2. Under **Azure**, click **Use Azure Private Link**.
3. Deselect **Private Link connection between Cloud Volumes ONTAP and storage accounts**.
4. Click **Save**.

After you finish

If you disabled Azure Private Links and the Connector uses a proxy server, you must enable direct API traffic.

Learn how to enable direct API traffic on the Connector

Work with Azure Private Links

In most cases, there's nothing that you need to do to set up Azure Private links with Cloud Volumes ONTAP. BlueXP manages Azure Private Links for you. But if you use an existing Azure Private DNS zone, then you'll need to edit a configuration file.

Requirement for custom DNS

Optionally, if you work with custom DNS, you need to create a conditional forwarder to the Azure private DNS zone from your custom DNS servers. To learn more, refer to Azure's documentation on using a DNS forwarder.

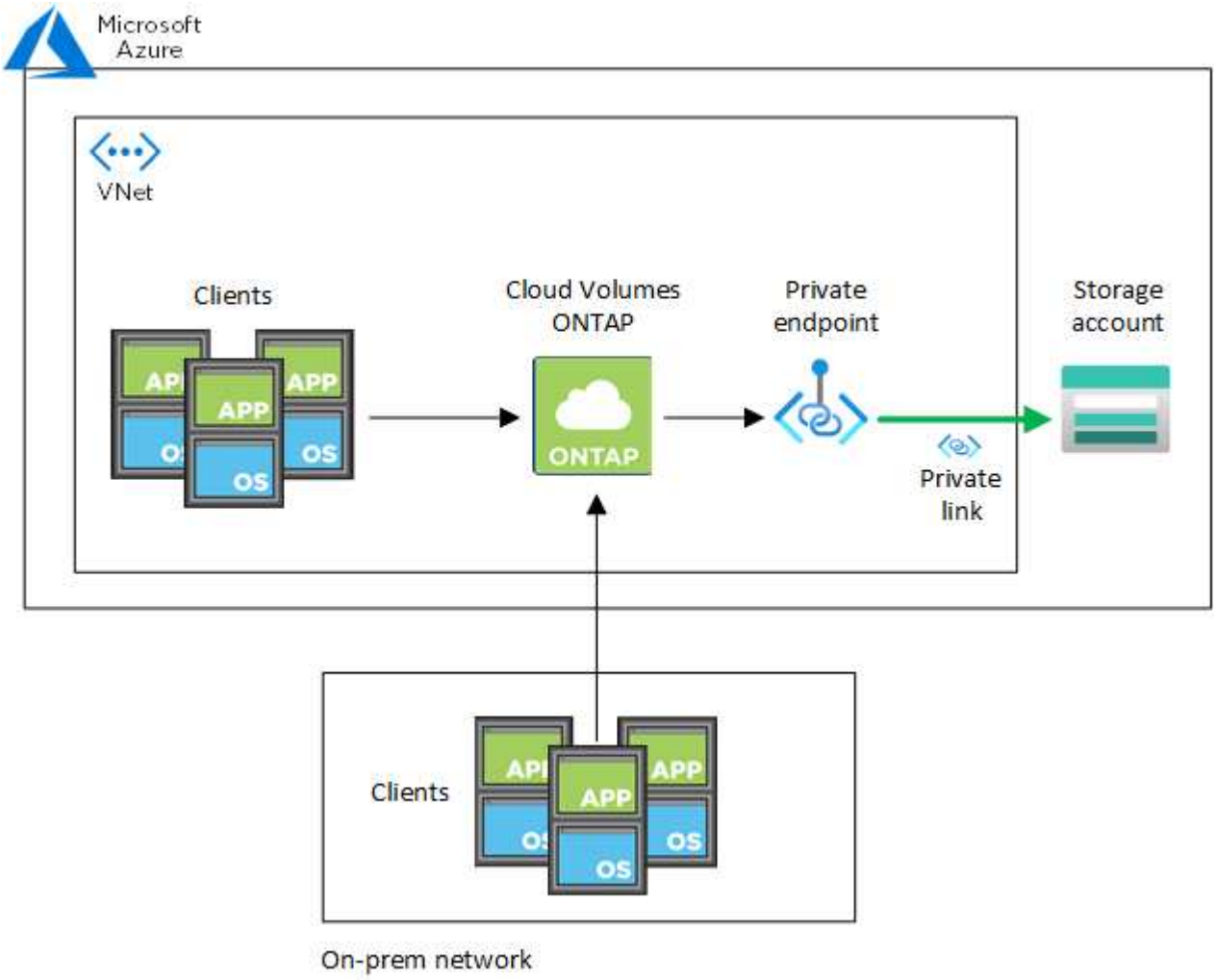
How Private Link connections work

When BlueXP deploys Cloud Volumes ONTAP in Azure, it creates a private endpoint in the resource group. The private endpoint is associated with storage accounts for Cloud Volumes ONTAP. As a result, access to Cloud Volumes ONTAP storage travels through the Microsoft backbone network.

Client access goes through the private link when clients are within the same VNet as Cloud Volumes ONTAP,

within peered VNets, or in your on-premises network when using a private VPN or ExpressRoute connection to the VNet.

Here's an example that shows client access over a private link from within the same VNet and from an on-prem network that has either a private VPN or ExpressRoute connection.



If the Connector and Cloud Volumes ONTAP systems are deployed in different VNets, then you must set up VNet peering between the VNet where the Connector is deployed and the VNet where the Cloud Volumes ONTAP systems are deployed.

Provide BlueXP with details about your Azure Private DNS

If you use Azure Private DNS, then you need to modify a configuration file on each Connector. Otherwise, BlueXP can't enable the Azure Private Link connection between Cloud Volumes ONTAP and its associated storage accounts.

Note that the DNS name must match Azure DNS naming requirements as shown in Azure documentation.

Steps

1. SSH to the Connector host and log in.
2. Navigate to the following directory: /opt/application/netapp/cloudmanager/docker_occm/data
3. Edit app.conf by adding the "user-private-dns-zone-settings" parameter with the following keyword-value

pairs:

```
"user-private-dns-zone-settings" : {
   "resource-group" : "<resource group name of the DNS zone>",
   "subscription" : "<subscription ID>",
   "use-existing" : true,
   "create-private-dns-zone-link" : true
}
```

The parameter should be entered at the same level as "system-id" like shown below:

```
"system-id" : "<system ID>",
"user-private-dns-zone-settings" : {
```

Note that the subscription keyword is required only if the Private DNS Zone exists in a different subscription than the Connector.

4. Save the file and log off the Connector.

   A reboot isn't required.

Enable rollback on failures

If BlueXP fails to create an Azure Private Link as part of specific actions, it completes the action without the Azure Private Link connection. This can happen when creating a new working environment (single node or HA pair), or when the following actions occur on an HA pair: creating a new aggregate, adding disks to an existing aggregate, or creating a new storage account when going above 32 TiB.

You can change this default behavior by enabling rollback if BlueXP fails to create the Azure Private Link. This can help to ensure that you're fully compliant with your company's security regulations.

If you enable rollback, BlueXP stops the action and rolls back all resources that were created as part of the action.

You can enable rollback through the API or by updating the app.conf file.

Enable rollback through the API

Step

1. Use the `PUT /occm/config` API call with the following request body:

```
{ "rollbackOnAzurePrivateLinkFailure": true }
```

Enable rollback by updating app.conf

Steps

1. SSH to the Connector host and log in.

2. Navigate to the following directory: /opt/application/netapp/cloudmanager/docker_occm/data
3. Edit app.conf by adding the following parameter and value:

```
"rollback-on-private-link-failure": true
```

4. Save the file and log off the Connector.

   A reboot isn't required.

Moving resource groups

Cloud Volumes ONTAP supports Azure resource groups moves but the workflow happens in the Azure console only.

You can move a working environment from one resource group to a different resource group in Azure within the same Azure subscription. Moving resource groups between different Azure subscriptions is not supported.

Steps

1. Remove the working environment from **Canvas**.

   To learn how to remove a working environment, see Removing Cloud Volumes ONTAP working environments.

2. Execute the resource group move in the Azure console.

   To complete the move, refer to Move resources to a new resource group or subscription in Microsoft Azure's documentation.

3. In **Canvas**, discover the working environment.
4. Look for the new resource group in the information for the working environment.

Result

The working environment and its resources (VMs, disks, storage accounts, network interfaces, snapshots) are in the new resource group.

Segregate SnapMirror traffic in Azure

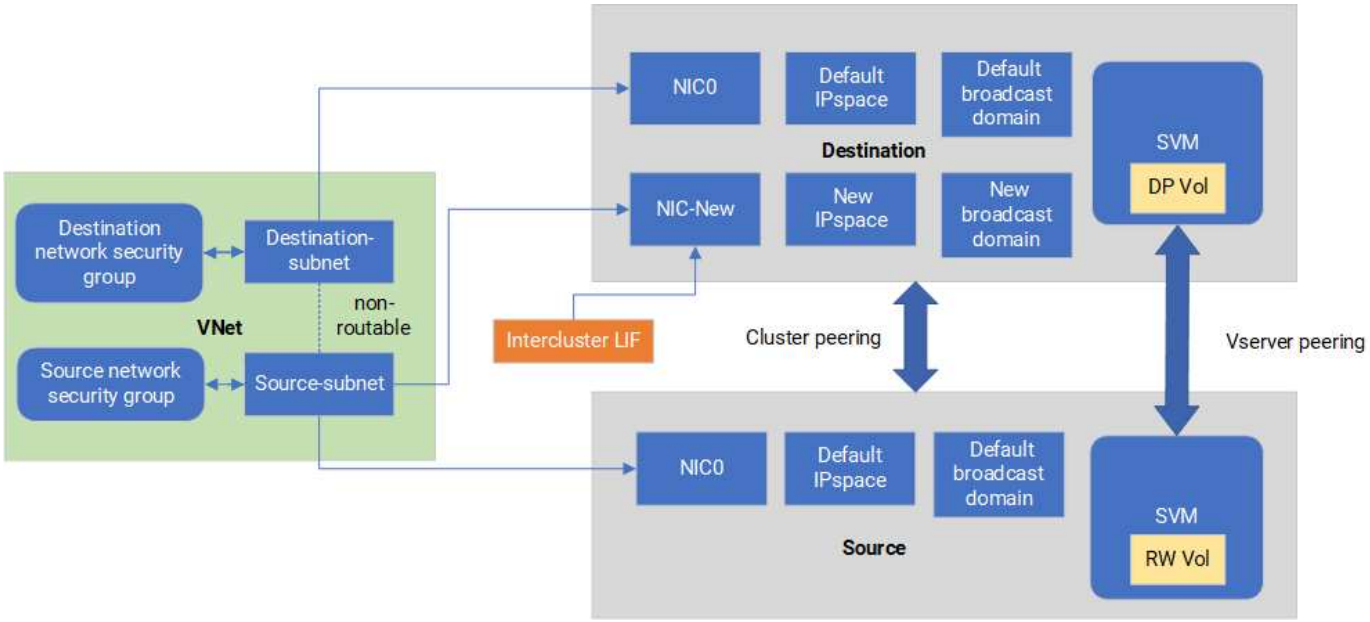
With Cloud Volumes ONTAP in Azure, you can segregate SnapMirror replication traffic from data and management traffic. To segregate SnapMirror replication traffic from your data traffic, you'll add a new network interface card (NIC), an associated intercluster LIF and a non-routable subnet.

About SnapMirror traffic segregation in Azure

By default, BlueXP configures all NICs and LIFs in a Cloud Volumes ONTAP deployment on the same subnet. In such configurations, SnapMirror replication traffic and data and management traffic use the same subnet. Segregating SnapMirror traffic leverages an additional subnet that isn't routable to the existing subnet used for data and management traffic.

Figure 1

The following diagrams show the segregation of SnapMirror replication traffic with an additional NIC, an associated intercluster LIF and a non-routable subnet in a single node deployment. An HA pair deployment differs slightly.



Before you begin

Review the following considerations:

- You can only add a single NIC to a Cloud Volumes ONTAP single node or HA-pair deployment (VM instance) for SnapMirror traffic segregation.
- To add a new NIC, the VM instance type you deploy must have an unused NIC.
- The source and destination clusters should have access to the same Virtual Network (VNet). The destination cluster is a Cloud Volumes ONTAP system in Azure. The source cluster can be a Cloud Volumes ONTAP system in Azure or an ONTAP system.

Step 1: Create an additional NIC and attach to the destination VM

This section provides instructions for how to create an additional NIC and attach it to the destination VM. The destination VM is the single node or HA-pair system in Cloud Volumes ONTAP in Azure where you want to set up your additional NIC.

Steps

1. In the ONTAP CLI, stop the node.

```
dest::> halt -node <dest_node-vm>
```

2. In the Azure portal, check that the VM (node) status is stopped.

```
az vm get-instance-view --resource-group <dest-rg> --name <dest-vm>
--query instanceView.statuses[1].displayStatus
```

3. Use the Bash environment in Azure Cloud Shell to stop the node.
   a. Stop the node.

```
az vm stop --resource-group <dest_node-rg> --name <dest_node-vm>
```

   b. Deallocate the node.

```
az vm deallocate --resource-group <dest_node-rg> --name <dest_node-
vm>
```

4. Configure network security group rules to make the two subnets (source cluster subnet and destination cluster subnet) non-routable to each other.
   a. Create the new NIC on the destination VM.
   b. Look up the subnet ID for the source cluster subnet.

```
az network vnet subnet show -g <src_vnet-rg> -n <src_subnet> --vnet
-name <vnet> --query id
```

   c. Create the new NIC on the destination VM with the subnet ID for the source cluster subnet. Here you enter the name for the new NIC.

```
az network nic create -g <dest_node-rg> -n <dest_node-vm-nic-new>
--subnet <id_from_prev_command> --accelerated-networking true
```

   d. Save the privateIPAddress. This IP address, <new_added_nic_primary_addr>, is used to create an intercluster LIF in broadcast domain, intercluster LIF for the new NIC.
5. Attach the new NIC to the VM.

```
az vm nic add -g <dest_node-rg> --vm-name <dest_node-vm> --nics
<dest_node-vm-nic-new>
```

6. Start the VM (node).

```
az vm start --resource-group <dest_node-rg>  --name <dest_node-vm>
```

7. In the Azure portal, go to **Networking** and confirm that the new NIC, e.g. nic-new, exists and accelerated networking is enabled.

```
az network nic list --resource-group azure-59806175-60147103-azure-rg
--query "[].{NIC: name, VM: virtualMachine.id}"
```

For HA-pair deployments, repeat the steps for the partner node.

Step 2: Create a new IPspace, broadcast domain, and intercluster LIF for the new NIC

A separate IPspace for intercluster LIFs provides logical separation between networking functionality for replication between clusters.

Use the ONTAP CLI for the following steps.

Steps

1. Create the new IPspace (new_ipspace).

```
dest::> network ipspace create -ipspace <new_ipspace>
```

2. Create a broadcast domain on the new IPspace (new_ipspace) and add the nic-new port.

```
dest::> network port show
```

3. For single node systems, the newly added port is *e0b*. For HA-pair deployments with managed disks, the newly added port is *e0d*. For HA-pair deployments with page blobs, the newly added port is *e0e*. Use the node name not the VM name. Find the node name by running `node show`.

```
dest::> broadcast-domain create -broadcast-domain <new_bd> -mtu 1500
-ipspace <new_ipspace> -ports <dest_node-cot-vm:e0b>
```

4. Create an intercluster LIF on the new broadcast-domain (new_bd) and on the new NIC (nic-new).

```
dest::> net int create -vserver <new_ipspace> -lif <new_dest_node-ic-
lif> -service-policy default-intercluster -address
<new_added_nic_primary_addr> -home-port <e0b> -home-node <node> -netmask
<new_netmask_ip> -broadcast-domain <new_bd>
```

5. Verify creation of the new intercluster LIF.

```
dest::> net int show
```

For HA-pair deployments, repeat the steps for the partner node.

Step 3: Verify cluster peering between the source and destination systems

This section provides instructions for how to verify peering between the source and destination systems.

Use the ONTAP CLI for the following steps.

Steps

1. Verify that the intercluster LIF of the destination cluster can ping the intercluster LIF of the source cluster. Because the destination cluster executes this command, the destination IP address is the intercluster LIF IP address on the source.

```
dest::> ping -lif <new_dest_node-ic-lif> -vserver <new_ipspace>
-destination <10.161.189.6>
```

2. Verify that the intercluster LIF of the source cluster can ping the intercluster LIF of the destination cluster. The destination is the IP address of the new NIC created on the destination.

```
src::> ping -lif <src_node-ic-lif> -vserver <src_svm> -destination
<10.161.189.18>
```

For HA-pair deployments, repeat the steps for the partner node.

Step 4: Create SVM peering between the source and destination system

This section provides instructions for how to create SVM peering between the source and destination system.

Use the ONTAP CLI for the following steps.

Steps

1. Create cluster peering on the destination using the source intercluster LIF IP address as the `-peer -addrs`. For HA pairs, list the source intercluster LIF IP address for both nodes as the `-peer-addrs`.

```
dest::> cluster peer create -peer-addrs <10.161.189.6> -ipspace
<new_ipspace>
```

2. Enter and confirm the passphrase.
3. Create cluster peering on the source using the destination cluster LIF IP address as the `peer-addrs`. For HA pairs, list the destination intercluster LIF IP address for both nodes as the `-peer-addrs`.

```
src::> cluster peer create -peer-addrs <10.161.189.18>
```

4. Enter and confirm the passphrase.
5. Check that the cluster peered.

```
src::> cluster peer show
```

Successful peering shows **Available** in the availability field.

6. Create SVM peering on the destination. Both source and destination SVMs should be data SVMs.

```
dest::> vserver peer create -vserver <dest_svm> -peer-vserver <src_svm>
-peer-cluster <src_cluster> -applications snapmirror``
```

7. Accept SVM peering.

```
src::> vserver peer accept -vserver <src_svm> -peer-vserver <dest_svm>
```

8. Check that the SVM peered.

```
dest::> vserver peer show
```

Peer state shows **`peered`** and peering applications shows **`snapmirror`**.

Step 5: Create a SnapMirror replication relationship between the source and destination system

This section provides instructions for how to create a SnapMirror replication relationship between the source and destination system.

To move an existing SnapMirror replication relationship, you must first break the existing SnapMirror replication relationship before you create a new SnapMirror replication relationship.

Use the ONTAP CLI for the following steps.

Steps

1. Create a data protected volume on the destination SVM.

```
dest::> vol create -volume <new_dest_vol> -vserver <dest_svm> -type DP
-size <10GB> -aggregate <aggr1>
```

2. Create the SnapMirror replication relationship on the destination which includes the SnapMirror policy and schedule for the replication.

```
dest::> snapmirror create -source-path src_svm:src_vol  -destination
-path  dest_svm:new_dest_vol -vserver dest_svm -policy
MirrorAllSnapshots -schedule 5min
```

3. Initialize the SnapMirror replication relationship on the destination.

```
dest::> snapmirror initialize -destination-path  <dest_svm:new_dest_vol>
```

4. In the ONTAP CLI, validate the SnapMirror relationship status by running the following command:

```
dest::> snapmirror show
```

   The relationship status is `Snapmirrored` and the health of the relationship is `true`.

5. Optional: In the ONTAP CLI, run the following command to view the actions history for the SnapMirror relationship.

```
dest::> snapmirror show-history
```

Optionally, you can mount the source and destination volumes, write a file to the source, and verify the volume is replicating to the destination.

Administer Cloud Volumes ONTAP using the Advanced View

If you need to perform advanced management of Cloud Volumes ONTAP, you can do so using ONTAP System Manager, which is a management interface that's provided with an ONTAP system. We have included the System Manager interface directly inside BlueXP so that you don't need to leave BlueXP for advanced management.

Features

The Advanced View in BlueXP gives you access to additional management features:

- Advanced storage management

  Manage consistency groups, shares, qtrees, quotas, and Storage VMs.

- Networking management

  Manage IPspaces, network interfaces, portsets, and ethernet ports.

- Events and jobs

  View event logs, system alerts, jobs, and audit logs.

- Advanced data protection

  Protect storage VMs, LUNs, and consistency groups.

- Host management

  Set up SAN initiator groups and NFS clients.

Supported configurations

Advanced management through System Manager is supported with Cloud Volumes ONTAP 9.10.0 and later in standard cloud regions.

System Manager integration is not supported in GovCloud regions or in regions that have no outbound internet access.

Limitations

A few features that appear in the System Manager interface are not supported with Cloud Volumes ONTAP:

- BlueXP tiering

  The BlueXP tiering service is not supported with Cloud Volumes ONTAP. Tiering data to object storage must be set up directly from BlueXP's Standard View when creating volumes.

- Tiers

  Aggregate management (including local tiers and cloud tiers) is not supported from System Manager. You must manage aggregates directly from BlueXP's Standard View.

- Firmware upgrades

  Automatic firmware updates from the **Cluster > Settings** page is not supported with Cloud Volumes ONTAP.

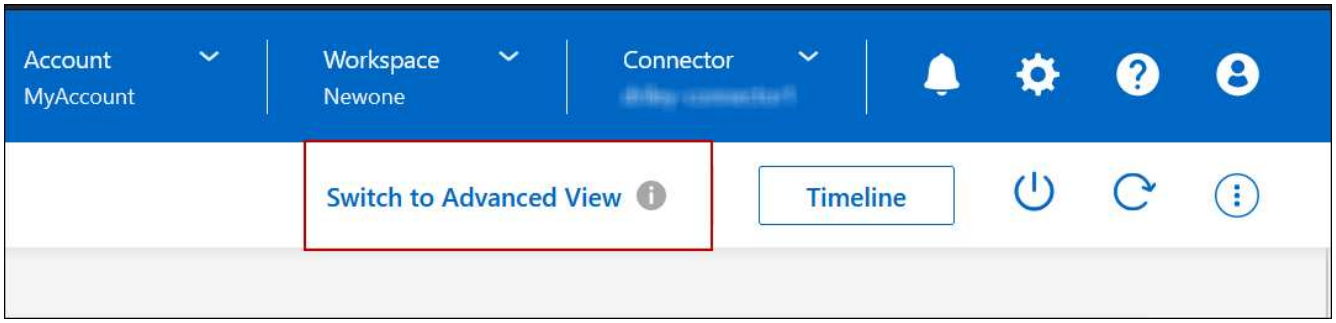
In addition, role-based access control from System Manager is not supported.

How to get started

Open a Cloud Volumes ONTAP working environment and click the Advanced View option.

Steps

1. From the left navigation menu, select **Storage > Canvas**.
2. On the Canvas page, double-click the name of a Cloud Volumes ONTAP system.
3. In the top-right, click **Switch to Advanced View**.



4. If the confirmation message appears, read through it and click **Close**.
5. Use System Manager to manage Cloud Volumes ONTAP.
6. If needed, click **Switch to Standard View** to return to standard management through BlueXP.

Help with using System Manager

If you need help using System Manager with Cloud Volumes ONTAP, you can refer to ONTAP documentation for step-by-step instructions. Here are a few links that might help:

- Volume and LUN management
- Network management
- Data protection

Administer Cloud Volumes ONTAP from the CLI

The Cloud Volumes ONTAP CLI enables you to run all administrative commands and is a good choice for advanced tasks or if you are more comfortable using the CLI. You can connect to the CLI using Secure Shell (SSH).

Before you begin

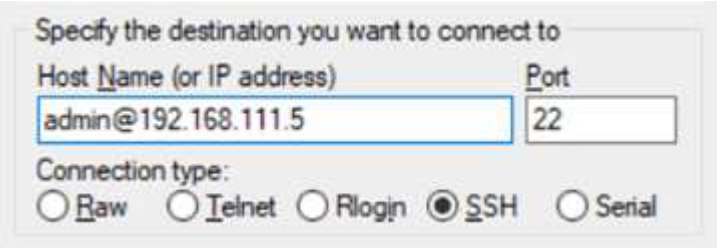
The host from which you use SSH to connect to Cloud Volumes ONTAP must have a network connection to Cloud Volumes ONTAP. For example, you might need to SSH from a jump host that's in your cloud provider network.

Steps

1. In BlueXP, identify the IP address of the cluster management interface:
   a. From the left navigation menu, select **Storage > Canvas**.
   b. On the Canvas page, select the Cloud Volumes ONTAP system.
   c. Copy the cluster management IP address that appears in the right pane.
2. Use SSH to connect to the cluster management interface IP address using the admin account.

   Example

   The following image shows an example using PuTTY:

3. At the login prompt, enter the password for the admin account.

   Example

   ```
   Password: ********
   COT2::>
   ```

System health and events

Verify AutoSupport setup

AutoSupport proactively monitors the health of your system and sends messages to NetApp technical support. By default, AutoSupport is enabled on each node to send messages to technical support using the HTTPS transport protocol. It's best to verify that AutoSupport can send these messages.

The only required configuration step is to ensure that Cloud Volumes ONTAP has outbound internet connectivity. For details, refer to the networking requirements for your cloud provider.

AutoSupport requirements

Cloud Volumes ONTAP nodes require outbound internet access for NetApp AutoSupport, which proactively monitors the health of your system and sends messages to NetApp technical support.

Routing and firewall policies must allow HTTP/HTTPS traffic to the following endpoints so Cloud Volumes ONTAP can send AutoSupport messages:

- https://support.netapp.com/aods/asupmessage
- https://support.netapp.com/asupprod/post/1.0/postAsup

If an outbound internet connection isn't available to send AutoSupport messages, BlueXP automatically configures your Cloud Volumes ONTAP systems to use the Connector as a proxy server. The only requirement is to ensure that the Connector's security group allows *inbound* connections over port 3128. You'll need to open this port after you deploy the Connector.

If you defined strict outbound rules for Cloud Volumes ONTAP, then you'll also need to ensure that the Cloud Volumes ONTAP security group allows *outbound* connections over port 3128.

After you've verified that outbound internet access is available, you can test AutoSupport to ensure that it can send messages. For instructions, refer to ONTAP docs: Set up AutoSupport.

Troubleshoot your AutoSupport configuration

If an outbound connection isn't available and BlueXP can't configure your Cloud Volumes ONTAP system to use the Connector as a proxy server, you'll receive a notification from BlueXP titled "<working environment name> is unable to send AutoSupport messages."

You're most likely receiving this message because of networking issues.

Follow these steps to address this problem.

Steps

1. SSH to the Cloud Volumes ONTAP system so that you can administer the system from the CLI.

   Learn how to SSH to Cloud Volumes ONTAP.

2. Display the detailed status of the AutoSupport subsystem:

   ```
   autosupport check show-details
   ```

The response should be similar to the following:

```
    Category: smtp
          Component: mail-server
             Status: failed
             Detail: SMTP connectivity check failed for destination:
                     mailhost. Error: Could not resolve host -
'mailhost'
            Corrective Action: Check the hostname of the SMTP server

    Category: http-https
        Component: http-put-destination
            Status: ok
            Detail: Successfully connected to:
                    <https://support.netapp.com/put/AsupPut/>.

          Component: http-post-destination
             Status: ok
             Detail: Successfully connected to:

https://support.netapp.com/asupprod/post/1.0/postAsup.

    Category: on-demand
          Component: ondemand-server
             Status: ok
             Detail: Successfully connected to:
                     https://support.netapp.com/aods/asupmessage.

    Category: configuration
            Component: configuration
                Status: ok
                Detail: No configuration issues found.
    5 entries were displayed.
```

If the status of the http-https category is "ok" then it means AutoSupport is configured properly and messages can be sent.

3. If the status is not ok, verify the proxy URL for each Cloud Volumes ONTAP node:

   ```
   autosupport show -fields proxy-url
   ```

4. If the proxy URL parameter is empty, configure Cloud Volumes ONTAP to use the Connector as a proxy:

   ```
   autosupport modify -proxy-url http://<connector private ip>:3128
   ```

5. Verify AutoSupport status again:

   ```
   autosupport check show-details
   ```

6. If the status is still is failed, validate that there is connectivity between Cloud Volumes ONTAP and the Connector over port 3128.
7. If the status ID is still failed after verifying that there is connectivity, SSH to the Connector.

   Learn more about Connecting to the Linux VM for the Connector

8. Go to `/opt/application/netapp/cloudmanager/docker_occm/data/`
9. Open the proxy configuration file `squid.conf`

   The basic structure of the file is as follows:

```
http_port 3128
acl localnet src 172.31.0.0/16
acl azure_aws_metadata dst 169.254.169.254

http_access allow localnet
http_access deny azure_aws_metadata
http_access allow localhost
http_access deny all
```

   The localnet src value is the CIDR of the Cloud Volumes ONTAP system.

10. If the CIDR block of the Cloud Volumes ONTAP system isn't in the range that's specified in the file, either update the value or add a new entry as follows:

```
acl cvonet src <cidr>
```

   If you add this new entry, don't forget to also add an allow entry:

```
http_access allow cvonet
```

   Here's an example:

```
http_port 3128
acl localnet src 172.31.0.0/16
acl cvonet src 172.33.0.0/16
acl azure_aws_metadata dst 169.254.169.254

http_access allow localnet
http_access allow cvonet
http_access deny azure_aws_metadata
http_access allow localhost
http_access deny all
```

11. After editing the config file, restart the proxy container as sudo:

```
docker restart squid
```

12. Go back to the Cloud Volumes ONTAP CLI and verify that Cloud Volumes ONTAP can send AutoSupport messages:

    ```
    autosupport check show-details
    ```

Configure EMS

The Event Management System (EMS) collects and displays information about events that occur on ONTAP systems. To receive event notifications, you can set event destinations (email addresses, SNMP trap hosts, or syslog servers) and event routes for a particular event severity.

You can configure EMS using the CLI. For instructions, refer to ONTAP docs: EMS configuration overview.

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